Jonathan D. Cohen
Curriculum Vitae

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SUMMARY

I am a computational cognitive neuroscientist, with over three decades of research experience in computational and mathematical modeling as well as empirical studies of human brain function and behavior, focusing on the neural mechanisms responsible for cognitive control and human intelligence, and how our growing understanding of these can be brought to bear in the design of computational architectures with more human-like cognitive capabilities. My work lies at the points of contact between neuroscience, psychology, computer science and mathematics, as well as behavioral economics and psychiatry, and involves collaborations with investigators in each of these fields. I also have considerable experience in the coordination and administration of scientific research, as one of the two founding Co-Directors of the Princeton Neuroscience Institute, leading multi-institutional projects (NIHM Conte Center; Templeton Center Grant; NSF Convergence Accelerator grant; PNI-Intel Labs collaboration), and a number of open source software development projects.

Theoretical contributions. Some of the contributions that have emerged from the theoretical work of my colleagues and I are: the first computationally-explicit models of how cognitive control may be implemented in the brain [5] and the role of prefrontal cortex in control [64]; the role of dopaminergic function in the gating and updating of information in prefrontal cortex [4][56], noradrenergic regulation of the explore/exploit tradeoff [47][115][129], and the interaction of these modulatory systems in adaptive regulation of exploration in reinforcement learning [108]; how these mechanisms may be disturbed in psychiatric disorders [9][49][109]; the role of anterior cingulate cortex in performance monitoring [68][84][180] and the optimal allocation of control [172][202]; mathematical analysis of optimal control of simple decision making processes [124][148]; normative approaches to understanding capacity constraints associated with working memory [61][133] and cognitive control [181][240][203]; and how the brain regulates the balance between flexible control-dependent and efficient automatic processing [215]. Increasingly, our work has come to focus on how these mechanisms contribute to higher cognitive functions and human intelligence, such as the control of memory, planning, and abstract reasoning [227][250][249][259], including ways in which the human brain achieves the flexibility of symbolic forms of computation [104][174][237][253] while preserving the efficiency of computation in neural networks, and how this can be used to inform research in machine learning and artificial intelligence [256][269][286].

Empirical and methodological contributions. The theoretical work summarized above has served as the foundation for a number of empirical and methodological contributions. Empirical contributions include: the first demonstrations in humans of sustained activity in PFC associated with working memory performance [17][31]; the distinction between the roles of dorsolateral PFC (in the regulatory functions of control) and anterior cingulate cortex (monitoring and evaluative functions of control [45][51][57][94]; and the role of the locus coeruleus / norepinephrine system in regulating the explore-exploit tradeoff [157][208]. We have also made influential contributions to advances in quantitative methods in cognitive neuroscience, including: the introduction of cluster size correction into the analysis of fMRI data [43]; the use of fMRI to directly study midbrain neuromodulatory nuclei [144]; the design of systems for realtime fMRI analysis [19][251] and closed-loop feedback designs [191][242]; and whole brain, full correlation analysis of fMRI data and its use in realtime analysis [184]. Finally, I have lead or co-lead several large software development projects, including: PsyScope [14], the first graphical environment for the design and execution of cognitive behavioral experiments; BrainIAK [245][251] (in collaboration with Intel Labs), an open-source, python-based toolbox for the implementation and optimization of advanced methods of brain image analysis; PsyNeuLink, an open-source, python-based environment for the design and exchange of computational models of brain and cognitive function; SweetPea [255], a framework for specifying empirical experimental designs and machine learning training environments using factorial structure, and generating maximally unbiased sampling of trials; and a model description format [269] for expressing models of brain and cognitive function as computational graphs in machine readable form for exchange across modeling environments.

BIOGRAPHICAL

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EDUCATION and TRAINING

UNDERGRADUATE:

1973-77 Yale University — B.A., 1977 Biology and Philosophy

GRADUATE:

1979-83 University of Pennsylvania — M.D., 1983 Medicine

1987-90 Carnegie Mellon University — Ph.D., 1990 Cognitive Psychology

POST-GRADUATE:

1983-89 Internship in General Medicine, Neurology and Psychiatry
Residency in Psychiatry
Stanford University School of Medicine

1985-87 NIMH Research Training Fellowship
Department of Psychiatry and Behavioral Sciences
Stanford University School of Medicine

APPOINTMENTS and POSITIONS

ACADEMIC:

1989-2005	Assistant to Full Professor of Psychiatry Western Psychiatric Institute and Clinic University of Pittsburgh
1990-98	Assistant to Associate Professor of Psychology Carnegie Mellon University
1992-2005	Director, Clinical Cognitive Neuroscience Laboratory University of Pittsburgh
1998-2005	Professor of Psychology, Princeton University
1999-2007	Founding Director, Center for the Study of Brain, Mind and Behavior Princeton University
2000-2008	Director, Program in Neuroscience Princeton University
2005-2012	Eugene Higgins Professor of Psychology, Princeton University
2005-2022	Founding Co-Director, Princeton Neuroscience Institute Princeton University
2007-present	Director, Scully Center for the Neuroscience of Mind and Behavior, Princeton University
2012-present	Robert Bendheim and Lynn Bendheim Thoman Professor in Neuroscience, Princeton University
2023-present	Director, Graduate Certificate Program in Statistics and Machine Learning, Princeton University
2025-present	Senior Advisor for Computing and Data, Office of the Dean for Research, Princeton University

HONORS and AWARDS

B.A. Cum Laude Distinction in the Biology Major Distinction in the Philosophy Major Yale University	1977
Miller Foundation Prize for Research in Psychiatry	1986
Department of Psychiatry and Behavioral Sciences Stanford University School of Medicine	
Annual Resident Research Award Northern California Psychiatric Society	1986
Joseph Zubin Memorial Fund Award for Research in Psychopathology	1993
Kempf Fund Award for Research Development in Psychobiological Psychiatry, American Psychiatric Association	2000
James McKeen Cattell Fund Sabbatical Fellowship Award	2003
Eugene Higgins Chaired Professorship, Princeton University	2005
Salmon Award Lecturer, New York Academy of Medicine	2006
Fellow, Association for Psychological Science	2007
Edward J. Sachar Award, Columbia University School of Medicine	2007
American Psychological Association Distinguished Scientific Contribution Award	2010
Fellow, American Association for the Advancement of Science	2012
William James Fellow Award, Association for Psychological Science	2018
Fellow, Cognitive Science Society	2019
Vannevar Bush Faculty Fellowship, Office of the Under Secretary of Defense for Research & Engineering	2021
Member, American Academy of the Arts & Sciences	2022
Lifetime Achievement Award, Society for Experimental and Cognitive Science, Division 3 of the American Psychological Association	2022
Fellow, Society of Experimental Psychologists	2025
Mercator Fellow, Stiftun Mercator	2026

PUBLICATIONS

1. Peer-Reviewed Articles

[302] S. M. Frankland, R. Marjieh, M. Nurisso, J. Fluegemann, T. W., G. Petri, R. L. Lewis, and J. D. Cohen (in press). Capacity limits in cognitive processing reflect the curse of generalization. *Psychological Review*. URL: https://osf.io/preprints/psyarxiv/cjuxb%5C_v2.

[301] T. Giallanza, T. T. Rogers, and J. D. Cohen (in press). An integrated model of semantics and control, Part 2: Solving the similarity paradox through context inference. *Psychological Review*. URL: https://osf.io/preprints/psyarxiv/fxc87.

[300] G. Petri, S. Musslick, and J. D. Cohen (in press). An information-theoretic approach to reward rate optimization in the tradeoff between controlled and automatic processing in neural network architectures. *eLife*. URL: https://doi.org/10.1101/2023.09.18.558214.

[299] J. Tromp, S. Nieuwenhuis, J. D. Cohen, and B. J. Jongkees (in press). A normative account of the trade-off between cognitive stability and flexibility in task switching. *Psychological Review*.

[298] U. Haputhanthrige, D. Campbell, R. Assouel, J. D. Cohen, and T. W. Webb (2027). Binding visual features point by point. *Advances in Neural Information Processing Systems (NeurIPS)*. URL: https://arxiv.org/abs/2605.25427.

[297] H. C. Conklin, T. Hosking, T. Yi-Chern, J. D. Cohen, S.-J. Leslie, T. L. Griffiths, M. Bartolo, and S. Goldfarb-Tarrant (2026). Learning is forgetting; LLM training as lossy compression. *ICLR 2026: Proceedings of the International Conference on Learning Representations*.

[296] J. Fluegemann, J. Huang, M. L. Rosendahl, J. Busemeyer, and J. D. Cohen (2026). A multiple-well framework for human perceptual decision-making. *Entropy* 28, p. 232. URL: https://www.mdpi.com/1099-4300/28/2/232.

[295] J. Masís, Y. Strittmatter, and J. D (2026). An episodic memory model can account for reinforcement learning in humans. *CogSci 2026: Proceedings of the Annual Meeting of the Cognitive Science Society*.

[294] A. Nam, H. Conklin, Y. Yang, T. Griffiths, J. D. Cohen, and S.-J. Leslie (2026). Causal head gating: A framework for interpreting roles of attention heads in transformers. *Advances in Neural Information Processing Systems (NeurIPS)* 38, pp. 115609–115631.

[293] M. Nurisso, J. Fernando, R. Deshpande, A. Perotti, R. Marjieh, S. M. Frankland, R. L. Lewis, T. W. Webb, D. Campbell, F. Vaccarino, J. D. Cohen, and G. Petri (2026). Bound by semanticity: universal laws governing the generalization-identification tradeoff. *ICLR 2026: Proceedings of the International Conference on Learning Representations*. URL: https://arxiv.org/abs/2506.14797.

[292] H. Ritz, A. Jha, N. D. Daw, and J. D. Cohen (2026). Inter-trial convergence of neural task states supports cognitive flexibility. *Current Biology*. URL: https://doi.org/10.1016/j.cub.2026.07.028.

[291] D. Savietto, D. Campbell, A. Panisson, M. Nurisso, G. Petri, J. D. Cohen, and A. Perotti (2026). The geometry of representational failures in vision language models. *ICML 2026: Proceedings of the International Conference on Machine Learning*. URL: https://arxiv.org/pdf/2602.07025.

[290] Z. Dulberg, R. Dubey, and J. D. Cohen (2025). Adapting to loss: A normative account of grief. *Psychological Review*. URL: https://www.biorxiv.org/content/10.1101/2024.02.06.578702v3.

[289] J. A. M. Obando, S. Musslick, and J. D. Cohen (2025). Learning expectations shape cognitive control allocation. *Proceedings of the National Academy of Sciences* 122.44, e2416720122. URL: https://doi.org/10.31234/osf.io/d2cbg.

[288] S. Suresh, K. Mukherjee, T. Giallanza, X. Yu, P. Patil, J. D. Cohen, and T. S. Rogers (2025). AI-enhanced semantic feature norms for 786 concepts. *ICLR 2025 and CHI 2025*. URL: https://openreview.net/forum?id=m7bkC8Lxoh.

[287] Y. Yang, D. I. Campbell, K. Huang, M. Wang, J. D. Cohen, and T. W. Webb (2025). Emergent symbolic mechanisms support abstract reasoning in large language models. *ICML 2025: Proceedings of the International Conference on Machine Learning*. URL: https://arxiv.org/pdf/2502.20332.

[286] A. Altabaa, T. Webb, J. D. Cohen, and J. Lafferty (2024). Abstractors: Transformer modules for symbolic message passing and relational reasoning. *ICLR 2024: Proceedings of the International Conference on Learning Representations*. URL: https://arxiv.org/abs/2304.00195.

[285] A. O. Beukers, M. Hamin, K. Norman, and J. D. Cohen (2024). When working memory may just be working, not memory. *Psychological Review*. URL: https://pubmed.ncbi.nlm.nih.gov/37956060/.

[284] L. A. Bustamante, D. M. Barch, J. Solis, T. Oshinowo, I. Grahek, A. B. Konova, N. D. Daw, and J. D. Cohen (2024). Major depression symptom severity associations with willingness to exert effort and patch foraging strategy. *Psychological Medicine*, pp. 1–12. URL: https://pubmed.ncbi.nlm.nih.gov/39618329/.

[283] D. Campbell and J. D. Cohen (2024). A relational inductive bias for dimensional abstraction in neural networks. *CogSci 2024: Proceedings of the Annual Meeting of the Cognitive Science Society* 46. URL: https://arxiv.org/abs/2402.18426.

[282] D. Campbell, S. Kumar, T. Giallanza, T. L. Griffiths, and J. D. Cohen (2024). Geometric abstraction in large foundation models. *CogSci 2024: Proceedings of the Annual Meeting of the Cognitive Science Society* 46. URL: https://arxiv.org/abs/2309.17363.

[281] D. I. Campbell, S. Rane, T. Giallanza, N. De Sabbata, K. Ghods, A. Joshi, A. Ku, T. L. Griffiths, and J. D. Cohen (2024). Understanding the limits of vision language models through the lens of the binding problem. *Advances in Neural Information Processing Systems (NeurIPS)*. URL: https://arxiv.org/abs/2411.00238.

[280] V. Geadah, J. Arbelaiz, H. Ritz, N. D. Daw, J. D. Cohen, and J. W. Pillow (2024). Inferring system and optimal control parameters of closed-loop systems from partial observations, pp. 8006–8013.

[279] T. Giallanza, D. Campbell, J. D. Cohen, and T. T. Rogers (2024). An integrated model of semantics and control. *Psychological Review* 132.5, pp. 1128–1177. URL: https://pubmed.ncbi.nlm.nih.gov/39052340/.

[278] T. Gillanza, D. Campbell, and J. D. Cohen (2024). Toward the emergence of intelligent control: Episodic generalization and optimization. *OpenMind*. URL: https://osf.io/preprints/psyarxiv/dzvpy/.

[277] M. C. Iordan, V. J. Ritvo, K. A. Norman, N. B. Turk-Browne, and J. D. Cohen (2024). Sculpting new visual categories into the human brain. *Proceedings of the National Academy of Sciences, USA*. URL: https://www.pnas.org/doi/10.1073/pnas.2410445121.

[276] S. Mondal, J. D. Cohen, and T. W. Webb (2024). Slot abstractors: Toward scalable abstract visual reasoning. *ICML 2024: Proceedings of the 41th International Conference on Machine*. URL: https://arxiv.org/pdf/2403.03458.

[275] T. Webb, S. Mondal, and J. D. Cohen (2024). Systematic visual reasoning through object-centric relational abstraction. *Advances in Neural Information Processing Systems (NeurIPS)*. URL: https://arxiv.org/abs/2306.02500.

[274] T. W. Webb, S. M. Frankland, A. Altabaa, S. Segert, K. Krishnamurthy, D. Campbell, J. Russin, T. Giallanza, Z. Dulberg, R. O'Reilly, J. Lafferty, and J. D. Cohen (2024). The relational bottleneck as an inductive bias for efficient abstraction. *Trends in Cognitive Science*. URL: https://arxiv.org/abs/2309.06629.

[273] M. Agrawal, J. C. Peterson, J. D. Cohen, and T. L. Griffiths (2023). Stress, intertemporal choice, and mitigation behavior during the COVID-19 pandemic. *Journal of Experimental Psychology: General*. URL: https://dx.doi.org/10.1037/xge0001417.

[272] A. M. Bornstein, M. Aly, S. F. Feng, N. B. Turk-Browne, K. A. Norman, and J. D. Cohen (2023). Associative memory retrieval modulates upcoming perceptual decisions. *Cognitive, Affective, and Behavioral Neuroscience*. URL: https://pubmed.ncbi.nlm.nih.gov/37316611/.

[271] L. A. Bustamante, T. Oshinowo, J. Lee, E. Tong, A. R. Burton, A. S. Shenav, N. Daw, and J. D. Cohen (2023). Effort Foraging Task reveals positive correlation between individual differences in the cost of cognitive and physical effort in humans. *Proceedings of the National Academy of Sciences, USA*. URL: https://pubmed.ncbi.nlm.nih.gov/38064507/.

[270] Z. Dulberg, R. Dubey, I. M. Berwian, and J. D. Cohen (2023). Having multiple selves helps learning agents explore and adapt in complex changing worlds. *Proceedings of the National Academy of Sciences, USA* 120.28, e2221180120. URL: https://www.pnas.org/doi/10.1073/pnas.2221180120.

[269] P. Gleeson, S. Crook, D. Turner, K. Mantel, M. Raunak, and J. D. Cohen (2023). Integrating model development across computational neuroscience, cognitive science and machine learning. *Neuron* 111.10, pp. 1526–1530. URL: https://www.sciencedirect.com/science/article/pii/S0896627323002611?via%3Dihub.

[268] M. Ho, J. D. Cohen, and T. L. Griffiths (2023). Rational simplification and rigidity in human planning. *Psychological Science* 34.11, pp. 1281–1292. URL: https://pubmed.ncbi.nlm.nih.gov/37878525/.

[267] S. Kumar, I. Dasgupta, R. Marjieh, N. D. Daw, J. D. Cohen, and T. L. Griffiths (2023). Disentangling abstraction from statistical pattern matching in human and machine learning. *PLOS Computational Biology* 19.8. URL: https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1011316.

[266] J. Masís, D. Melnikoff, L. Barrett, and J. D. Cohen (2023). When to choose: Information seeking in the speed-accuracy tradeoff. *Cogsci 2023: Proceedings of the Annual Meeting of the Cognitive Science Society* 45. URL: https://escholarship.org/uc/item/46j4g3p4#main.

[265] S. S. Mondal, S. M. Frankland, T. W. Webb, and J. D. Cohen (2023). Determinantal Point Process Attention Over Grid Codes Supports Extrapolation. *eLife*. URL: http://arxiv.org/abs/2305.18417.

[264] S. S. Mondal, T. W. Webb, and J. D. Cohen (2023). Learning to reason over visual objects. *ICLR 2023: Proceedings of the International Conference on Learning Representations*. URL: https://arxiv.org/abs/2303.02260.

[263] L. Rier, S. Michelmann, H. Ritz, V. Shah, R. M. Hill, J. Osborne, C. Doyle, N. Holmes, R. Bowtell, M. J. Brookes, K. Norman, U. Hasson, J. D. Cohen, and E. Boto (2023). Test-Retest reliability of the human connectome: An OPM-MEG study. *Imaging Neuroscience* 1, pp. 1–20. URL: https://zenodo.org/records/7525341.

[262] H. Ritz, W. Wolf, and J. D. Cohen (2023). Continuous and discrete transitions during task-switching. *CogSci 2023: Proceedings of the Annual Meeting of the Cognitive Science Society* 45. URL: https://escholarship.org/uc/item/9kq1p2rt.

[261] S. Segert and J. D. Cohen (2023). Beyond transformers for function learning. *CogSci 2023: Proceedings of the Annual Meeting of the Cognitive Science Society* 45. URL: https://escholarship.org/uc/item/4wg2h9g1#main.

[260] T. W. Webb, S. S. Mondal, and J. D. Cohen (2023). Systematic visual reasoning through object-centric relational abstraction. *ICLR 2023: Proceedings of the International Conference on Learning Representations*. URL: https://arxiv.org/abs/2306.02500.

[259] M. Ho, D. Abel, C. G. Correa, M. L. Littman, J. D. Cohen, and T. L. Griffiths (2022). People construct simple representations to plan. *Nature* 606.7912, pp. 129–136. URL: https://rdcu.be/cNVkp.

[258] C. Iordan, T. Giallanza, C. T. Ellis, N. M. Beckage, and J. D. Cohen (2022). Context matters: Recovering human semantic structure from machine learning analysis of large-scale text corpora. *Cognitive Science* 46.2, e13085. URL: https://onlinelibrary.wiley.com/doi/full/10.1111/cogs.13085.

[257] G. A. Kane, M. H. James, A. Shenhav, N. D. Daw, J. D. Cohen, and G. Aston-Jones (2022). Rat anterior cingulate cortex continuously signals decision variables in a patch foraging task. *Journal of Neuroscience* 42.29, pp. 5730–5744. URL: https://doi.org/10.1523/JNEUROSCI.1940-21.2022.

[256] S. Kumar, C. G. Correa, I. Dasgupta, R. Marjieh, M. Y. Hu, R. D. Hawkins, N. D. Daw, J. D. Cohen, K. Narasimhan, and T. L. Griffiths (2022). Using natural language and program abstractions to instill human inductive biases in machines. *Advances in Neural Information Processing Systems (NeurIPS)*. URL: https://arxiv.org/abs/2205.11558.

[255] S. Musslick, A. Cherkaev, B. Draut, A. Butt, V. Srikumar, M. Flatt, and J. D. Cohen (2022). SweetPea: A standard language for factorial experimental design. *Behavioral Methods in Research* 54.2, pp. 805–829. URL: https://doi.org/10.3758/s13428-021-01598-2.

[254] S. Segert and J. D. Cohen (2022). A Self-Supervised Framework for Function Learning and Extrapolation. *Transactions on Machine Learning Research*. URL: https://openreview.net/pdf?id=ILPFasEaHA.

[253] S. Segert and J. D. Cohen (2022). Maximum entropy function learning. *CogSci 2022: Proceedings of the Annual Meeting of the Cognitive Science Society* 44. URL: https://escholarship.org/uc/item/2tb2f84r.

[252] J. Vesely, R. Pothukuchi, S. Gupta, J. Ketaki, J. D. Cohen, and A. Bhattacharjee (2022). Distill: Domain-specific compilation for cognitive models. *2022 IEEE/ACM International Symposium on Code Generation and Optimization (CGO)*, pp. 301–312. URL: https://doi.org/10.1109/cgo53902.2022.9741278.

[251] G. Wallace, S. Polcyn, P. P. Brooks, A. Mennen, K. Zhao, P. S. Scotti, S. Michelmann, K. Li, N. B. TurkBrowne, J. D. Cohen, and K. A. Norman (2022). RT-Cloud: A cloud-based software framework to simplify and standardize real-time fMRI. *NeuroImage* 257, p. 119295. URL: https://doi.org/10.1016/j.neuroimage.2022.119295.

[250] M. Agrawal, M. Mattar, N. D. Daw, and J. D. Cohen (2021). The temporal dynamics of opportunity costs: A normative account of cognitive fatigue and boredom. *Psychological Review* 129.3, p. 564. URL: https://doi.org/10.1037/rev0000309.

[249] A. O. Beukers, T. J. Buschman, J. D. Cohen, and K. A. Norman (2021). Is activity silent working memory simply episodic memory? *Trends in Cognitive Sciences* 25.4, pp. 284–293. URL: https://doi.org/10.1016/j.tics.2021.01.003.

[248] L. Bustamante, F. Lieder, S. Musslick, A. Shenhav, and J. D. Cohen (2021). Learning to overexert cognitive control in a Stroop task. *Cognitive, Affective, and Behavioral Neuroscience* 21.3, pp. 453–471. URL: https://link.springer.com/article/10.3758/s13415-020-00845-x.

[247] Z. Dulberg, T. Webb, and J. D. Cohen (2021). Modeling the development of counting with memory-augmented neural networks. *CogSci 2020: Proceedings of the Annual Meeting of the Cognitive Science Society* 43. URL: https://arxiv.org/abs/2105.10577.

[246] E. Eldar, V. Felso, J. D. Cohen, and Y. Niv (2021). A pupillary index of susceptibility to decision biases. *Nature Human Behaviour* 5, pp. 653–662. URL: https://doi.org/10.1038/s41562-020-01006-3.

[245] M. Kumar, M. Anderson, J. Antony, C. Baldassano, P. B. Brooks, M. B. Cai, et al. (2021). BrainIAK: The Brain Imaging Analysis Kit. *Aperture Neuro*. URL: https://doi.org/10.52294/31bb5b68-2184-411b-8c00-a1dacb61e1da.

[244] S. Kumar, I. Dasgupta, J. D. Cohen, N. D. Daw, and T. L. Griffiths (2021). Meta-learning of structured task distributions in humans and machines. *ICLR 2021: Proceedings of the International Conference on Learning Representations*. URL: https://openreview.net/forum?id=--gvHfE3Xf5.

[243] J. Masis, S. Musslick, and J. D. Cohen (2021). The value of learning and cognitive control allocation. *CogSci 2021: Proceedings of the Annual Meeting of the Cognitive Science Society* 43. URL: https://escholarship.org/uc/item/7w0223v0.

[242] A. C. Mennen, N. B. Turk-Browne, G. Wallace, D. Seok, A. Jaganjac, J. Stock, and Y. I. Sheline (2021). Cloud-based fMRI neurofeedback to reduce the negative attentional bias in depression: a proof-of-concept study. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging* 6.4, pp. 490–497. URL: https://www.researchgate.net/publication/346510051_Cloud-based_fMRI_neurofeedback_to_reduce_the_negative_attentional_bias_in_depression_a_proof-of-concept_study.

[241] I. Momennejad, J. Lewis-Peacock, K. A. Norman, J. D. Cohen, S. Singh, and R. L. Lewis (2021). Rational use of episodic and working memory: A normative account of prospective memory. *Neuropsychologia* 158, p. 107657. URL: https://doi.org/10.1016/j.neuropsychologia.2020.107657.

[240] S. Musslick and J. D. Cohen (2021). Rationalizing constraints on the capacity for cognitive control. *Trends in Cognitive Sciences* 25.9, pp. 757–775. URL: https://www.sciencedirect.com/science/article/pii/S1364661321001480?via%3Dihub.

[239] G. Petri, S. Musslick, B. Dey, K. Özcimder, N. K. Ahmed, T. L. Willke, and J. D. Cohen (2021). Topological limits to parallel processing capability of network architectures. *Nature Physics* 17.5, pp. 646–651. URL: https://doi.org/10.1038/s41567-021-01170-x.

[238] A. Prat-Carrabin, R. C. Wilson, J. D. Cohen, and R. A. da Silveira (2021). Human inference in changing environments with temporal structure. *Psychological Review* 128.5, pp. 879–912. URL: https://doi.org/10.1037/rev0000276.

[237] T. W. Webb, I. Sinha, and J. D. Cohen (2021). Emergent symbols through binding in external memory. *ICLR 2021: Proceedings of the International Conference on Learning Representations*. URL: https://doi.org/10.48550/arXiv.2012.14601.

[236] J. D. Cohen, K. M. Ericson, D. Laibson, and J. M. White (2020). Measuring Time Preferences. *Journal of Economic Literature* 58.2, pp. 299–347. URL: https://doi.org/10.1257/jel.20191074.

[235] C. T. Ellis, C. Baldassano, A. Schapiro, M. B. Cai, and J. D. Cohen (2020). Facilitating open-science with realistic fMRI simulation: Validation and application. *PeerJ* 8, e8564. URL: https://peerj.com/articles/8564/.

[234] S. M. Frankland and J. D. Cohen (2020). Determinantal point processes for memory and structured inference. *CogSci 2020: Proceedings of the Annual Meeting of the Cognitive Science Society* 42. URL: https://cognitivesciencesociety.org/cogsci20/papers/0812/index.html.

[233] M. K. Ho, D. Abel, J. D. Cohen, M. L. Littman, and T. L. Griffiths (2020). The Efficiency of human cognition reflects planned information processing. *Proceedings of the 34th AAAI Conference on Artificial Intelligence*. URL: https://arxiv.org/abs/2002.05769.

[232] M. Mosleh, K. Kyker, J. D. Cohen, and D. G. Rand (2020). Globalization and the Rise and Fall of Cognitive Control. *Nature Communications* 11, p. 3099. URL: https://doi.org/10.1038/s41467-020-16850-0.

[231] M. Rosendahl, A. Bizyaeva, and J. D. Cohen (2020). A novel quantum approach to the dynamics of decision making. *CogSci 2020: Proceedings of the Annual Meeting of the Cognitive Science Society* 42. URL: https://cognitivesciencesociety.org/cogsci20/papers/0508/0508.pdf.

[230] S. Segert, M. Tepper, J. Turek, and J. D. Cohen (2020). Relaxed graph matching for analogical reasoning. *ICLR 2020: Proceedings of the International Conference on Learning Representations, Workshop on Bridging AI and Cognitive Science*. URL: https://baicsworkshop.github.io/pdf/BAICS_4.pdf.

[229] T. W. Webb, Z. Dulberg, S. M. Frankland, A. A. Petrov, R. C. O'Reilly, and J. D. Cohen (2020). Learning representations that support extrapolation. *ICML 2020: Proceedings of the International Conference on Machine Learning* 119. URL: https://arxiv.org/abs/2007.05059.

[228] C. T. Ellis, M. Lesnick, G. Henselman-Petrusek, B. Keller, and J. D. Cohen (2019). Feasibility of topological data analysis for event-based fMRI. *Network Neuroscience*, pp. 1–12. URL: https://pubmed.ncbi.nlm.nih.gov/31410374/.

[227] S. M. Frankland, T. W. Webb, A. A. Petrov, R. C. O'Reilly, and J. D. Cohen (2019). Extracting and utilizing abstract, structured representations for analogy. *CogSci 2019: Proceedings of the Annual Meeting of the Cognitive Science Society* 41. URL: https://www.alexpetrov.com/pub/cogsci19/.

[226] A. N. Hoskin, A. M. Bornstein, K. A. Norman, and J. D. Cohen (2019). Refresh my memory: Episodic memory reinstatements intrude on working memory maintenance. *Cognitive, Affective, and Behavioral Neuroscience* 19, pp. 338–354. URL: https://doi.org/10.3758/s13415-018-00674-z.

[225] G. A. Kane, A. M. Bornstein, A. Shenhav, R. C. Wilson, N. D. Daw, and J. D. Cohen (2019). Rats exhibit similar biases in foraging and intertemporal choice tasks. *eLife* 8, e48429. URL: http://www.biorxiv.org/content/10.1101/497321v1.

[224] S. Musslick, A. Bizyaeva, S. Aragon, N. Leonard, and J. D. Cohen (2019). Stability-flexibility dilemma in cognitive control: A dynamical system perspective. *CogSci 2019: Proceedings of the Annual Meeting of the Cognitive Science Society* 41. URL: https://par.nsf.gov/biblio/10125021.

[223] S. Musslick and J. D. Cohen (2019). A mechanistic account of constraints on control-dependent processing: Shared representation, conflict and persistence. *CogSci 2019: Proceedings of the Annual Meeting of the Cognitive Science Society* 41. URL: https://collaborate.princeton.edu/en/publications/a-mechanistic-account-of-constraints-on-control-dependent-process.

[222] S. Musslick, J. D. Cohen, and A. Shenhav (2019). Decomposing individual differences in cognitive control: A model-based approach. *CogSci 2019: Proceedings of the Annual Meeting of the Cognitive Science Society* 41. URL: https://static1.squarespace.com/static/56e19a6d04426223be32a8b9/t/5ce835e653450ad74d48ee0c/1558722022988/Musslick_5_9.pdf.

[221] R. Poldrack, F. Feingold, M. Frank, P. Gleeson, G. de Hollander, Q. J. M. Huys, and J. D. ... Cohen (2019). The importance of standards for sharing computational models and data. *Computational Brain & Behavior* 2, pp. 229–232. URL: https://link.springer.com/article/10.1007/s42113-019-00062-x.

[220] M. Spitzer, S. Musslick, M. Shvartsman, A. Shenhav, and J. D. Cohen (2019). Asymmetric switch costs as a function of task strength. *CogSci 2019: Proceedings of the Annual Meeting of the Cognitive Science Society* 41. URL: https://static1.squarespace.com/static/56e19a6d04426223be32a8b9/t/5ce835a64e17b62b88216749/1558721958757/Spitzer_5_9.pdf.

[219] R. C. Wilson, A. Shenhav, M. Straccia, and J. D. Cohen (2019). The eighty-five percent rule for optimal learning. *Nature Communications* 10, p. 4646. URL: https://www.nature.com/articles/s41467-019-12552-4.

[218] M. C. Iordan, C. T. Ellis, M. Lesnick, D. N. Osherson, and J. D. Cohen (2018). Feature ratings and empirical dimension-specific similarity explain distinct aspects of semantic similarity judgments. *CogSci 2018: Proceedings of the Annual Meeting of the Cognitive Science Society* 40. URL: https://collaborate.princeton.edu/en/publications/feature-ratings-and-empirical-dimension-specific-similarity-expla.

[217] S. Musslick, S. J. Jang, M. Shvartsman, A. Shenhav, and J. D. Cohen (2018). Constraints associated with cognitive control and the stability-flexibility dilemma. *CogSci 2018: Proceedings of the Annual Meeting of the Cognitive Science Society* 40. URL: https://www.researchgate.net/publication/325324409_Constraints_associated_with_cognitive_control_and_the_stability-flexibility_dilemma.

[216] S. Musslick, J. D. Cohen, and A. Shenhav (2018). Estimating the costs of cognitive control from task performance: Theoretical validation and potential pitfalls. *CogSci 2018: Proceedings of the Annual Meeting of the Cognitive Science Society* 40. URL: https://collaborate.princeton.edu/en/publications/estimating-the-costs-of-cognitive-control-from-task-performance-t.

[215] Y. Sagiv, S. Musslick, Y. Niv, and J. D. Cohen (2018). Efficiency of learning vs. processing: Towards a normative theory of multitasking. *CogSci 2018: Proceedings of the Annual Meeting of the Cognitive Science Society* 40. URL: http://arxiv.org/abs/2007.03124.

[214] A. Shenhav, M. A. Straccia, S. Musslick, J. D. Cohen, and M. M. Botvinick (2018). Dissociable neural mechanisms track evidence accumulation for selection of attention versus action. *Nature Communications* 9.1, p. 2485. URL: https://www.nature.com/articles/s41467-018-04841-1.

[213] N. Alon, D. Reichman, I. Shinkar, T. Wagner, S. Musslick, J. D. Cohen, T. Griffiths, B. Dey, and K. Ozcimder (2017). A graph-theoretic approach to multitasking. *Advances in Neural Information Processing Systems (NeurIPS)* 30, pp. 2097–2106. URL: https://arxiv.org/abs/1611.02400.

[212] I. C. Ballard, B. Kim, A. Liatsis, G. Aydogan, J. D. Cohen, and S. M. McClure (2017). More is meaningful: the magnitude effect in intertemporal choice depends on self-control. *Psychological Science*. DOI: 10.1177/0956797617711455. URL: https://pubmed.ncbi.nlm.nih.gov/28858559/.

[211] V. R. Bejjanki, R. A. daSilveira, J. D. Cohen, and N. B. Turk-Browne (2017). Noise correlations in the human brain and their impact on pattern classification. *PLoS computational biology* 13.8, e1005674. URL: https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1005674.

[210] J. D. Cohen, N. Daw, B. Engelhardt, U. Hasson, K. Li, Y. Niv, K. A. Norman, J. Pillow, P. Ramadge, N. B. Turk-Browne, and T. L. Willke (2017). Computational Approaches to fMRI Analysis. *Nature Neuroscience* 20.3, pp. 304–313. URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5457304/.

[209] G. Kane, A. Bornstein, N. D. Daw, J. D. Cohen, R. C. Wilson, and A. Shenhav (2017). Mechanisms of overharvesting in patch foraging. *CogSci 2017: Proceedings of the Annual Meeting of the Cognitive Science Society* 39. URL: http://aaron.bornstein.org/cv/pubs/2017_kbswdc_cs.pdf.

[208] G. A. Kane, E. M. Vazey, R. C. Wilson, A. Shenhav, N. D. Daw, G. Aston-Jones, and J. D. Cohen (2017). Increased locus coeruleus tonic activity causes disengagement from a patch foraging task. *Cognitive, Affective, and Behavioral Neuroscience* 17.6, pp. 1073–1083. URL: https://pubmed.ncbi.nlm.nih.gov/28900892/.

[207] P. M. Krueger, M. K. van Vugt, P. Simen, L. Nystrom, P. Holmes, and J. D. Cohen (2017). Evidence accumulation detected in BOLD signal using slow perceptual decision making. *Journal of Neuroscience Methods* 281, pp. 1–12. URL: https://pubmed.ncbi.nlm.nih.gov/28131862/.

[206] P. M. Krueger, R. C. Wilson, and J. D. Cohen (2017). Strategies for exploration in the domain of losses. *Judgment and Decision Making* 12.2, pp. 104–117. URL: https://psycnet.apa.org/record/2017-17480-002.

[205] S. Musslick, A. M. Saxe, K. Ozcimder, B. Dey, G. Henselman, and J. D. Cohen (2017). Multitasking capability versus learning efficiency in neural network architectures. *CogSci 2017: Proceedings of the Annual Meeting of the Cognitive Science Society* 39. URL: https://www.researchgate.net/publication/317019423_Multitasking_Capability_Versus_Learning_Efficiency_in_Neural_Network_Architectures.

[204] K. Ozcimder, B. Dey, S. Musslick, G. Petri, N. Ahmed, T. L. Willke, and J. D. Cohen (2017). A formal approach to modeling the cost of cognitive control. *CogSci 2017: Proceedings of the Annual Meeting of the Cognitive Science Society* 39. URL: https://arxiv.org/abs/1706.00085.

[203] D. G. Rand, D. Tomlin, A. Bear, E. Ludvig, and J. D. Cohen (2017). Cyclical population dynamics of automatic versus controlled processing: An evolutionary pendulum. *Psychological Review* 124.5, pp. 626–642. URL: https://pubmed.ncbi.nlm.nih.gov/28703606/.

[202] A. Shenhav, S. Musslick, F. Lieder, W. Kool, T. L. Griffiths, J. D. Cohen, and M. M. Botvinick (2017). Toward a rational and mechanistic account of mental effort. *Annual Review of Neuroscience* 40, pp. 99–124. URL: https://pubmed.ncbi.nlm.nih.gov/28375769/.

[201] V. Srivastava, S. F. Feng, J. D. Cohen, A. E. Leonard, and A. Shenhav (2017). A Martingale Analysis of First Passage Times of Time-Independent Wiener Diffusion Models. *Journal of Mathematical Psychology* 77, pp. 94–110. URL: https://pubmed.ncbi.nlm.nih.gov/28630524/.

[200] D. Tomlin, A. Nedic, D. A. Prentice, P. Holmes, and J. D. Cohen (2017). The integration of social influence and reward: Computational approaches and neural evidence. *Cognitive, Affective, and Behavioral Neuroscience* 17.4, pp. 784–808. URL: https://link.springer.com/article/10.3758/s13415-017-0512-1.

[199] E. Eldar, Y. Niv, and J. D. Cohen (2016). Do you see the forest or the tree? Neural gain and breadth versus focus in perceptual processing. *Psychological Science* 27.12, pp. 1632–1643. URL: https://pubmed.ncbi.nlm.nih.gov/28195019/.

[198] A. Geana, R. C. Wilson, N. Daw, and J. D. Cohen (2016). Boredom, Information-Seeking and Exploration. *CogSci 2016: Proceedings of the Annual Meeting of the Cognitive Science Society* 38. URL: https://www.princeton.edu/~ndaw/gwdc16.pdf.

[197] A. Geana, R. C. Wilson, N. Daw, and J. D. Cohen (2016). Information-Seeking, Learning and the Marginal Value Theorem: A Normative Approach to Adaptive Exploration. *CogSci 2016: Proceedings of the Annual Meeting of the Cognitive Science Society* 38. URL: https://collaborate.princeton.edu/en/publications/information-seeking-learning-and-the-marginal-value-theorem-a-nor.

[196] J. A. Lewis-Peacock, J. D. Cohen, and K. A. Norman (2016). Neural evidence of the strategic choice between working memory and episodic memory in prospective remembering. *Neuropsychologia* 93, pp. 280–288. URL: https://www.sciencedirect.com/science/article/pii/S0028393216304055.

[195] S. Musslick, B. Dey, K. Ozcimder, M. M. A. Patwary, T. L. Willke, and J. D. Cohen (2016). Controlled vs. automatic processing: A graph-theoretic approach to the analysis of serial vs. parallel processing in neural network architectures. *CogSci 2016: Proceedings of the Annual Meeting of the Cognitive Science Society* 38. URL: https://www.researchgate.net/profile/Theodore-Willke/publication/301202504_Controlled_vs_Automatic_Processing_A_Graph-Theoretic_Approach_to_the_Analysis_of_Serial_vs_Parallel_Processing_in_Neural_Network_Architectures/links/578670e308ae36ad40a692d0/Controlled-vs-Automatic-Processing-A-Graph-Theoretic-Approach-to-the-Analysis-of-Serial-vs-Parallel-Processing-in-Neural-Network-Architectures.pdf.

[194] A. Shenhav, J. D. Cohen, and M. M. Botvinick (2016). Dorsal Anterior Cingulate Cortex and the Expected Value of Control. *Nature Neuroscience* 19, pp. 1286–1291. URL: https://www.nature.com/articles/nn.4384.

[193] S. Shenhav, M. A. Straccia, M. M. Botvinick, and J. D. Cohen (2016). Dorsal anterior cingulate and ventromedial prefrontal cortex have inverse roles in both foraging and economic choice. *Cognitive, Affective, and Behavioral Neuroscience* 16.6, pp. 1127–1139. URL: https://link.springer.com/article/10.3758/s13415-016-0458-8.

[192] C. Warren, E. Eldar, R. Van den Brink, K.-D. Tona, N. van der Wee, E. Giltay, M. van Noorden, J. Bosch, R. Wilson, J. D. Cohen, and S. Nieuwenhuis (2016). Catecholamine-mediated increases in gain enhance the precision of cortical representations. *Journal of Neuroscience* 36.21, pp. 5699–5708. URL: https://www.jneurosci.org/content/36/21/5699.short.

[191] M. T. deBettencourt, J. D. Cohen, R. F. Lee, K. A. Norman, and N. B. Turk-Browne (2015). Closed-loop training of attention with real-time brain imaging. *Nature Neuroscience* 18.3, pp. 470–474. URL: https://www.nature.com/articles/nn.3940.

[190] K. M. Ericson, J. M. White, D. Laibson, and J. D. Cohen (2015). Money earlier or later? Simple heuristics explain intertemporal choices about money better than delay discounting. *Psychological Science* 26.6, pp. 826–833. URL: https://pubmed.ncbi.nlm.nih.gov/25911124/.

[189] B. A. Field, C. L. Buck, S. M. McClure, L. E. Nystrom, D. Kahneman, and J. D. Cohen (2015). Attentional modulation of brain responses to primary appetitive and aversive stimuli. *PLoS ONE* 10.7, e0130880. URL: https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0130880.

[188] D. M. Schnyer, C. G. Beevers, M. T. deBettencourt, S. M. Sherman, J. D. Cohen, K. A. Norman, and N. B. Turk-Browne (2015). Neurocognitive therapeutics: From concept to application in the treatment of negative attention bias. *Biology of Mood & Anxiety Disorders* 5, pp. 1–4. URL: https://biolmoodanxietydisord.biomedcentral.com/articles/10.1186/s13587-015-0016-y.

[187] M. Shvartsman, V. Srivastava, and J. D. Cohen (2015). A theory of decision making under dynamically changing context. *Advances in Neural Information Processing Systems (NeurIPS)* 28. URL: https://proceedings.neurips.cc/paper/2015/file/4e8412ad48562e3c9934f45c3e144d48-Paper.pdf.

[186] D. Tomlin, D. G. Rand, E. A. Ludvig, and J. D. Cohen (2015). The evolution and devolution of cognitive control: The costs of deliberation in a competitive world. *Scientific Reports* 5.11002. URL: https://pubmed.ncbi.nlm.nih.gov/26078086/.

[185] D. F. Toupo, S. H. Strogatz, J. D. Cohen, and D. G. Rand (2015). Evolutionary game dynamics of controlled and automatic decision-making. *Chaos: An Interdisciplinary Journal of Nonlinear Science* 25.7, p. 073120. URL: https://aip.scitation.org/doi/full/10.1063/1.4927488.

[184] Y. Wang, K. Li, J. D. Cohen, and N. B. Turk-Browne (2015). Full correlation matrix analysis (FCMA): An unbiased method for task-related functional connectivity. *Journal of Neuroscience Methods* 251, pp. 108–119. URL: https://www.sciencedirect.com/science/article/pii/S0165027015001910.

[183] M. M. Botvinick and J. D. Cohen (2014). The computational and neural basis of cognitive control: Charted territory and new frontiers. *Cognitive Science* 38, pp. 1249–1285. URL: https://onlinelibrary.wiley.com/doi/full/10.1111/cogs.12126.

[182] J. D. Cohen and P. Holmes (2014). Optimality and some of its discontents. *Topics in Cognitive Science* 6, pp. 258–278. URL: https://onlinelibrary.wiley.com/doi/full/10.1111/tops.12084.

[181] S. F. Feng, M. Schwemmer, S. J. Gershman, and J. D. Cohen (2014). Multitasking vs. multiplexing: Toward a normative account of limitations in the simultaneous execution of control-demanding behaviors. *Cognitive, Affective, and Behavioral Neuroscience* 14.1, pp. 129–146. URL: https://link.springer.com/article/10.3758/s13415-013-0236-9.

[180] A. Shenhav, M. A. Straccia, J. D. Cohen, and M. M. Botvinick (2014). Anterior cingulate engagement in a foraging context reflects choice difficulty rather than foraging value. *Nature Neuroscience* 17, pp. 1249–1254. URL: https://www.nature.com/articles/nn.3771.

[179] M. K. van Vugt, P. Simen, L. E. Nystrom, P. J. Holmes, and J. D. Cohen (2014). Lateralized readiness potentials reveal properties of a neural mechanism for implementing a decision threshold. *PLOS ONE* 9.3, e90943. URL: https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0090943.

[178] R. C. Wilson, A. Geana, J. M. White, E. A. Ludvig, and J. D. Cohen (2014). Humans use directed and random exploration to solve the explore-exploit dilemma. *Journal of Experimental Psychology: General* 143.6, p. 2074. URL: https://psycnet.apa.org/record/2014-44776-001.

[177] E. Eldar, J. D. Cohen, and Y. Niv (2013). The effects of neural gain on attention and learning. *Nature Neuroscience* 16, pp. 1146–1153. URL: https://www.nature.com/articles/nn.3428.

[176] B. Eppinger, L. E. Nystrom, and J. D. Cohen (2013). Reduced sensitivity to immediate reward during decision-making in older than younger adults. *PLOS ONE* 7.5, e36953. URL: https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0036953.

[175] B. Eppinger, N. W. Schuck, L. E. Nystrom, and J. D. Cohen (2013). Reduced striatal responses to reward prediction errors in older compared to younger adults. *Journal of Neuroscience* 33.24, pp. 9905–9912. URL: https://www.jneurosci.org/content/33/24/9905.short.

[174] T. Kriete, D. C. Noelle, J. D. Cohen, and R. C. O'Reilly (2013). Indirection and symbol-like processing in the prefrontal cortex and basal ganglia. *Proceedings of the National Academy of Sciences, USA* 110.41, pp. 16390–16395. URL: https://www.pnas.org/doi/abs/10.1073/pnas.1303547110?cookieSet=1.

[173] A. E. Richard, C. S. Carter, J. D. Cohen, and R. Y. Cho (2013). Persistence, diagnostic specificity and genetic liability for context-processing deficits in schizophrenia. *Schizophrenia Research* 147.1, pp. 75–80. URL: https://www.sciencedirect.com/science/article/abs/pii/S0920996413001126.

[172] A. Shenhav, M. M. Botvinick, and J. D. Cohen (2013). The expected value of control: An integrative theory of anterior cingulate cortex function. *Neuron* 79, pp. 217–240. URL: https://www.sciencedirect.com/science/article/pii/S0896627313006077.

[171] M. T. Todd, L. E. Nystrom, and J. D. Cohen (2013). Confounds in multivariate pattern analysis: Theory and rule representation case study. *Neuroimage* 77, pp. 157–165. URL: https://www.sciencedirect.com/science/article/pii/S1053811913002887.

[170] D. Tomlin, A. Nedic, D. A. Prentice, P. Holmes, and J. D. Cohen (2013). The neural substrates of social influence on decision making. *PLOS ONE* 8.1, e52630. URL: https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0052630.

[169] K. D'Ardenne, N. Eshel, J. Luka, A. Lenartowicz, L. E. Nystrom, and J. D. Cohen (2012). The role of prefrontal cortex and the midbrain dopamine system in working memory updating. *Proceedings of the National Academy of Sciences, USA* 109.49. URL: https://www.pnas.org/doi/abs/10.1073/pnas.1116727109.

[168] A. Nedic, D. Tomlin, P. Holmes, D. A. Prentice, and J. D. Cohen (2012). A decision task in a social context: Human experiments, models, and analyses of behavioral data. *Proceedings of the IEEE* 100.3, pp. 713–733. URL: https://ieeexplore.ieee.org/abstract/document/6069518.

[167] J. R. Reynolds, R. C. O'Reilly, J. D. Cohen, and T. S. Braver (2012). The function and organization of lateral prefrontal cortex: A test of competing hypotheses. *PLOS ONE* 7.2, e30284. URL: https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0030284.

[166] M. K. van Vugt, P. Simen, L. E. Nystrom, P. Holmes, and J. D. Cohen (2012). EEG oscillations reveal neural correlates of evidence accumulation. *Frontiers in Decision Neuroscience* 6, p. 106. URL: https://www.frontiersin.org/articles/10.3389/fnins.2012.00106/full.

[165] F. Balci, D. Freestone, P. Simen, L. deSouza, J. D. Cohen, and P. Holmes (2011). Optimal temporal risk assessment. *Frontiers in Integrative Neuroscience* 5.56, pp. 1–15. URL: https://www.frontiersin.org/articles/10.3389/fnint.2011.00056/full.

[164] F. Balci, P. Simen, R. Niyogi, A. Saxe, J. A. Hughes, P. Holmes, and J. D. Cohen (2011). Acquisition of decision making criteria: Reward rate ultimately beats accuracy. *Attention, Perception, & Psychophysics* 73, pp. 640–657. URL: https://link.springer.com/article/10.3758/s13414-010-0049-7.

[163] C. S. Carter, D. M. Barch, E. Bullmore, J. Breiling, R. W. Buchanan, P. Butler, J. D. Cohen, M. Geyer, R. Gollub, M. F. Green, J. Jaeger, J. H. Krystal, H. Moore, K. Nuechterlein, T. Robbins, S. Silverstein, E. E. Smith, M. Strauss, and T. Wykes (2011). Cognitive Neuroscience Treatment Research to Improve Cognition in Schizophrenia II: Developing imaging biomarkers to enhance treatment development for schizophrenia and related disorders. *Biological Psychiatry* 70.1, pp. 7–12. URL: https://www.sciencedirect.com/science/article/pii/S0006322311002071.

[162] S. E. Forster, C. S. Carter, J. D. Cohen, and R. Y. Cho (2011). Parametric manipulation of the conflict signal and control-state adaptation. *Journal of Cognitive Neuroscience* 23.4, pp. 923–935. URL: https://direct.mit.edu/jocn/article/23/4/923/5064/Parametric-Manipulation-of-the-Conflict-Signal-and.

[161] A. Lenartowicz, N. Yeung, and J. D. Cohen (2011). No-go trials can modulate switch cost by interfering with effects of task preparation. *Psychological Research* 75, pp. 66–76. URL: https://link.springer.com/article/10.1007/s00426-010-0286-3.

[160] P. Simen, F. Balci, L. DeSouza, J. D. Cohen, and P. Holmes (2011). A model of interval timing by neural integration. *Journal of Neuroscience* 31, pp. 9238–9253. URL: https://www.jneurosci.org/content/31/25/9238.short.

[159] R. Bogacz, P. T. Hu, P. Holmes, and J. D. Cohen (2010). Do humans produce the speed-accuracy tradeoff that maximizes reward rate? *Quarterly Journal of Experimental Psychology* 63.5, pp. 863–891. URL: https://psycnet.apa.org/record/2010-08456-005.

[158] S. J. Gershman, J. D. Cohen, and Y. Niv (2010). Learning to selectively attend. *CogSci 2010: Proceedings of the Annual Meeting of the Cognitive Science Society* 32. URL: https://gershmanlab.com/pubs/GershmanCohenNiv10.pdf.

[157] M. S. Gilzenrat, S. Nieuwenhuis, and J. D. Cohen (2010). Pupil Diameter Tracks Changes in Control State Predicted by the Adaptive Gain Theory of Locus Coeruleus Function. *Cognitive, Affective, and Behavioral Neuroscience* 10.2, pp. 252–269. URL: https://link.springer.com/article/10.3758/CABN.10.2.252.

[156] A. Lenartowicz, R. Escobedo-Quiroz, and J. D. Cohen (2010). Updating of context in working memory: An event related potential study. *Cognitive, Affective, and Behavioral Neuroscience* 10.2, pp. 298–315. URL: https://link.springer.com/article/10.3758/CABN.10.2.298.

[155] A. Schurger, F. Pereira, A. Treisman, and J. D. Cohen (2010). Reproducibility of activity characterizes conscious from non-conscious neural representations. *Science* 327.5961, pp. 97–99. URL: https://www.science.org/doi/full/10.1126/science.1180029.

[154] K. F. Wong-Lin, P. Eckhoff, P. Holmes, and J. D. Cohen (2010). Optimal performance in a countermanding saccade task. *Brain Research* 1318, pp. 178–187. URL: https://www.sciencedirect.com/science/article/pii/S0006899309026523.

[153] D. M. Barch, C. S. Carter, A. Arnsten, R. W. Buchanan, J. D. Cohen, M. Geyer, ..., and R. Heinssen (2009). Selecting paradigms from cognitive neuroscience for translation into use in clinical trials: Proceedings of the third CNTRICS meeting. *Schizophrenia Bulletin* 35, pp. 109–114. URL: https://academic.oup.com/schizophreniabulletin/article/35/1/109/1933522.

[152] R. Y. Cho, J. M. Orr, J. D. Cohen, and C. S. Carter (2009). Generalized signaling for control: Evidence from post-conflict and post-error performance adjustments. *Journal of Experimental Psychology: Human Perception and Performance* 35, pp. 1161–1177. URL: https://psycnet.apa.org/record/2009-11357-012?doi=1.

[151] I. Daubechies, E. Roussos, S. Takerkart, M. Benharrosh, C. Golden, K. D'Ardenne, ..., and J. Haxby (2009). Independent component analysis for brain fMRI does NOT select for independence. *Proceedings of the National Academy of Sciences, USA* 106, pp. 10415–10422. URL: https://www.pnas.org/doi/abs/10.1073/pnas.0903525106.

[150] J. Gao, K. F. Wong-Lin, P. Holmes, P. Simen, and J. D. Cohen (2009). Sequential effects in two-choice reaction time tasks: Decomposition and synthesis of mechanisms. *Neural Computation* 21, pp. 2407–2436. URL: https://direct.mit.edu/neco/article/21/9/2407/7474/Sequential-Effects-in-Two-Choice-Reaction-Time.

[149] P. Simen and J. D. Cohen (2009). Explicit melioration by a neural diffusion model. *Brain Research* 1299, pp. 95–117. URL: https://www.sciencedirect.com/science/article/pii/S0006899309013900.

[148] P. Simen, D. Contreras-Ros, C. Buck, P. Hu, P. Holmes, and J. D. Cohen (2009). Reward rate optimization in two-alternative decision making: Empirical tests of theoretical predictions. *Journal of Experimental Psychology: Human Perception and Performance* 35, pp. 1865–1897. URL: https://psycnet.apa.org/record/2009-22869-017?doi=1.

[147] A. J. Yu, P. Dayan, and J. D. Cohen (2009). Dynamics of attentional selection under conflict: Toward a rational Bayesian account. *Journal of Experimental Psychology: Human Perception and Performance* 35.3, pp. 700–717. URL: https://pubmed.ncbi.nlm.nih.gov/19485686/.

[146] W. van den Bos, J. Li, T. Lau, E. Maskin, J. D. Cohen, P. R. Montague, and S. M. McClure (2008). The value of victory: social origins of the winner's curse in common value auctions. *Judgment and Decision Making* 3.7, pp. 483–492. URL: https://www.cambridge.org/core/journals/judgment-and-decision-making/article/value-of-victory-social-origins-of-the-winners-curse-in-common-value-auctions/5DDB727F9F2E599C7BA90106F5AAC3C0.

[145] C. S. Carter, D. M. Barch, R. W. Buchanan, E. Bullmore, J. H. Krystal, J. D. Cohen, ..., and R. Heinssen (2008). Identifying cognitive mechanisms targeted for treatment development in schizophrenia: An overview of the first meeting of the Cognitive Neuroscience Treatment Research to Improve Cognition in Schizophrenia (CNTRICS) initiative. *Biological Psychiatry* 64.1, pp. 4–10. URL: https://www.sciencedirect.com/science/article/pii/S0006322308003648.

[144] K. D'Ardenne, S. M. McClure, L. E. Nystrom, and J. D. Cohen (2008). BOLD responses in the dopaminergic ventral tegmental area during a classical conditioning task. *Science* 319.5867, pp. 1264–1267. URL: https://www.science.org/doi/full/10.1126/science.1150605.

[143] C. L. Fales, D. M. Barch, M. M. Rundle, M. A. Mintun, A. X. Snyder, J. D. Cohen, ..., and Y. I. Sheline (2008). Altered emotional interference processing in affective and cognitive-control brain circuitry in major depression. *Biological Psychiatry* 63.4, pp. 377–384. URL: https://www.sciencedirect.com/science/article/pii/S0006322307005653.

[142] J. D. Greene, F. A. Cushman, L. E. Stewart, K. Lowenberg, L. E. Nystrom, and J. D. Cohen (2008). Pushing moral buttons: The interaction between personal force and intention in moral judgment. *Cognition* 111, pp. 364–371. URL: https://www.sciencedirect.com/science/article/pii/S0010027709000420.

[141] J. D. Greene, S. A. Morelli, K. Lowenberg, L. E. Nystrom, and J. D. Cohen (2008). Cognitive load selectively interferes with utilitarian moral judgment. *Cognition* 107, pp. 1144–1154. URL: https://pubmed.ncbi.nlm.nih.gov/18158145/.

[140] J. K. Kroger, L. E. Nystrom, J. D. Cohen, and P. N. Johnson-Laird (2008). Distinct neural substrates for deductive and mathematical processing. *Brain Research* 1243, pp. 83–103. URL: https://pubmed.ncbi.nlm.nih.gov/18760263/.

[139] Y. Liu, P. Holmes, and J. D. Cohen (2008). A neural network model of the Eriksen task: Reduction, analysis, and data fitting. *Neural Computation* 20.2, pp. 345–373. URL: https://pubmed.ncbi.nlm.nih.gov/18045022/.

[138] G. Loewenstein, S. Rick, and J. D. Cohen (2008). Neuroeconomics. *Annual Review of Psychology* 59, pp. 18.1–18.26. URL: https://pubmed.ncbi.nlm.nih.gov/17883335/.

[137] K. H. Nuechterlein, M. F. Green, R. S. Kern, L. E. Baade, D. Barch, J. D. Cohen, ..., and S. R. Marder (2008). The MATRICS Consensus Cognitive Battery: Part 1. Test selection, reliability, and validity. *American Journal of Psychiatry* 165.2, pp. 203–213. URL: https://ajp.psychiatryonline.org/doi/full/10.1176/appi.ajp.2007.07010042.

[136] J.-P. Pochon, J. Riis, A. Sanfey, L. Nystrom, and J. D. Cohen (2008). Functional imaging of decision conflict. *Journal of Neuroscience* 28.13, pp. 3468–3473. URL: https://www.jneurosci.org/content/28/13/3468.short.

[135] A. Schurger, A. Cowey, J. D. Cohen, A. Treisman, and C. Tallon-Baudry (2008). Distinct and independent correlates of attention and awareness in a hemianopic patient. *Neuropsychologia* 46.8, pp. 2189–2197. URL: https://www.sciencedirect.com/science/article/pii/S0028393208000833.

[134] E. T. Shea-Brown, M. S. Gilzenrat, and J. D. Cohen (2008). Optimization of decision making in multilayer networks: The role of Locus Coeruleus. *Neural Computation* 20, pp. 2863–2894. URL: https://direct.mit.edu/neco/article/20/12/2863/7374/Optimization-of-Decision-Making-in-Multilayer.

[133] M. Todd, Y. Niv, and J. D. Cohen (2008). Learning to use working memory in partially observable environments through dopaminergic reinforcement. *Advances in Neural Information Processing Systems (NeurIPS)* 20, pp. 1689–1696. URL: https://dl.acm.org/doi/abs/10.5555/2981780.2981991.

[132] A. Yu and J. D. Cohen (2008). Sequential effects: Superstition or rational behavior? *Advances in Neural Information Processing Systems (NeurIPS)* 21, pp. 1873–1880. URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4580342/.

[131] R. Bogacz, S. M. McClure, J. Li, J. D. Cohen, and P. R. Montague (2007). Short-term memory traces for action bias in human reinforcement learning. *Brain Research* 1153, pp. 111–121. URL: https://www.sciencedirect.com/science/article/pii/S000689930700666X.

[130] W. van den Bos, S. M. McClure, L. T. Harris, S. T. Fiske, and J. D. Cohen (2007). Dissociating affective evaluation and social cognitive processes in the ventral medial prefrontal cortex. *Cognitive, Affective, and Behavioral Neuroscience* 7.4, pp. 337–346. URL: https://pubmed.ncbi.nlm.nih.gov/18189007/.

[129] J. D. Cohen, S. M. McClure, and A. J. Yu (2007). Should I stay or should I go? How the human brain manages the tradeoff between exploitation and exploration. *Philosophical Transactions of the Royal Society of London Series B (Biological Sciences)* 362.1481, pp. 933–942. URL: https://royalsocietypublishing.org/doi/full/10.1098/rstb.2007.2098.

[128] L. T. Harris, S. M. McClure, W. van den Bos, J. D. Cohen, and S. T. Fiske (2007). Regions of MPFC differentially tuned to social and non-social affective evaluation. *Cognitive, Affective, and Behavioral Neuroscience* 7.4, pp. 309–316. URL: https://link.springer.com/article/10.3758/CABN.7.4.309.

[127] S. M. McClure, K. M. Ericson, D. I. Laibson, G. Loewenstein, and J. D. Cohen (2007). Time discounting for primary rewards. *Journal of Neuroscience* 27.21, pp. 5796–5804. URL: https://www.jneurosci.org/content/27/21/5796.

[126] N. Yeung, R. Bogacz, C. B. Holroyd, S. Nieuwenhuis, and J. D. Cohen (2007). Theta phase resetting and the error-related negativity. *Psychophysiology* 44, pp. 39–49. URL: https://onlinelibrary.wiley.com/doi/full/10.1111/j.1469-8986.2006.00482.x.

[125] S. J. Bishop, J. D. Cohen, J. Fossella, B. J. Casey, and M. J. Farah (2006). COMT genotype influences prefrontal response to emotional distraction. *Cognitive, Affective, and Behavioral Neuroscience* 6.1, pp. 62–70. URL: https://link.springer.com/article/10.3758/CABN.6.1.62.

[124] R. Bogacz, E. T. Brown, J. Moehlis, P. Holmes, and J. D. Cohen (2006). The physics of optimal decision making: A formal analysis of models of performance in two-alternative forced choice tasks. *Psychological Review* 113.4, pp. 700–765. URL: https://psycnet.apa.org/record/2006-12689-002?doi=1.

[123] P. R. Montague, B. King-Casas, and J. D. Cohen (2006). Imaging valuation models in human choice. *Annual Review of Neuroscience* 29, pp. 417–448. URL: https://www.annualreviews.org/doi/abs/10.1146/annurev.neuro.29.051605.112903.

[122] J. B. Nitschke, G. E. Dixon, I. Sarinopoulos, S. J. Short, J. D. Cohen, E. E. Smith, and R. J. Davidson (2006). Altering expectancy dampens neural response to aversive taste in primary taste cortex. *Nature Neuroscience* 9.3, pp. 435–442. URL: https://www.nature.com/articles/nn1645.

[121] D. A. Pizzagalli, L. A. Peccoralo, R. J. Davidson, and D. Cohen (2006). Resting anterior cingulate activity and abnormal responses to errors in subjects with elevated depressive symptoms: A 128-channel EEG study. *Human Brain Mapping* 27, pp. 185–201. URL: https://onlinelibrary.wiley.com/doi/full/10.1002/hbm.20172.

[120] A. G. Sanfey, G. Loewenstein, S. M. McClure, and J. D. Cohen (2006). Neuroeconomics: cross-currents in research on decision-making. *Trends in Cognitive Sciences* 10.3, pp. 108–116. URL: https://www.sciencedirect.com/science/article/pii/S1364661306000271.

[119] P. Simen, J. D. Cohen, and P. Holmes (2006). Rapid decision threshold modulation by reward rate in a neural network. *Neural Networks* 19, pp. 1013–1026. URL: https://www.sciencedirect.com/science/article/pii/S0893608006001626.

[118] N. Yeung and J. D. Cohen (2006). The impact of cognitive deficits on conflict monitoring: Predictable dissociations between the ERN and N2. *Psychological Science* 17.2, pp. 164–171. URL: https://psycnet.apa.org/record/2006-02115-012.

[117] N. Yeung, L. E. Nystrom, J. A. Aronson, and J. D. Cohen (2006). Between-task competition and cognitive control in task switching. *Journal of Neuroscience* 26.5, pp. 1429–1438. URL: https://www.jneurosci.org/content/26/5/1429.

[116] G. Aston-Jones and J. D. Cohen (2005). Adaptive gain and the role of the locus coeruleus-norepinephrine system in optimal performance. *Journal of Comparative Neurology* 493.1, pp. 99–110. URL: https://onlinelibrary.wiley.com/doi/full/10.1002/cne.20723.

[115] G. Aston-Jones and J. D. Cohen (2005). An integrative theory of locus coeruleus-norepinephrine function: Adaptive gain and optimal performance. *Annual Review of Neuroscience* 28, pp. 403–450. URL: https://www.annualreviews.org/doi/abs/10.1146/annurev.neuro.28.061604.135709.

[114] E. T. Brown, J. Gao, P. Holmes, R. Bogacz, M. Gilzenrat, and J. D. Cohen (2005). Simple neural networks that optimize decisions. *International Journal of Bifurcation and Chaos* 15.3, pp. 803–826. URL: https://www.worldscientific.com/doi/abs/10.1142/S0218127405012478.

[113] J. D. Cohen (2005). The vulcanization of the human brain: A neural perspective on interactions between cognition and emotion. *Journal of Economic Perspectives* 19.4, pp. 3–24. URL: https://www.aeaweb.org/articles?id=10.1257/089533005775196750.

[112] P. Holmes, E. Brown, J. Moehlis, R. Bogacz, J. Gao, G. Aston-Jones, E. Clayton, J. Rajkowski, and J. D. Cohen (2005). Optimal decisions: From neural spikes, through stochastic differential equations, to behavior. *IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences* 88.10, pp. 2496–2503. URL: https://www.researchgate.net/publication/31011977_Optimal_Decisions_From_Neural_Spikes_through_Stochastic_Differential_Equations_to_Behavior.

[111] C. B. Holroyd, N. Yeung, M. G. H. Coles, and J. D. Cohen (2005). A mechanism for error detection in speeded response time tasks. *Journal of Experimental Psychology: General* 134.2, pp. 163–191. URL: https://psycnet.apa.org/record/2005-04168-003?doi=1.

[110] J. G. Kerns, J. D. Cohen, A. W. MacDonald III, M. K. Johnson, V. A. Stenger, H. Aizenstein, and C. S. Carter (2005). Decreased conflict- and error-related activity in the anterior cingulate cortex in subjects with schizophrenia. *American Journal of Psychiatry* 162.10, pp. 1833–1839. URL: https://ajp.psychiatryonline.org/doi/full/10.1176/appi.ajp.162.10.1833.

[109] A. MacDonald III, C. S. Carter, J. Kerns, S. Ursu, D. M. Barch, A. Holmes, V. A. Stenger, and J. D. Cohen (2005). Specificity of prefrontal dysfunction and context processing deficits to schizophrenia in never-medicated patients with first-episode psychosis. *American Journal of Psychiatry* 162, pp. 475–484. URL: https://ajp.psychiatryonline.org/doi/full/10.1176/appi.ajp.162.3.475.

[108] S. McClure, M. Gilzenrat, and J. Cohen (2005). An exploration-exploitation model based on norepinephrine and dopamine activity. *Advances in Neural Information Processing Systems (NeurIPS)* 18,

pp. 867–874. URL: https://collaborate.princeton.edu/en/publications/an-exploration-exploitation-model-based-on-norepinepherine-and-do.

[107] S. Nieuwenhuis, G. Aston-Jones, and J. D. Cohen (2005). Decision making, the P3, and the locus coeruleus-norepinephrine system. *Psychological Bulletin* 131.4, pp. 510–532. URL: https://psycnet.apa.org/record/2005-08334-002.

[106] S. Nieuwenhuis, M. S. Gilzenrat, B. D. Holmes, and J. D. Cohen (2005). The role of the locus coeruleus in mediating the attentional blink: A neurocomputational theory. *Journal of Experimental Psychology: General* 134, pp. 291–307. URL: https://psycnet.apa.org/record/2005-09704-001.

[105] S. M. Polyn, V. S. Natu, J. D. Cohen, and K. A. Norman (2005). Category-specific cortical activity precedes retrieval during memory search. *Science* 310.5756, pp. 1963–1966. URL: https://www.science.org/doi/full/10.1126/science.1117645.

[104] N. P. Rougier, D. C. Noelle, T. S. Braver, J. D. Cohen, and R. C. O'Reilly (2005). Prefrontal cortex and the flexibility of cognitive control: Rules without symbols. *Proceedings of the National Academy of Sciences, USA* 102.20, pp. 7338–7343. URL: https://www.pnas.org/doi/abs/10.1073/pnas.0502455102.

[103] N. Yeung, C. B. Holroyd, and J. D. Cohen (2005). ERP correlates of feedback and reward processing in the presence and absence of response choice. *Cerebral Cortex* 15, pp. 535–544. URL: https://academic.oup.com/cercor/article/15/5/535/442121.

[102] R. Bogacz and J. D. Cohen (2004). Parameterization of connectionist models. *Behavioral Research Methods, Instruments & Computers* 36.4, pp. 732–741. URL: https://link.springer.com/article/10.3758/BF03206554.

[101] M. M. Botvinick, J. D. Cohen, and C. S. Carter (2004). Conflict monitoring and anterior cingulate cortex: An update. *Trends in Cognitive Sciences* 8.12, pp. 539–546. URL: https://www.sciencedirect.com/science/article/pii/S1364661304002657.

[100] E. C. Clayton, J. Rajkowski, J. D. Cohen, and G. Aston-Jones (2004). Phasic activation of monkey locus coeruleus neurons by simple decisions in a forced choice task. *Journal of Neuroscience* 24.44, pp. 9914–9920. URL: https://www.jneurosci.org/content/24/44/9914.short.

[99] M. F. Green, K. H. Nuechterlein, J. M. Gold, D. M. Barch, J. D. Cohen, S. Essock, ..., and S. R. Marder (2004). Approaching a consensus cognitive battery for clinical trials in schizophrenia: The NIMH-MATRICS conference to select cognitive domains and test criteria. *Biological Psychiatry* 56.5, pp. 301–307. URL: https://www.sciencedirect.com/science/article/pii/S0006322304007012.

[98] J. D. Greene and J. D. Cohen (2004). For the law, neuroscience changes nothing and everything. *Philosophical Transactions of the Royal Society of London Series B* 359.1451, pp. 1775–1785. URL: https://pubmed.ncbi.nlm.nih.gov/15590618/.

[97] J. D. Greene, L. E. Nystrom, A. D. Engell, J. M. Darley, and J. D. Cohen (2004). The neural bases of cognitive conflict and control in moral judgment. *Neuron* 44.2, pp. 389–400. URL: https://pubmed.ncbi.nlm.nih.gov/15473975/.

[96] C. B. Holroyd, J. T. Larsen, and J. D. Cohen (2004). Context dependence of the event-related brain potential associated with reward and punishment. *Psychophysiology* 41.2, pp. 245–253. URL: https://pubmed.ncbi.nlm.nih.gov/15032989/.

[95] C. B. Holroyd, S. Nieuwenhuis, N. Yeung, L. E. Nystrom, R. B. Mars, M. G. H. Coles, and J. D. Cohen (2004). Dorsal anterior cingulate cortex shows fMRI response to internal and external error signals. *Nature Neuroscience* 7.5, pp. 497–498. URL: https://www.nature.com/articles/nn1238.

[94] J. G. Kerns, J. D. Cohen, A. W. MacDonald III, R. Y. Cho, V. A. Stenger, and C. S. Carter (2004). Anterior cingulate conflict monitoring and adjustments in control. *Science* 303, pp. 1023–1026. URL: https://pubmed.ncbi.nlm.nih.gov/14963333/.

[93] J. G. Kerns, J. D. Cohen, V. A. Stenger, and C. S. Carter (2004). Prefrontal cortex guides context-appropriate responding during language production. *Neuron* 43.2, pp. 283–291. URL: https://www.sciencedirect.com/science/article/pii/S0896627304004192.

[92] S. M. McClure, D. I. Laibson, G. Loewenstein, and J. D. Cohen (2004). Separate neural systems value immediate and delayed monetary rewards. *Science* 306, pp. 503–507. URL: https://www.science.org/doi/full/10.1126/science.1100907.

[91] P. R. Montague, S. E. Hyman, and J. D. Cohen (2004). Computational roles for dopamine in behavioral control. *Nature* 431.7010, pp. 760–767. URL: https://www.nature.com/articles/nature03015.

[90] S. Nieuwenhuis, N. Yeung, and J. D. Cohen (2004). Stimulus modality, perceptual overlap, and the Go/No-Go N2. *Psychophysiology* 41, pp. 157–160. URL: https://onlinelibrary.wiley.com/doi/full/10.1046/j.1469-8986.2003.00128.x.

[89] S. Nieuwenhuis, N. Yeung, C. B. Holroyd, A. Schurger, and J. D. Cohen (2004). Sensitivity of electrophysiological activity from medial frontal cortex to utilitarian and performance feedback. *Cerebral Cortex* 14, pp. 741–747. URL: https://academic.oup.com/cercor/article/14/7/741/375871.

[88] J. K. Rilling, A. G. Sanfey, J. A. Aronson, L. E. Nystrom, and J. D. Cohen (2004). Opposing BOLD responses to reciprocated and unreciprocated altruism in putative reward pathways. *NeuroReport* 15.16, pp. 2539–2243. URL: https://journals.lww.com/neuroreport/Fulltext/2004/11150/Opposing_BOLD_responses_to_reciprocated_and.22.aspx.

[87] J. K. Rilling, A. G. Sanfey, J. A. Aronson, L. E. Nystrom, and J. D. Cohen (2004). The neural correlates of theory of mind within interpersonal interactions. *Neuroimage* 22.4, pp. 1694–1703. URL: https://www.sciencedirect.com/science/article/pii/S1053811904002241.

[86] T. D. Wager, J. K. Rilling, E. E. Smith, A. Sokolik, K. L. Casey, R. J. Davidson, S. K. Kosslyn, R. M. Rose, and J. D. Cohen (2004). Placebo-induced changes in fMRI in the anticipation and experience of pain. *Science* 303, pp. 1162–1167. URL: https://www.science.org/doi/full/10.1126/science.1093065.

[85] N. Yeung, R. Bogacz, C. Holroyd, and J. D. Cohen (2004). Detection of synchronized oscillations in the electroencephalogram: An evaluation of methods. *Psychophysiology* 41, pp. 822–832. URL: https://onlinelibrary.wiley.com/doi/full/10.1111/j.1469-8986.2004.00239.x.

[84] N. Yeung, M. M. Botvinick, and J. D. Cohen (2004). The neural basis of error detection: conflict monitoring and the error-related negativity. *Psychological Review* 111.4, pp. 931–959. URL: https://psycnet.apa.org/record/2004-19012-005.

[83] D. M. Barch, C. S. Carter, A. MacDonald, T. S. Braver, and J. D. Cohen (2003). Context-processing deficits in schizophrenia: Diagnostic specificity, 4-week course, and relationships to clinical symptoms. *Journal of Abnormal Psychology* 112, pp. 132–143. URL: https://psycnet.apa.org/record/2003-01662-014.

[82] R. Condray, G. J. Siegle, J. D. Cohen, D. P. van Kammen, and S. R. Steinhauer (2003). Automatic activation of the semantic network in schizophrenia: N400 elicited by a brief inter-stimulus interval. *Biological Psychiatry* 54.11, pp. 1134–1148. URL: https://www.sciencedirect.com/science/article/pii/S0006322303006991.

[81] C. B. Holroyd, S. Nieuwenhuis, N. Yeung, and J. D. Cohen (2003). Errors in reward prediction are reflected in the event-related brain potential. *NeuroReport* 14.18, pp. 2481–2484. URL: https://journals.lww.com/neuroreport/Fulltext/2003/12190/Errors_in_reward_prediction_are_reflected_in_the.37.aspx.

[80] W. M. Perlstein, N. K. Dixit, C. S. Carter, D. C. Noll, and J. D. Cohen (2003). Prefrontal cortex dysfunction mediates deficits in working memory and prepotent responding in schizophrenia. *Biological Psychiatry* 53.1, pp. 25–38. URL: https://pubmed.ncbi.nlm.nih.gov/12513942/.

[79] A. G. Sanfey, J. K. Rilling, J. A. Aronson, L. E. Nystrom, and J. D. Cohen (2003). The neural basis of economic decision-making in the ultimatum game. *Science* 300, pp. 1755–1757. URL: https://www.science.org/doi/full/10.1126/science.1082976.

[78] R. Y. Cho, L. E. Nystrom, E. Brown, A. D. Jones, T. S. Braver, P. Holmes, and J. D. Cohen (2002). Mechanisms underlying performance dependencies on stimulus history in a two-alternative forced choice task. *Cognitive, Affective, and Behavioral Neuroscience* 2.4, pp. 283–299. URL: https://link.springer.com/article/10.3758/CABN.2.4.283.

[77] J. D. Cohen, T. S. Braver, and J. W. Brown (2002). Computational perspectives on dopamine function in prefrontal cortex. *Current Opinion in Neurobiology* 12, pp. 223–229. URL: https://www.sciencedirect.com/science/article/pii/S0959438802003148.

[76] R. J. Davidson, D. A. Lewis, L. Alloy, D. Amaral, G. Bush, J. D. Cohen, ..., and B. Peterson (2002). Neural and behavioral substrates of mood and mood regulation. *Biological Psychiatry* 52.6, pp. 478–502. URL: https://www.sciencedirect.com/science/article/pii/S0006322302014580.

[75] K. Fissell, E. Tseytlin, D. Cunningham, I. Karthickeyan, C. S. Carter, W. Schneider, and J. D. Cohen (2002). Fiswidgets: A graphical computing environment for neuroimaging analysis. *Neuroinformatics* 1, pp. 111–125. URL: https://link.springer.com/article/10.1385/NI:1:1:111.

[74] M. S. Gilzenrat, B. D. Holmes, P. J. Holmes, J. Rajkowski, G. Aston-Jones, and J. D. Cohen (2002). Simplified dynamics in a model of noradrenergic modulation of cognitive performance. *Neural Networks* 15, pp. 647–663. URL: https://www.sciencedirect.com/science/article/pii/S0893608002000552.

[73] M. Graziano, J. D. Cohen, and M. M. Botvinick (2002). How the brain represents the body. *Attention and Performance XIX: Common Mechanisms in Perception and Action*, pp. 136–157. URL: https://web.math.princeton.edu/~sswang/autism/graziano_botvininck03.pdf.

[72] A. D. Jones, R. Cho, L. E. Nystrom, J. D. Cohen, and T. S. Braver (2002). A computational model of anterior cingulate function in speeded response tasks: Effects of frequency, sequence and conflict. *Cognitive, Affective, and Behavioral Neuroscience* 2.4, pp. 300–317. URL: https://link.springer.com/article/10.3758/CABN.2.4.300.

[71] P. R. Montague, G. S. Berns, J. D. Cohen, S. M. McClure, G. Pagnoni, M. Dhamala, ..., and R. E. Fisher (2002). Hyperscanning: Simultaneous fMRI during linked social interactions. *NeuroImage* 16.4, pp. 1159–1164. URL: https://pubmed.ncbi.nlm.nih.gov/12202103/.

[70] R. C. O'Reilly, D. C. Noelle, T. S. Braver, and J. D. Cohen (2002). Prefrontal cortex in dynamic categorization tasks: Representational organization and neuromodulatory control. *Cerebral Cortex* 12, pp. 246–257. URL: https://academic.oup.com/cercor/article/12/3/246/278120.

[69] D. M. Barch, C. S. Carter, T. S. Braver, F. W. Sabb, A. MacDonald, D. C. Noll, and J. D. Cohen (2001). Selective deficits in prefrontal cortex function in medication-naïve patients with schizophrenia. *Archives of General Psychiatry* 58, pp. 280–288. URL: https://jamanetwork.com/journals/jamapsychiatry/article-abstract/481734.

[68] M. M. Botvinick, T. S. Braver, C. S. Carter, D. M. Barch, and J. D. Cohen (2001). Conflict monitoring and cognitive control. *Psychological Review* 108.3, pp. 624–652. URL: https://psycnet.apa.org/record/2001-07628-005.

[67] T. S. Braver, D. M. Barch, B. A. Keys, C. S. Carter, J. D. Cohen, J. A. Kaye, ..., and B. Reed (2001). Context processing in older adults: Evidence for a theory relating cognitive control to neurobiology in healthy aging. *Journal of Experimental Psychology: General* 130, pp. 746–763. URL: https://psycnet.apa.org/record/2001-05320-009.

[66] T. S. Braver and J. D. Cohen (2001). Working memory, cognitive control, and the prefrontal cortex: Computational and empirical studies. *Cognitive Processing* 2, pp. 25–55. URL: https://bpb-us-w2.wpmucdn.com/sites.wustl.edu/dist/1/1008/files/2017/10/2001CogProc25-55-2hvgefy.pdf.

[65] J. D. Greene, R. B. Sommerville, L. E. Nystrom, J. M. Darley, and J. D. Cohen (2001). An fMRI investigation of emotional engagement in moral judgment. *Science* 293, pp. 2105–2108. URL: https://www.science.org/doi/full/10.1126/science.1062872.

[64] E. K. Miller and J. D. Cohen (2001). An integrative theory of prefrontal cortex function. *Annual Review of Neuroscience* 24, pp. 167–202. URL: https://www.annualreviews.org/doi/abs/10.1146/annurev.neuro.24.1.167.

[63] G. C. Organization of Human Brain Mapping (2001). Neuroimaging databases. *Science* 292, pp. 1673–1676. URL: https://www.science.org/doi/full/10.1126/science.1061041.

[62] W. M. Perlstein, C. S. Carter, D. C. Noll, and J. D. Cohen (2001). Relation of prefrontal cortex dysfunction to working memory and symptoms in schizophrenia. *American Journal of Psychiatry* 158, pp. 1105–1113. URL: https://ajp.psychiatryonline.org/doi/full/10.1176/appi.ajp.158.7.1105.

[61] M. Usher, H. Haarmann, J. D. Cohen, and D. Horn (2001). Neural mechanism for the magical number 4: Competitive interactions and non-linear oscillations. *Behavioral and Brain Sciences* 24, p. 151. URL: https://www.cambridge.org/core/journals/behavioral-and-brain-sciences/article/neural-mechanism-for-the-magical-number-4-competitive-interactions-and-nonlinear-oscillation/3C2898F832DAF8F836728B6A3698890E.

[60] V. van Veen, J. D. Cohen, M. M. Botvinick, V. A. Stenger, and C. S. Carter (2001). Anterior cingulate cortex, conflict monitoring, and levels of processing. *Neuroimage* 14, pp. 1302–1308. URL: https://www.sciencedirect.com/science/article/pii/S1053811901909231.

[59] T. S. Braver and J. D. Cohen (2000). On the control of control: The role of dopamine in regulating prefrontal function and working memory. *Attention and Performance XVIII: Control of Cognitive Processes*, pp. 713–737. URL: https://profiles.wustl.edu/en/publications/on-the-control-of-control-the-role-of-dopamine-in-regulating-pref.

[58] C. S. Carter, A. M. Macdonald, M. Botvinick, L. L. Ross, V. A. Stenger, D. Noll, and J. D. Cohen (2000). Parsing executive processes: Strategic vs. evaluative functions of the anterior cingulate cortex. *Proceedings of the National Academy of Sciences, USA* 97.4, pp. 1944–1948. URL: https://www.pnas.org/doi/abs/10.1073/pnas.97.4.1944.

[57] A. W. MacDonald III, J. D. Cohen, V. A. Stenger, and C. S. Carter (2000). Dissociating the role of dorsolateral prefrontal cortex and anterior cingulate cortex in cognitive control. *Science* 288, pp. 1835–1837. URL: https://www.science.org/doi/full/10.1126/science.288.5472.1835.

[56] L. E. Nystrom, T. S. Braver, F. W. Sabb, M. R. Delgado, D. C. Noll, and J. D. Cohen (2000). Working memory for letters, shapes and locations: fMRI evidence against stimulus-based regional organization of human prefrontal cortex. *Neuroimage* 11, pp. 424–446. URL: https://www.sciencedirect.com/science/article/pii/S105381190090572X.

[55] G. Aston-Jones, J. Rajkowski, and J. Cohen (1999). Role of locus coeruleus in attention and behavioral flexibility. *Biological Psychiatry* 46, pp. 1309–1320. URL: https://www.sciencedirect.com/science/article/pii/S0006322399001407.

[54] D. M. Barch, C. S. Carter, T. S. Braver, F. W. Sabb, D. C. Noll, and J. D. Cohen (1999). Overt verbal responding during fMRI scanning: Empirical investigations of problems and potential solutions. *Neuroimage* 10.6, pp. 642–657. URL: https://www.sciencedirect.com/science/article/pii/S1053811999905001.

[53] D. M. Barch, C. S. Carter, P. C. Hachten, and J. D. Cohen (1999). The "benefits" of distractibility: Mechanisms underlying increased Stroop effects in schizophrenia. *Schizophrenia Bulletin* 24.4, pp. 749–762. URL: https://academic.oup.com/schizophreniabulletin/article/25/4/749/1939774.

[52] D. M. Barch, C. S. Carter, W. M. Perlstein, J. Baird, J. D. Cohen, and N. Schooler (1999). Increased Stroop facilitation effects in schizophrenia are not due to increased automatic spreading activation. *Schizophrenia Research* 39.1, pp. 51–64. URL: https://www.sciencedirect.com/science/article/pii/S0920996499000250.

[51] M. M. Botvinick, L. Nystrom, K. Fissell, C. S. Carter, and J. D. Cohen (1999). Conflict monitoring versus selection-for-action in anterior cingulate cortex. *Nature* 402.6758, pp. 179–181. URL: https://www.nature.com/articles/46035.

[50] T. S. Braver, D. M. Barch, and J. D. Cohen (1999). Cognition and control in schizophrenia: A computational model of dopamine and prefrontal function. *Biological Psychiatry* 46.3, pp. 312–328. URL: https://www.sciencedirect.com/science/article/pii/S000632239900116X.

[49] J. D. Cohen, D. M. Barch, C. S. Carter, and D. Servan-Schreiber (1999). Context-processing deficits in schizophrenia: converging evidence from three theoretically motivated cognitive tasks. *Journal of Abnormal Psychology* 108, pp. 120–133. URL: https://psycnet.apa.org/record/1999-00603-011.

[48] R. Condray, S. R. Steinhauer, J. D. Cohen, D. P. van Kammen, and A. Kasparek (1999). Modulation of language processing in schizophrenia: Effects of context and haloperidol on the event-related potential. *Biological Psychiatry* 45.10, pp. 1336–1355. URL: https://www.sciencedirect.com/science/article/pii/S0006322398003175.

[47] M. Usher, J. D. Cohen, D. Servan-Schreiber, J. Rajkowski, P. Kubiak, and G. Aston-Jones (1999). The role of locus coeruleus in the regulation of cognitive performance. *Science* 283, pp. 549–554. URL: https://www.science.org/doi/full/10.1126/science.283.5401.549.

[46] M. M. Botvinick and J. D. Cohen (1998). Rubber hands 'feel' touch that eyes see. *Nature* 391, p. 756. URL: https://www.nature.com/articles/35784.

[45] C. S. Carter, T. S. Braver, D. M. Barch, M. M. Botvinick, D. C. Noll, and J. D. Cohen (1998). Anterior cingulate cortex, error detection and the on-line monitoring of performance. *Science* 280, pp. 747–749. URL: https://pubmed.ncbi.nlm.nih.gov/9563953/.

[44] C. S. Carter, W. Perlstein, R. Ganguli, J. Brar, M. Mintun, and J. D. Cohen (1998). Functional hypofrontality and working memory dysfunction in schizophrenia. *American Journal of Psychiatry* 155.9, pp. 1285–1287. URL: https://ajp.psychiatryonline.org/doi/full/10.1176/ajp.155.9.1285.

[43] B. J. Casey, J. D. Cohen, K. O'Craven, R. J. Davidson, W. Irwin, C. A. Nelson, ..., and P. A. Turski (1998). Reproducibility of fMRI results across four institutions using a spatial working memory task. *Neuroimage* 8.3, pp. 249–261. URL: https://www.sciencedirect.com/science/article/pii/S1053811998903603.

[42] J. D. Cohen, M. Usher, and J. L. McClelland (1998). A PDP approach to set size effects within the Stroop task: Reply to Kanne, Balota, Spieler, and Faust (1998). *Psychological Review* 105.1, pp. 188–194. URL: https://psycnet.apa.org/record/1997-42747-009.

[41] D. Servan-Schreiber, R. Bruno, C. Carter, and J. D. Cohen (1998). Dopamine and the mechanisms of cognition. Part I: A neural network model predicting dopamine effects on selective attention. *Biological Psychiatry* 43, pp. 713–722. URL: https://www.sciencedirect.com/science/article/pii/S0006322397004484.

[40] D. Servan-Schreiber, C. Carter, R. Bruno, and J. D. Cohen (1998). Dopamine and the mechanisms of cognition. Part II: D-Amphetamine effects in human subjects performing a selective attention task. *Biological Psychiatry* 43, pp. 723–729. URL: https://www.sciencedirect.com/science/article/pii/S0006322397004496.

[39] D. Servan-Schreiber, W. M. Perlstein, J. D. Cohen, and M. Mintun (1998). Selective pharmacological activation of limbic structures in human volunteers: A positron emission tomography study. *The Journal of Neuropsychiatry and Clinical Neurosciences* 10, pp. 148–159. URL: https://neuro.psychiatryonline.org/doi/full/10.1176/jnp.10.2.148.

[38] J. L. Armony, D. Servan-Schreiber, J. D. Cohen, and J. E. LeDoux (1997). Computational modeling of emotion: Explorations through the anatomy and physiology of fear conditioning. *Trends in Cognitive Sciences* 1, pp. 28–34. URL: https://pubmed.ncbi.nlm.nih.gov/21223850/.

[37] J. L. Armony, D. Servan-Schreiber, L. M. Romanski, J. D. Cohen, and J. E. LeDoux (1997). Stimulus generalization of fear responses: Effects of auditory cortex lesions in a computational model and in rats. *Cerebral Cortex* 7, pp. 157–165. URL: https://psycnet.apa.org/record/1997-05543-001.

[36] D. M. Barch, T. S. Braver, L. E. Nystrom, S. D. Forman, D. C. Noll, and J. D. Cohen (1997). Dissociating working memory from task difficulty in human prefrontal cortex. *Neuropsychologia* 35, pp. 1373–1380. URL: https://pubmed.ncbi.nlm.nih.gov/9347483/.

[35] G. S. Berns, J. D. Cohen, and M. A. Mintun (1997). Brain regions responsive to novelty in the absence of awareness. *Science* 276, pp. 1272–1275. URL: https://www.science.org/doi/full/10.1126/science.276.5316.1272.

[34] T. S. Braver, J. D. Cohen, J. Jonides, E. E. Smith, and D. C. Noll (1997). A parametric study of prefrontal cortex involvement in human working memory. *NeuroImage* 5.1, pp. 49–62. URL: https://www.sciencedirect.com/science/article/pii/S1053811996902475.

[33] C. S. Carter, M. Mintun, T. Nichols, and J. D. Cohen (1997). Anterior cingulate gyrus dysfunction and selective attention dysfunction in schizophrenia: An 15OH20 PET study during Stroop task performance. *American Journal of Psychiatry* 154, pp. 1670–1675. URL: https://ajp.psychiatryonline.org/doi/full/10.1176/ajp.154.12.1670.

[32] B. J. Casey, R. J. Trainor, J. L. Orendi, A. B. Schubert, L. E. Nystrom, J. Giedd, X. Castellanos, J. Haxby, D. C. Noll, J. D. Cohen, S. D. Forman, R. E. Dahl, and J. L. Rapoport (1997). A developmental functional MRI study of prefrontal activation during performance of a Go-No-Go task. *Journal of Cognitive Neuroscience* 9, pp. 835–847. URL: https://direct.mit.edu/jocn/article/9/6/835/3403/A-Developmental-Functional-MRI-Study-of-Prefrontal.

[31] J. D. Cohen, W. M. Perlstein, T. S. Braver, L. E. Nystrom, D. C. Noll, J. Jonides, and E. E. Smith (1997). Temporal dynamics of brain activation during a working memory task. *Nature* 386, pp. 604–608. URL: https://www.nature.com/articles/386604a0.

[30] C. R. Genovese, D. C. Noll, J. D. Cohen, and W. F. Eddy (1997). Estimating test-retest reliability in functional MR imaging I: Statistical methodology. *Magnetic Resonance in Medicine* 38, pp. 497–507. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/mrm.1910380319.

[29] N. Goddard, G. Hood, J. D. Cohen, W. F. Eddy, C. R. Genovese, and D. C. Noll (1997). Online analysis of functional MRI datasets on parallel platforms. *Journal of Supercomputing* 11, pp. 295–318. URL: https://link.springer.com/article/10.1023/A:1007964009986.

[28] B. MacWhinney, J. Cohen, and J. Provost (1997). The PsyScope experiment-building system. *Spatial Vision* 11, pp. 99–101. URL: https://europepmc.org/article/med/9304758.

[27] D. C. Noll, C. R. Genovese, L. Nystrom, A. Vazquez, S. D. Forman, W. F. Eddy, and J. D. Cohen (1997). Estimating test-retest reliability in functional MR imaging II: Application to motor and cognitive

activation studies. *Magnetic Resonance in Medicine* 38, pp. 508–517. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/mrm.1910380320.

[26] D. Barch, J. D. Cohen, D. Servan-Schreiber, S. Steingard, S. Steinhauer, and D. van Kammen (1996). Semantic priming in schizophrenia: An examination of spreading activation using word pronunciation and multiple SOAs. *Journal of Abnormal Psychology* 105, pp. 592–601. URL: https://pubmed.ncbi.nlm.nih.gov/8952192/.

[25] J. D. Cohen, T. S. Braver, and R. C. O'Reilly (1996). A computational approach to prefrontal cortex, cognitive control, and schizophrenia: Recent developments and current challenges. *Philosophical Transactions of the Royal Society of London Series B (Biological Sciences)* 351.1346, pp. 1515–1527. URL: https://royalsocietypublishing.org/doi/abs/10.1098/rstb.1996.0138.

[24] D. Servan-Schreiber, J. D. Cohen, and S. Steingard (1996). Schizophrenic deficits in the processing of context: A test of a theoretical model. *Archives of General Psychiatry* 53, pp. 1105–1112. URL: https://jamanetwork.com/journals/jamapsychiatry/article-abstract/497698.

[23] J. L. Armony, D. Servan-Schreiber, J. D. Cohen, and J. E. LeDoux (1995). An anatomically-constrained neural network model of fear conditioning. *Behavioral Neuroscience* 109.2, pp. 246–256. URL: https://psycnet.apa.org/record/1995-28155-001.

[22] T. S. Braver, J. D. Cohen, and D. Servan-Schreiber (1995). A computational model of prefrontal cortex function. *Advances in Neural Information Processing Systems (NeurIPS)* 7, pp. 141–149. URL: https://dl.acm.org/doi/abs/10.5555/2998687.2998705.

[21] C. S. Carter, M. Mintun, and J. D. Cohen (1995). Interference and facilitation effects during selective attention: An [150]-H20 PET study of Stroop task performance. *NeuroImage* 2, pp. 264–272. URL: https://pubmed.ncbi.nlm.nih.gov/9343611/.

[20] B. J. Casey, J. D. Cohen, P. Jezzard, R. Turner, D. C. Noll, R. Trainor, J. Giedd, L. Pannier, D. Kaysen, and J. L. Rapoport (1995). Activation of prefrontal cortex in children during a non-spatial working memory task with functional MRI. *NeuroImage* 2, pp. 221–229. URL: https://pubmed.ncbi.nlm.nih.gov/9343606/.

[19] S. D. Forman, J. D. Cohen, M. Fitzgerald, W. F. Eddy, M. A. Mintun, and D. C. Noll (1995). Improved assessment of significant activation in functional magnetic resonance imaging (fMRI): Use of a cluster-size threshold. *Magnetic Resonance in Medicine* 33, pp. 636–647. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/mrm.1910330508.

[18] D. C. Noll, J. D. Cohen, C. H. Meyer, and W. Schneider (1995). Spiral k-space MR imaging of cortical activation. *Journal of Magnetic Resonance Imaging* 45, pp. 49–56. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/jmri.1880050112.

[17] J. D. Cohen, S. D. Forman, T. S. Braver, B. J. Casey, D. Servan-Schreiber, and D. C. Noll (1994). Activation of prefrontal cortex in a non-spatial working memory task with functional MRI. *Human Brain Mapping* 1, pp. 293–304. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/hbm.460010407.

[16] J. D. Cohen and T. A. Huston (1994). Progress in the use of interactive models for understanding attention and performance. *Attention and Performance XV: Conscious and Nonconscious Information Processing*, pp. 453–476. URL: https://psycnet.apa.org/record/1994-98352-018.

[15] J. D. Cohen, R. D. Romero, D. Servan-Schreiber, and M. J. Farah (1994). Mechanisms of spatial attention: The relation of macrostructure to microstructure in parietal neglect. *Journal of Cognitive Neuroscience* 6.4, pp. 377–387. URL: https://direct.mit.edu/jocn/article/6/4/377/3128/Mechanisms-of-Spatial-Attention-The-Relation-of.

[14] J. D. Cohen, B. MacWhinney, M. Flatt, and J. Provost (1993). PsyScope: A new graphic interactive environment for designing psychology experiments. *Behavioral Research Methods, Instruments & Computers* 25.2, pp. 257–271. URL: https://cir.nii.ac.jp/crid/1573105974099321728.

[13] J. D. Cohen, D. C. Noll, and W. Schneider (1993). Functional Magnetic Resonance Imaging: Overview and methods for psychological research. *Behavioral Research Methods, Instruments & Computers* 25.2, pp. 101–113. URL: https://link.springer.com/article/10.3758/BF03204483.

[12] J. D. Cohen and D. Servan-Schreiber (1993). A theory of dopamine function and its role in cognitive deficits in schizophrenia. *Schizophrenia Bulletin* 19.1, pp. 85–104. URL: https://academic.oup.com/schizophreniabulletin/article/19/1/85/1895784.

[11] S. D. Forman, J. D. Cohen, and M. H. Johnson (1993). Frontal eye fields: Inhibition through competition. *Behavioral and Brain Sciences* 6, p. 578. URL: https://www.cambridge.org/core/journals/behavioral-and-brain-sciences/article/frontal-eye-fields-inhibition-through-competition/C904DB0715E247CABA44C47

[10] W. Schneider, D. C. Noll, and J. D. Cohen (1993). Functional topographic mapping of the cortical ribbon in human vision with conventional MRI scanners. *Nature* 365, pp. 150–153. URL: https://www.nature.com/articles/365150a0.

[9] J. D. Cohen and D. Servan-Schreiber (1992). Context, cortex and dopamine: A connectionist approach to behavior and biology in schizophrenia. *Psychological Review* 99, pp. 45–77. URL: https://psycnet.apa.org/record/1992-16573-001.

[8] J. D. Cohen, D. Servan-Schreiber, and J. L. McClelland (1992). A parallel distributed processing approach to automaticity. *American Journal of Psychology* 105, pp. 239–269. URL: https://www.jstor.org/stable/1423029.

[7] D. Servan-Schreiber and J. D. Cohen (1991). Models of neuromodulation and information processing deficits in schizophrenia. *Revue Internationale de Psychopathologie* 1, pp. 113–134.

[6] D. Servan-Schreiber and J. D. Cohen (1991). Network models of neuromodulation and information processing deficits in schizophrenia. *CogSci 1991: Proceedings of the Annual Meeting of the Cognitive Science Society* 13. URL: https://www.taylorfrancis.com/chapters/edit/10.4324/9781315807249-39/models-neuromodulation-information-processing-deficits-schizophrenia-david-servan-schreiber-jonathan-cohen.

[5] J. D. Cohen, K. Dunbar, and J. L. McClelland (1990). On the control of automatic processes: A parallel distributed processing model of the Stroop effect. *Psychological Review* 97.3, pp. 332–361. URL: https://psycnet.apa.org/record/1990-27437-001.

[4] D. Servan-Schreiber, H. Printz, and J. D. Cohen (1990). A network model of catecholamine effects: Gain, signal-to-noise ratio, and behavior. *Science* 249, pp. 892–895. URL: https://www.science.org/doi/abs/10.1126/science.2392679.

[3] D. Servan-Schreiber, H. Printz, and J. D. Cohen (1989). The effect of catecholamines on performance: From unit to system behavior. *Advances in Neural Information Processing Systems (NeurIPS)* 2. URL: https://proceedings.neurips.cc/paper/1989/hash/6f3ef77ac0e3619e98159e9b6febf557-Abstract.html.

[2] J. D. Cohen, T. Van Putten, S. Marder, P. A. Berger, and S. M. Stahl (1987). The efficacy of piquindone, a new atypical neuroleptic, in the treatment of the positive and the negative symptoms of schizophrenia. *Journal of Clinical Psychopharmacology* 7.5, pp. 324–329. URL: https://journals.lww.com/psychopharmacology/Abstract/1987/10000/BRIEF_REPORTS_The_Efficacy_of_Piquindone,_a_New.6.aspx.

[1] J. D. Cohen, T. Van Putten, S. Marder, and S. M. Stahl (1987). Treatment of the symptoms of schizophrenia with piquindone, a new atypical neuroleptic. *Psychopharmacology Bulletin* 23.3, pp. 514–518. URL: https://books.google.com/books?hl=en&lr=&id=krRJmzbbZ5UC&oi=fnd&pg=PA514&dq=Cohen+JD,+Van+Putten+T,+Marder+S,+Berger+PA+%26+Stahl+SM+(1987).+Treatment+of+the+symptoms+of+schizophrenia+with+piquindone,+a+new+atypical+neuroleptic.+Psychopharmacology+Bulletin,+23(3),+514-518.&ots=I_cr42q2hI&sig=7zbmaqu_d5KJtDSFiQcjqJBYlRg#v=onepage&q&f=false.

2. Invited Reviews, Commentary, Chapters, Edited Volumes & Technical Reports

[63] S. Musslick, J. D. Cohen, and T. Goschke (2024). Meta-control. *Reference Module in Neuroscience and Biobehavioral Psychology*. Elsevier, 2024. DOI: 10.1016/B978-0-12-820480-1.00058-9.

[62] C. Baru, M. Pozmantier, I. Altintas, S. Baek, J. Cohen, L. Condon, G. Fanti, I. Foster, E. Jackson, U. Lall, B. Landman, H. Li, C. Marin, B. M. Lopez, D. Metaxas, B. Olsen, G. Page, J. Shang, Y. Turkan, and P. Zhang (2021). Enabling AI Innovation via Data and Model Sharing: An Overview of the NSF Convergence Accelerator Track D. *AI Magazine*.

[61] J. D. Cohen (2017). Cognitive Control: Core Constructs and Current Considerations. *The Wiley Handbook of Cognitive Control*. Ed. by T. Egner. Chichester, West Sussex, UK: John Wiley & Sons, 2017.

[60] J. D. Cohen (2016). Functional MRI (fMRI): A window into the working brain. *Handbook of Experimental Economics*. Ed. by J. Kagel and A. Roth. 2016.

[59] J. D. Cohen, B. Dey, T. Griffiths, S. Musslick, K. Ozcimder, D. Reichman, I. Shinkar, and T. Wagner (2016). A graph-theoretic approach to multitasking. 2016. arXiv: 1611.02400.

[58] W. Keung, D. Osherson, and J. D. Cohen (2015). Influence of cognitive control on semantic representation. 2015. URL: https://www.biorxiv.org/content/10.1101/067553v1.

[57] Y. Wang, M. Anderson, J. D. Cohen, A. Heinecke, K. Li, N. Satish, N. Sundaram, N. B. Turk-Browne, and T. Willke (2015). Optimizing full correlation matrix analysis of fMRI data on Intel Xeon Phi coprocessors. *SC '15: Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*. Austin, TX, 2015, pp. 1–11. DOI: 10.1145/2807591.2807631.

[56] J. T. McGuire, J. D. Cohen, and M. M. Botvinick (2013). Mental Effort. *Encyclopedia of the Mind*. Ed. by H. Pashler. Thousand Oaks, CA: Sage Publications, 2013.

[55] R. C. O'Reilly, A. A. Petrov, J. D. Cohen, C. Lebiere, S. A. Herd, T. Kriete, and J. Symons (2013). How limited systematicity emerges: A computational cognitive neuroscience approach. *The architecture of cognition: Rethinking fodor and Pylyshyn's systematicity challenge*. 2013.

[54] S. E. Hyman and J. D. Cohen (2012). Disorders of mood and anxiety. *Principles of Neural Science V*. Ed. by E. R. Kandel, J. H. Schwartz, and T. M. Jessell. New York: McGraw-Hill, 2012.

[53] S. E. Hyman and J. D. Cohen (2012). Disorders of thought and volition: Schizophrenia. *Principles of Neural Science V*. Ed. by E. R. Kandel, J. H. Schwartz, and T. M. Jessell. New York: McGraw-Hill, 2012.

[52] N. Yeung, J. D. Cohen, and M. M. Botvinick (2011). Errors of interpretation and modeling: A reply to Grinband et al. *NeuroImage* 57.2, pp. 316–319.

[51] P. Holmes, P. Eckhoff, K. Wong-Lin, R. Bogacz, M. Zacksenhouse, and J. D. Cohen (2010). The physics of decision making: stochastic differential equations as models for neural dynamics and evidence accumulation in cortical circuits. *XVIth International Congress on Mathematical Physics*. 2010, pp. 123–142.

[50] J. D. Cohen and T. R. Insel (2008). Cognitive neuroscience and schizophrenia: Translational research in need of a translator. *Biological Psychiatry* 64.1, pp. 2–3.

[49] C. Keysers, R. Boyd, J. Cohen, M. Donald, W. Güth, E. Johnson, R. Kurzban, L. Schooler, J. Schooler, E. Spelke, and J. Trommershäuser (2008). Explicit and implicit strategies in decision making. *Strüngmann Forum Report: Better than Conscious? Decision Making, the Human Mind, and Implications*. Ed. by C. Engel and W. Singer. Cambridge, MA: MIT Press, 2008, pp. 225–258.

[48] P. Simen, P. Holmes, and J. D. Cohen (2008). On the neural implementation of optimal decisions. *Oxford Handbook of Human Action*. Ed. by E. Morsella, J. A. Bargh, and P. M. Gollwitzer. Oxford, UK: Oxford University Press, 2008, pp. 534–549.

[47] G. S. Aston-Jones, M. Iba, E. Clayton, J. Rajkowski, and J. Cohen (2007). The locus coeruleus and regulation of behavioral flexibility and attention: Clinical implications. *Brain Norepinephrine: Neurobiology and Therapeutics*. Ed. by G. A. Ordway, M. A. Schwartz, and A. Frazer. Cambridge, UK: Cambridge University Press, 2007, pp. 196–235.

[46] S. M. McClure, M. M. Botvinick, N. Yeung, J. D. Greene, and J. D. Cohen (2007). Conflict monitoring in cognition-emotion competition. *Handbook of Emotion Regulation*. Ed. by J. J. Gross. New York: Guilford Press, 2007.

[45] C. B. Holroyd, N. Yeung, M. G. H. Coles, and J. D. Cohen (2006). *Internalizing representations of correct behavior from feedback*. Tech. rep. 06-02. Princeton University, 2006.

[44] J. D. Cohen and G. Aston-Jones (2005). Cognitive neuroscience: Decision amid uncertainty. *Nature* 436.7050, pp. 471–472.

[43] P. J. Holmes, R. Bogacz, J. D. Cohen, and J. I. Gold (2005). Letter to the Editor, in response to Hunter PW (2004), Connections, context, and community: Abraham Wald and the sequential probability ratio test. *Mathematical Intelligencer* 27.1, pp. 4–5.

[42] M. Botvinick, T. S. Braver, N. Yeung, C. S. Carter, and J. D. Cohen (2004). Conflict monitoring: Computational and empirical studies. *Cognitive Neuroscience of Attention*. Ed. by M. I. Posner. New York: Guilford Press, 2004.

[41] C. S. Carter, J. G. Kerns, and J. D. Cohen (2004). Cognitive neuroscience: Bridging thinking and feeling to the brain, and its implications for psychiatry. *Neurobiology of Mental Illness*. Ed. by D. S. Charney and E. J. Nestler. Oxford: Oxford University Press, 2004.

[40] J. D. Cohen, G. Aston-Jones, and M. S. Gilzenrat (2004). A systems-level perspective on attention and cognitive control: Guided activation, adaptive gating, conflict monitoring, and exploitation vs. exploration. *Cognitive Neuroscience of Attention*. Ed. by M. I. Posner. New York: Guilford Press, 2004, pp. 71–90.

[39] P. Holmes, E. Brown, J. Moehlis, R. Bogacz, J. Gao, P. Hu, G. Aston-Jones, E. Clayton, J. Rajkowski, and J. D. Cohen (2004). Optimal decisions: From neural spikes, through stochastic differential equations, to behavior. *Proceedings of the International Symposium on Nonlinear Theory and its Applications (NOLTA2004)*. 2004.

[38] A. G. Sanfey and J. D. Cohen (2004). Is knowing always feeling? *Proceedings of the National Academy of Sciences of the United States of America* 101.48, pp. 16709–16710.

[37] S. Nieuwenhuis, N. Yeung, and J. D. Cohen (2003). *A computational simulation of electrophysiological markers of anterior cingulate function in a go/nogo task*. Tech. rep. CSBMB. 03.1. Princeton NJ: Princeton University, 2003.

[36] T. Paus, J. V. Haxby, and J. D. Cohen (2003). Special Issue. *NeuroImage*.

[35] G. Aston-Jones, J. Rajkowski, and J. D. Cohen (2002). Role of the locus coeruleus-norepinephrine system in attention and behavioral flexibility. *Catecholamine Research: From Molecular Insights to Clinical Medicine*. Ed. by T. Nagatsu, T. Nabeshima, R. McCarty, and D. Goldstein. New York: Kluwer Academic/Plenum Publishing, 2002, pp. 357–361.

[34] R. Bogacz and J. D. Cohen (2002). *Parameterization of connectionist models*. Tech. rep. CSBMB.Conte.02.1. Princeton NJ: Princeton University, 2002.

[33] J. D. Cohen (2002). Inhibition and control. *The Dana Guide To Brain Health*. Ed. by F. E. Bloom, M. F. Beal, and D. J. Kupfer. New York: Free Press, 2002, pp. 185–189.

[32] J. D. Cohen (2002). Neural network models of prefrontal cortex and cognitive control. *Handbook of Neuropsychology, 2nd Edition, Vol. 7: The Frontal Lobes*. Ed. by J. Grafman. New York: Elsevier, 2002, pp. 195–213.

[31] J. D. Cohen and K. I. Blum (2002). Overview: Reward and decision. Introduction to special issue. *Neuron* 36.2, pp. 193–198.

[30] J. D. Cohen and D. Servan-Schreiber (2002). Context, cortex and schizophrenia: A connectionist approach to behavior and biology in schizophrenia. *Cognitive Modeling*. Ed. by T. A. Polk and C. M. Seifert. Cambridge: MIT Press, 2002.

[29] J. D. Cohen and F. Tong (2001). The face of controversy: Modular vs. distributed representations in the brain. *Science* 293, pp. 2405–2407.

[28] G. Aston-Jones, J. Rajkowski, and J. Cohen (2000). Locus coeruleus and regulation of behavioral flexibility and attention. *Progress in Brain Research*. 2000.

[27] J. D. Cohen (2000). Special Issue: Functional topography of prefrontal cortex. *NeuroImage* 11, pp. 378–379.

[26] J. D. Cohen, M. M. Botvinick, and C. S. Carter (2000). Anterior cingulate and prefrontal cortex: who's in control? *Nature Neuroscience* 3.5, pp. 421–423.

[25] T. S. Braver, D. M. Barch, and J. D. Cohen (1999). *Mechanisms of cognitive control: Active memory, inhibition, and the prefrontal cortex*. Tech. rep. PDP.CNS.99.1. Pittsburgh PA: Carnegie Mellon University, 1999.

[24] T. S. Braver and J. D. Cohen (1999). Dopamine, cognitive control, and schizophrenia: The gating model. *Disorders of Brain, Behavior and Cognition: The Neurocomputational Perspective*. Ed. by J. Reggia, E. Ruppin, and D. Glanzman. Elsevier, 1999.

[23] T. S. Braver and J. D. Cohen (1999). Dopamine, cognitive control, and schizophrenia: The gating model. *Progress in Brain Research* 121, pp. 327–349.

[22] C. S. Carter, M. M. Botvinick, and J. D. Cohen (1999). The contribution of the anterior cingulate cortex to executive processes in cognition. *Reviews in the Neurosciences* 10, pp. 49–57.

[21] R. C. O'Reilly, T. S. Braver, and J. D. Cohen (1999). A biologically-based neural network model of working memory. *Models of Working Memory*. Ed. by P. Shah and A. Miyake. Cambridge University Press, 1999, pp. 375–411.

[20] M. Usher and J. D. Cohen (1999). Short term memory and selection processes in a frontal-lobe model. *Connectionist Models in Cognitive Neuroscience*. Ed. by D. Heinke, G. W. Humphries, and A. Olsen. Springer-Verlag, 1999.

[19] G. Aston-Jones, J. Rajkowski, P. Kubiak, S. Ivanova, M. Usher, and J. D. Cohen (1998). Neuromodulation and cognitive performance: Recent studies of noradrenergic locus coeruleus neurons in behaving monkeys. *Catecholamines: Bridging Basic Science with Clinical Medicine (Advances in Pharmacology, Vol. 42)*. Ed. by D. Goldstein, G. Eisenhofer, and R. McCarty. New York: Academic Press, 1998, pp. 755–759.

[18] M. M. Botvinick, T. S. Braver, C. S. Carter, D. M. Barch, and J. D. Cohen (1998). *Toward a non-homuncular account of control: Cognitive neuroscientific evidence for conflict monitoring*. Tech. rep. PDP.CNS.98.1. Pittsburgh PA: Carnegie Mellon University, 1998.

[17] J. D. Cohen (1997). Brain Terrain. *Nature* 389, p. 142.

[16] J. D. Cohen (1997). Dimensions of consciousness: A commentary on Kinsbourne and Hobson. *Scientific Approaches to Consciousness: 25th Carnegie Symposium on Cognition*. Ed. by J. D. Cohen and J. W. Schooler. New Jersey: Lawrence Erlbaum Associates, 1997.

[15] J. D. Cohen, K. O. Dunbar, D. M. Barch, and T. S. Braver (1997). Reply to Schooler et al: Issues concerning relative speed of processing hypotheses, schizophrenic performance deficits, and prefrontal function in the Stroop task. *Journal of Experimental Psychology: General* 126, pp. 37–41.

[14] J. D. Cohen and J. W. Schooler (1997). Science and sentience: Some questions regarding the scientific investigation of consciousness. *Scientific Approaches to Consciousness: 25th Carnegie Symposium on Cognition*. Ed. by J. D. Cohen and J. W. Schooler. New Jersey: Lawrence Erlbaum Associates, 1997.

[13] J. D. Cohen and E. E. Smith (1997). Response to Owen AM, Tuning in to the temporal dynamics of brain activation using functional magnetic resonance imaging (fMRI). *Trends in Cognitive Sciences* 1, p. 125.

[12] B. J. Casey, J. D. Cohen, D. Noll, W. Schneider, J. Giedd, and J. L. Rapoport (1996). Functional magnetic resonance imaging: Studies of Cognition. *Handbook of Human Brain Function, Neuroimaging II: Clinical Applications*. Ed. by E. D. Bigler. New York: Plenum Press, 1996.

[11] J. D. Cohen and R. C. O'Reilly (1996). A preliminary theory of the interactions between prefrontal cortex and hippocampus that contribute to planning and prospective memory. *Prospective Memory: Theory and Applications*. Ed. by M. Brandimonte, G. O. Einstein, and M. A. McDaniel. Hillsdale, NJ: Erlbaum, 1996, pp. 267–295.

[10] M. S. Keshavan and J. D. Cohen (1996). Magnetic resonance spectroscopy and functional MRI. *Brain Imaging in Psychiatry*. Ed. by S. Lewis and N. Higgins. Oxford, UK: Blackwell Science, Ltd., 1996.

[9] D. Servan-Schreiber and J. D. Cohen (1996). Dopamine, frontal cortex, and schizophrenia: Model and data. *Neural Modeling of Brain and Cognitive Disorders*. Ed. by J. A. Reggia, E. Ruppin, and R. S. Berndt. Singapore: World Scientific Publishing Co., 1996.

[8] M. Usher, J. D. Cohen, D. Servan-Schreiber, J. Rajkowski, P. Kubiak, and G. Aston-Jones (1995). *A computational model of locus coeruleus function and its influence on cognitive performance*. Tech. rep. PDP.CNS.95.1. Pittsburgh PA: Carnegie Mellon University, 1995.

[7] J. D. Cohen, S. D. Forman, T. S. Braver, B. J. Casey, D. Servan-Schreiber, and D. C. Noll (1993). *Activation of prefrontal cortex in a non-spatial working memory task with functional MRI*. Tech. rep. PDP.CNS.93.3. Pittsburgh PA: Carnegie Mellon University, 1993.

[6] J. D. Cohen, R. Romero, D. Servan-Schreiber, and M. J. Farah (1993). *Disengaging from the disengage function: The relation of macrostructure to microstructure in parietal attentional deficits*. Tech. rep. PDP.CNS.93.2. Pittsburgh PA: Carnegie Mellon University, 1993.

[5] B. J. Baars, J. D. Cohen, G. H. Bower, and J. W. Berry (1992). Some caveats on testing the Freudian slip hypothesis: Problems in systematic replication. *Experimental slips and human error, Exploring the architecture of volition*. Ed. by B. J. Baars. New York: Plenum Press, 1992.

[4] J. D. Cohen and D. Servan-Schreiber (1992). Introduction to neural networks in psychiatry. *Psychiatric Annals* 22.3, pp. 113–118.

[3] J. D. Cohen, D. Servan-Schreiber, E. Targ, and D. Spiegel (1992). The fabric of thought disorder: A cognitive neuroscience approach to disturbances in the processing of context in schizophrenia. *Cognitive Science and Clinical Disorders*. Ed. by D. Stein and J. Young. NY: Academic Press, 1992.

[2] D. Servan-Schreiber and J. D. Cohen (1992). A neural network model of catecholaminergic modulation of behavior. *Psychiatric Annals* 22.3, pp. 125–130.

[1] D. Servan-Schreiber and J. D. Cohen (1992). Issue Editors, Neural network models in Psychiatry. *Psychiatric Annals* 22.3.

3. Books

[1] J. D. Cohen and J. W. Schooler, eds. *Scientific Approaches to Consciousness: 25th Carnegie Symposium on Cognition*. New Jersey: Lawrence Erlbaum Associates, 1997. URL: https://www.taylorfrancis.com/books/edit/10.4324/9781315806563/scientific-approaches-consciousness-jonathan-cohen-jonathan-schooler.

4. Published Abstracts

[178] A. Pandey, D. Turner, and J. Cohen (2026). Fitting cognitive simulators to people: Distributed and hierarchical parameter estimation in PsyNeuLink. *USRSE'26 Conference Proceedings*. URL: https://zenodo.org/records/23402909.

[177] Y. Yang, M. Wang, J. D. Cohen, and T. W. Webb (2026). How large reasoning models solve problems: A mechanistic study on Tower of Hanoi. *Mechanistic Interpretability Workshop at ICML 2026*. 2026. URL: https://openreview.net/forum?id=SFazK589Xm.

[176] L. Bustamante, S. Grunevski, J. Solis, J. Lee, N. Daw, J. Cohen, D. Barch, and A. Konova (2024). Reduced food craving induction response associated with behavioral apathy and cognitive function symptoms of major depression. 2024.

[175] D. Campbell, S. Kumar, T. Giallanza, J. D. Cohen, and T. L. Griffiths (2024). Human-like geometric abstraction in large pre-trained neural networks. *International Conference on Learning and Representation (ICLR) 2024, Workshop on Representational Alignment*. 2024. URL: https://openreview.net/forum?id=h15aZUyxjw.

[174] L. Bustamante, M. Bogdanov, J. Solis, T. Oshinowo, S. Sheldon, S. Devine, J. Lee, E. Tong, A. Burton, I. Grahek, M. P. Fahey, A. Konova, R. Otto, D. Barch, A. Shenhav, J. Cohen, and N. Daw (2023). Quantifying individual differences in cognitive and physical effort cost with the Effort Foraging Task; applications in large online, clinical depression, and transcranial direct stimulation studies. *International conference on Motivational and Cognitive Control*. 2023.

[173] L. A. Bustamante, M. P. Fahey, J. Solis, T. Oshinowo, E. Tong, J. R. Lee, A. R. Burton, I. Grahek, A. Shenhav, D. M. Barch, N. D. Daw, A. B. Konova, and J. D. Cohen (2023). Anxiety in major depression associated with increased willingness to exert cognitive, but not physical effort. *Society for Biological Psychiatry Abstract*. Abstract.

[172] D. Campbell, S. Kumar, T. Giallanza, J. D. Cohen, and T. L. Griffiths (2023). Unraveling geometric reasoning: A neural network model of regularity biases. *International Symposium on the Mathematics of Neuroscience*. Spotlight talk. 2023.

[171] T. Giallanza, D. Campbell, and J. D. Cohen (2023). Adapting to a changing environment with controlled retrieval of episodic memories. *Fourth Annual Symposium on the Mathematics of Neuroscience*. 2023.

[170] B. J. Jongkees, S. Musslick, and J. D. Cohen (2022). A mechanistic account of the stability-flexibility trade-off. *The NVP Spring Conference on Cognition, Brain, and Behaviour of the Dutch Psychonomic Society*. Conference Abstract. 2022.

[169] B. J. Jongkees, S. Musslick, and J. D. Cohen (2022). A mechanistic account of the stability-flexibility trade-off. *The 22nd Conference of the European Society for Cognitive Psychology (ESCOP)*. Conference Abstract. 2022.

[168] Z. Dulberg and J. D. Cohen (2020). Learning canonical transformations. *BabyMind 2020, NeurIPS Workshop*. 2020.

[167] S. Kumar, I. Dasgupta, J. D. Cohen, N. Daw, and T. Griffiths (2020). Meta-Learning of compositional task-distributions in humans and machines. *MetaLearn 2020, NeurIPS Workshop*. 2020.

[166] V. van Veen, J. D. Cohen, M. M. Botvinick, V. A. Stenger, and C. S. Carter (2020). Conflict related activity and the anterior cingulate: An event-related fMRI analysis of stimulus vs. response based conflict. *Journal of Cognitive Neuroscience*, S114.

[165] M. Agrawal, M. G. Mattar, N. D. Daw, and J. D. Cohen (2019). Rational arbitration of hippocampal replay. *CCN 2019: Proceedings of the Annual Conference on Cognitive Computational Neuroscience*. 2019.

[164] A. Beukers, K. A. Norman, and J. D. Cohen (2019). Working with episodic memory: The n-back task. *CCN 2019: Proceedings of the Annual Conference on Cognitive Computational Neuroscience*. 2019.

[163] B. Ebitz, J. D. Cohen, T. Buschman, T. Moore, and B. Hayden (2019). Exploration via disrupted sensorimotor control dynamics. *Computational and Systems Neuroscience (CoSyNe)*. 2019.

[162] G. Henselman-Petrusek, S. Segert, B. Keller, M. Tepper, and J. D. Cohen (2019). Geometry of Shared Representations. *CCN 2019: Proceedings of the Annual Conference on Cognitive Computational Neuroscience*. 2019. DOI: 10.32470/CCN.2019.1418-0.

[161] T. L. Willke, S. B. M. Yoo, M. Capotă, S. Musslick, B. Y. Hayden, and J. D. Cohen (2019). A Comparison of non-human primate and deep reinforcement learning agent performance in a virtual pursuit-avoidance task. *RLDM 2019: Proceedings of the Multi-disciplinary Conference on Reinforcement Learning and Decision Making*. Preprint available at https://www.biorxiv.org/content/10.1101/567545v2. 2019.

[160] L. A. Bustamante, A. R. Burton, A. L. Baker, A. L. Shenhav, N. D. Daw, and J. D. Cohen (2018). The cost of cognitive control and the balance of random versus directed exploration. *Society for Neuroscience Abstracts*. 2018.

[159] L. A. Bustamante, F. Lieder, S. Musslick, A. Shenhav, and J. D. Cohen (2018). Learning to overexert cognitive control in the Stroop task. *CCN 2018: Proceedings of the Annual Conference on Cognitive Computational Neuroscience*. 2018.

[158] M. C. Iordan, V. J. H. Ritvo, K. A. Norman, N. B. Turk-Browne, and J. D. Cohen (2018). Using closed loop real-time fMRI neurofeedback to induce neural plasticity and influence perceptual similarity. *Society for Neuroscience Abstracts*. 2018.

[157] G. Kane, M. James, A. Shenhav, R. C. Wilson, N. Daw, G. Aston-Jones, and J. D. Cohen (2018). Does the anterior cingulate contribute to foraging decisions? *Computational and Systems Neuroscience (CoSyNe)*. 2018.

[156] S. Musslick, J. D. Cohen, and A. Shenhav (2018). Estimating the costs of cognitive control: Theoretical validation and potential pitfalls. *Society for Neuroscience Abstracts*. 2018.

[155] A. Novick, S. Musslick, M. Iordan, and J. D. Cohen (2018). Why we struggle to multitask: Converging evidence from computational modeling, human behavior, and neuroimaging. *Society for Neuroscience Abstracts*. 2018.

[154] M. Shvartsman, A. Charles, J. D. Cohen, M. Aoi, N. Sundaram, and T. Wilke (2018). Matrix-normal models for fMRI analysis. *Computational and Systems Neuroscience (CoSyNe)*. 2018.

[153] M. Shvartsman, N. Sundaram, M. Aoi, A. Charles, T. Willke, and J. D. Cohen (2018). Matrix-normal models for fMRI analysis. *Proceedings of the International Conference on Artificial Intelligence and Statistics (AISTATS)*. 2018.

[152] R. C. Wilson and J. D. Cohen (2018). Deep exploration explains the tradeoff between directed and random exploration. *Society for Neuroscience Abstracts*. 2018.

[151] M. Capotă, T. L. Willke, K. A. Norman, J. D. Cohen, and N. B. Turk-Browne (2017). Brain imaging analysis kit: Advanced fMRI analysis at scale. *Society for Neuroscience Abstracts*. 2017.

[150] A. Cherkaev, S. Musslick, J. D. Cohen, V. Srikumar, and M. Flatt (2017). SweetPea: A Language for Designing Experiments. *The 45th Symposium on Principles of Programming Languages (POPL)*. 2017.

[149] J. D. Cohen, M. Lesnick, B. Keller, C. Baldassano, A. Schapiro, and C. Ellis (2017). Using realistic, synthetic fMRI data to validate Topological Data Analysis as a tool for fMRI. *Society for Neuroscience Abstracts*. 2017.

[148] B. R. Ebitz, J. D. Cohen, and T. Buschman (2017). Mechanisms for generating flexibility in a changing world. *Society for Neuroscience Abstracts*. 2017.

[147] M. C. Iordan, V. J. H. Ritvo, K. A. Norman, N. B. Turk-Browne, and J. D. Cohen (2017). Inducing neural plasticity and perceptual similarity via real-time fMRI neurofeedback. *Society for Neuroscience Abstracts*. 2017.

[146] S. Musslick, S. G. Jang, M. Panichello, L. Bustamante, A. Shenhav, and J. D. Cohen (2017). Constraints associated with cognitive control and the stability-flexibility dilemma. *Society for Neuroscience Abstracts*. 2017.

[145] A. S. Novick, A. M. Bornstein, K. A. Norman, and J. D. Cohen (2017). Drift diffusion modeling of interactions between episodic and working memory. *Society for Neuroscience Abstracts*. 2017.

[144] D. C. Suo, J. B. Hutchinson, M. deBettencourt, A. Mennen, Y. Wang, T. Willke, N. B. Turk-Browne, K. A. Norman, J. D. Cohen, and K. Li (2017). Real-time fMRI analysis in the cloud. *Society for Neuroscience Abstracts*. 2017.

[143] J. D. Cohen, B. Dey, T. Griffiths, S. Musslick, K. Ozcimder, D. Reichman, I. Shinkar, and T. Wagner (2016). A graph-theoretic approach to multitasking. 2016. arXiv: 1611.02400.

[142] O. Lositsky, R. C. Wilson, M. Shvartsman, and J. D. Cohen (2016). Adaptive task representations in context-based decision making. *Society for Neuroscience Abstracts*. 2016.

[141] I. Momennejad, M. Tomov, K. A. Norman, and J. D. Cohen (2016). The strategic allocation of working memory and episodic memory in prospective remembering: A neural network model. *Society for Neuroscience Abstracts*. 2016.

[140] S. Musslick, B. Dey, K. Özcimder, M. Patwary, P. Krieger, T. L. Willke, and J. D. Cohen (2016). Multitasking capacity versus efficiency of representation in neural network architectures. *Society for Neuroscience Abstracts, Nanosymposium*. 2016.

[139] A. S. Novick, A. M. Bornstein, K. A. Norman, and J. D. Cohen (2016). Blast from the past: Episodic memory supports working memory maintenance. *Society for Neuroscience Abstracts*. 2016.

[138] S. Wang, J. D. Cohen, and R. C. Wilson (2016). Blink different! Blink rate reflects individual differences in directed exploration. *Computational and Systems Neuroscience (CoSyNe)*. 2016.

[137] Y. Wang, B. Keller, M. Capota, M. J. Anderson, N. Sundaram, J. D. Cohen, K. Li, N. B. Turk-Browne, and T. L. Willke (2016). Real-time full correlation matrix analysis of fMRI data. *BigD237: 2016 IEEE International Conference on Big Data*. 2016.

[136] R. C. Wilson and J. D. Cohen (2016). A unifying theory of explore-exploit decisions. *Computational and Systems Neuroscience (CoSyNe)*. 2016.

[135] G. A. Kane, E. M. Vazey, R. C. Wilson, A. Shenhav, N. Daw, G. Aston-Jones, and J. D. Cohen (2015). Examining the Role of the Locus Coeruleus in Foraging and Exploration. *Society for Neuroscience Abstracts*. 2015.

[134] W. Keung, D. Osherson, and J. D. Cohen (2015). Influence of cognitive control on semantic representation. 2015. URL: https://www.biorxiv.org/content/10.1101/067553v1.

[133] P. Krueger, R. C. Wilson, and J. D. Cohen (2015). Strategies for exploration in the domain of losses. *Computational and Systems Neuroscience (CoSyNe)*. 2015.

[132] O. Lositsky, R. C. Wilson, M. Shvartsman, and J. D. Cohen (2015). A drift diffusion model of proactive and reactive control in a context-dependent two-alternative forced choice task. *RLDM 2015: Proceedings of the Multi-disciplinary Conference on Reinforcement Learning and Decision Making*. 2015.

[131] A. S. Novick, A. M. Bornstein, K. A. Norman, and J. D. Cohen (2015). Refresh my memory: Context information from episodic memory affects working memory maintenance. *Society for Neuroscience Abstracts*. 2015.

[130] Y. Wang, M. Anderson, J. D. Cohen, A. Heinecke, K. Li, N. Satish, N. Sundaram, N. B. Turk-Browne, and T. Willke (2015). Optimizing full correlation matrix analysis of fMRI data on Intel Xeon Phi coprocessors. *SC '15: Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*. Austin, TX, 2015, pp. 1–11. DOI: 10.1145/2807591.2807631.

[129] Y. Wang, M. Anderson, J. D. Cohen, A. Heinecke, K. Li, N. Satish, N. Sundaram, N. B. Turk-Browne, and T. L. Willke (2015). Full correlation matrix analysis of fMRI data on Intel® Xeon Phi™ coprocessors. *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*. ACM. 2015, p. 23.

[128] A. Geana, R. C. Wilson, and J. D. Cohen (2014). Risk, ambiguity and decision horizon in human exploration: A Wheel of Fortune task. *Society for Neuroscience Abstracts*. 2014.

[127] I. Momennejad, J. D. Cohen, and K. A. Norman (2014). Imagine the future! How does episodic simulation enhance prospective memory? *Society for Neuroscience Abstracts*. 2014.

[126] R. C. Wilson and J. D. Cohen (2014). Humans tradeoff information seeking and randomness in explore-exploit decisions. *Society for Neuroscience Abstracts*. 2014.

[125] M. T. deBettencourt, R. F. Lee, J. D. Cohen, K. A. Norman, and N. B. Turk-Browne (2013). Externalizing internal states with real-time neurofeedback to train visual attention. *Journal of Vision Sciences*.

[124] A. Shenhav, M. M. Botvinick, and J. D. Cohen (2013). Decision costs may explain dACC activity in a "foraging" context. *Society for Neuroscience Abstracts*. 2013.

[123] Y. Wang, K. Li, M. Charikar, J. D. Cohen, and N. B. Turk-Browne (2013). What you find depends on how you look: Category selectivity in frontal cortex revealed by whole-brain correlation analysis. *Journal of Vision Sciences*.

[122] R. C. Wilson, J. M. White, and J. D. Cohen (2013). The role of adaptive decision noise in exploration. *Society for Neuroscience Abstracts*. 2013.

[121] J. D. Cohen, J. A. Lewis-Peacock, and K. A. Norman (2012). Neural evidence for the flexible use of working memory and episodic memory in prospective remembering. *Society for Neuroscience Abstracts*. 2012.

[120] M. T. deBettencourt, R. F. Lee, J. D. Cohen, K. A. Norman, and N. B. Turk-Browne (2012). Real-time decoding and training of sustained attention. *Society for Neuroscience Abstracts*. 2012.

[119] E. Eldar, A. Radulescu, Y. Niv, and J. D. Cohen (2012). Norepinephrine, neural gain, and "first one wins" network dynamics. *Computational and Systems Neuroscience (CoSyNe)*. 2012.

[118] D. Tomlin, A. Nedić, R. C. Wilson, P. Holmes, and J. D. Cohen (2012). Group foraging task reveals separable influences of individual experience and social information. *Society for Neuroscience Abstracts*. 2012.

[117] S. Feng, M. Schwemmer, and J. D. Cohen (2011). Computational Constraints on Cognitive Control. *Society for Neuroscience Abstracts*. 2011.

[116] A. Geana, R. C. Wilson, J. M. White, E. Ludvig, and J. D. Cohen (2011). The Separate Roles of Reward Magnitude and Uncertainty in the Explore/Exploit Dilemma. *Society for Neuroscience Abstracts*. 830.10/XX40. 2011.

[115] S. Goldfarb, P. Simen, C. Caiceido, P. Holmes, N. E. Leonard, and J. D. Cohen (2011). Optimal performance and the LBA model. *Abstracts of the Psychonomic Society*. 2011.

[114] J. A. Lewis-Peacock, M. R. Salesi, J. D. Cohen, and K. A. Norman (2011). Decoding the use of working memory and episodic memory in prospective remembering. *Society for Neuroscience Abstracts*. 829.04. 2011.

[113] M. T. Todd, M. M. Botvinick, M. A. Schwemmer, J. D. Cohen, and P. Dayan (2011). Normative analysis of task switching. *Society for Neuroscience Abstracts*. 2011.

[112] D. Tomlin, A. Nedić, M. T. Todd, R. C. Wilson, D. A. Prentice, P. Holmes, and J. D. Cohen (2011). Group foraging task reveals separable influences of individual experience and social information. *Society for Neuroscience Abstracts*. 2011.

[111] R. C. Wilson, A. Geana, J. Myles-White, E. Ludwig, and J. D. Cohen (2011). Why the grass is greener on the other side: Behavioral evidence for an ambiguity bonus in human exploratory decision making. *Society for Neuroscience Abstracts*. 830.13. 2011.

[110] B. Eppinger, L. Nystrom, and J. D. Cohen (2010). Age-related changes in reward sensitivity during decision-making and learning. *Society for Neuroscience Abstracts*. 911.2/LLL48. 2010.

[109] S. J. Gershman, J. D. Cohen, and Y. Niv (2010). Learning to selectively attend. *CogSci 2010: Proceedings of the 32nd Annual Meeting of the Cognitive Science Society*. 2010.

[108] S. J. Getz, D. Tomlin, L. E. Nystrom, J. D. Cohen, and A. R. A. Conway (2010). Executive control of intertemporal choice: Effects of cognitive load on impulsive decision-making. *Abstracts of the Psychonomic Society*. Abstract 3067. 2010.

[107] M. T. Todd, Y. Niv, and J. D. Cohen (2010). Mental set representations in fMRI. *Society for Neuroscience Abstracts*. 603.8/KKK52. 2010.

[106] D. Tomlin, A. Nedić, D. A. Prentice, P. Holmes, and J. D. Cohen (2010). Group foraging task reveals neural substrates of social influence. *Society for Neuroscience Abstracts*. 403.9/LLL65. 2010.

[105] R. C. Wilson, J. D. Cohen, and Y. Niv (2010). Inferring relevance in a changing world. *Society for Neuroscience Abstracts*. 907.12/KKK45. 2010.

[104] T. M. Becker, R. Y. Cho, J. D. Cohen, J. G. Kerns, and C. S. Carter (2009). The association between impaired language production and prefrontal cortex goal maintenance deficits in medication-naive people with schizophrenia. *NeuroImage* 47.S1, p. 189.

[103] R. Cho, J. D. Cohen, A. Sacks, Y.-T. Yen, and C. S. Carter (2009). Disturbances in Frontal Cortical Gamma Oscillations in Association with Genetic Liability for Schizophrenia. *Abstracts of Panels and Posters, 48th Annual Meeting of the American College of Neuropsychopharmacology*. 2009.

[102] R. Y. Cho, S. E. Forster, J. D. Cohen, and C. S. Carter (2009). Impairments in prefrontal cortical gamma-band synchrony and cognitive control in first-episode schizophrenia. *Neuroimage*.

[101] S. E. Forster, R. Y. Cho, J. D. Cohen, and C. S. Carter (2009). Effects of parametric manipulations of conflict on N2 amplitude and cognitive control adjustments. *NeuroImage*.

[100] P. A. Simen, L. E. Nystrom, M. van Vugt, P. Krueger, and J. D. Cohen (2009). Event-related fMRI during slow decision making can reveal temporal structure in neural activity. *Society for Neuroscience Abstracts*. 576.9/FF111. 2009.

[99] D. Tomlin, A. Nedić, P. Holmes, and J. D. Cohen (2009). Neural and behavioral responses to social feedback during group decision-making. *Society for Neuroscience Abstracts*. 475.4/FF27. 2009.

[98] A. Nedić, D. Tomlin, P. Holmes, D. A. Prentice, and J. D. Cohen (2008). A simple decision task in a social context: Preliminary experiments and a model. *Proceedings of the 47th IEEE Conference on Decision and Control*. 2008.

[97] A. J. Yu and J. D. Cohen (2008). Sequential Effects: Annoying Quirk or Adaptive Behavior? *Computational and Systems Neuroscience*. Abstract III-58. 2008. URL: http://cosyne.org/cosyne08/posters/.

[96] W. van den Bos, J. Li, T. Lau, S. M. McClure, J. D. Cohen, and P. R. Montague (2007). Social influences on bidding in common value auctions and the Winner's Curse. *Abstract at meeting of the Society for Neuroeconomics*. Nantucket Beach, MA, 2007.

[95] K. D'Ardenne, S. M. McClure, L. E. Nystrom, and J. D. Cohen (2007). BOLD responses in the dopaminergic ventral tegmental area. *Society for Neuroscience Abstracts*. 704.21. 2007.

[94] J. D. Greene, J. M. Paxton, L. E. Nystrom, and J. D. Cohen (2007). Dissociation between affective and cognitive moral disapproval. *Society for Neuroscience Abstracts*. 126.4. 2007.

[93] S. M. McClure, M. M. Botvinick, N. Yeung, J. D. Greene, and J. D. Cohen (2007). Conflict monitoring in cognition-emotion competition. *Handbook of Emotion Regulation*. Ed. by J. J. Gross. New York: Guilford Press, 2007.

[92] P. A. Simen and J. D. Cohen (2007). A diffusion-based neural network model of interval timing and temporal discounting. *Society for Neuroscience Abstracts*. 637.13. 2007.

[91] M. T. Todd, K. F. Wong, and J. D. Cohen (2007). Competition, gating, and learning: A new computational model of task switching. *Society for Neuroscience Abstracts*. 634.10. 2007.

[90] G. J. Detre, S. M. Polyn, C. D. Moore, V. S. Natu, B. D. Singer, J. D. Cohen, J. V. Haxby, and K. A. Norman (2006). The Multi-Voxel Pattern Analysis (MVPA) toolbox. *Neuroimage* 31.S1, p. 128.

[89] K. D. McClure, S. M. McClure, L. E. Nystrom, and J. D. Cohen (2006). Functional MRI of midbrain dopamine nuclei. *Neuroimage* 31.S1, p. 2796.

[88] S. M. McClure, K. M. Ericson, D. I. Laibson, G. Loewenstein, and J. D. Cohen (2006). Time discounting for primary reward. *Neuroimage* 31.S1, p. 2749.

[87] J. D. Cohen, S. M. McClure, M. S. Gilzenrat, and G. Aston-Jones (2005). Dopamine-norepinephrine interactions: Exploitation vs. Exploration. *Neuropsychopharmacology* 30.S1, S28.

[86] J. D. Greene, K. Lowenberg, L. E. Nystrom, J. M. Darley, and J. D. Cohen (2005). Saving lives versus keeping promises: An fMRI investigation of consequentialist and deontological moral judgment. *Society for Neuroscience Abstracts*. 12.7. 2005.

[85] S. M. McClure, D. I. Laibson, G. Loewenstein, K. Ericson, K. D. McClure, and J. D. Cohen (2005). Neural mechanisms of time discounting for primary reward. *Society for Neuroeconomics*. 2005.

[84] S. M. Polyn, G. J. Detre, S. Takerkart, V. Natu, M. Benharrosh, B. Singer, J. D. Cohen, J. V. Haxby, and K. A. Norman (2005). A Matlab-based toolbox to facilitate multi-voxel pattern classification of fMRI data. *Neuroimage* 26.S1, p. 728.

[83] P. A. Simen, P. Holmes, and J. Cohen (2005). Threshold adaptation in decision making. *Society for Neuroscience Abstracts*. 768.15. 2005.

[82] B. E. Snitz, R. Y. Cho, G. Archer, J. D. Cohen, and C. S. Carter (2005). Lateral and medial hypofrontality in first-episode psychosis: Diagnostic specificity to schizophrenia. *Neuroimage* 26.S1, p. 1274.

[81] K. D'Ardenne, K. McClure, S. M. McClure, M. C. Richter, J. D. Cohen, and W. Richter (2004). Rest Matters: Kinetics of the BOLD response depend on inter-stimulus time. *Society for Neuroscience Abstracts*. 694.17. 2004.

[80] G. E. Dixon, J. B. Nitschke, S. J. Short, A. Lakshmanan, M. E. Carew, M. J. Anderle, H. S. Schaefer, T. Johnstone, J. D. Cohen, S. M. Kosslyn, E. E. Smith, and R. J. Davidson (2004). The neural correlates of anticipation: The functional neurobiology of gustatory expectancy in the human brain. *Society for Neuroscience Abstracts*. 179.10. 2004.

[79] M. S. Gilzenrat, E. T. Brown, G. Aston-Jones, and J. D. Cohen (2004). Locus Coeruleus, adaptive gain, and the optimization of decision tasks. *Society for Neuroscience Abstracts*. 899.6. 2004.

[78] S. M. McClure, D. I. Laibson, G. Loewenstein, and J. D. Cohen (2004). Separate neural systems represent immediate and delayed rewards. *Society for Neuroscience Abstracts*. 548.1. 2004.

[77] S. M. Polyn, J. D. Cohen, and K. A. Norman (2004). Detecting distributed patterns in an fMRI study of free recall. *Society for Neuroscience Abstracts*. 79.14. 2004.

[76] J. K. Rilling, A. G. Sanfey, J. A. Aronson, L. E. Nystrom, and J. D. Cohen (2004). Neural correlates of theory of mind within interpersonal interactions. *NeuroImage* 22.S1, TH129.

[75] M. Benharrosh, S. Takerkart, J. D. Cohen, I. Daubechies, and W. Richter (2003). Using ICA on fMRI data: does independence matter? *Neuroimage* 19.2, p. 778.

[74] S. J. Bishop, M. Farah, J. A. Fossella, B. J. Casey, and J. D. Cohen (2003). The MAOA-LPR polymorphism and prefrontal cortical activation to emotional distractors. *Neuroimage* 19.2, p. 6.

[73] R. Bogacz, J. Moehlis, E. Brown, P. Holmes, and J. D. Cohen (2003). Neural mechanisms for decision optimization. *Society for Neuroscience Abstracts*. 197.6. 2003.

[72] C. S. Carter, J. Kerns, M.-H. Sohn, and J. D. Cohen (2003). A conflict over conflict monitoring and the anterior cingulate cortex. *Society for Neuroscience Abstracts*. 343.12. 2003.

[71] M. S. Gilzenrat, J. D. Cohen, J. Rajkowski, and G. Aston-Jones (2003). Pupil dynamics predict changes in task engagement mediated by locus coeruleus. *Society for Neuroscience Abstracts*. 515.19. 2003.

[70] J. D. Greene, L. E. Nystrom, J. M. Darley, and J. D. Cohen (2003). Neural activity correlated with outcome of moral decision. *Cognitive Neuroscience Society Annual Meeting Program*. 2003, E81, p. 166.

[69] J. D. Greene, L. E. Nystrom, A. D. Engell, J. M. Darley, and J. D. Cohen (2003). Patterns of neural activity correlated with individual differences in moral judgment. *Society for Neuroscience Abstracts*. 443.3. 2003.

[68] S. Nieuwenhuis, C. B. Holroyd, N. Yeung, L. E. Nystrom, J. D. Cohen, R. B. Mars, and M. G. H. Coles (2003). Neural correlates of reinforcement learning and error processing: A functional magnetic resonance imaging study. *Society for Neuroscience Abstracts*. 197.11. 2003.

[67] J. K. Rilling, T. S. Wager, R. J. Davidson, S. M. Kosslyn, R. M. Rose, E. E. Smith, and J. D. Cohen (2003). An fMRI investigation of the neural mechanisms of placebo analgesia. *Neuroimage* 19.2, p. 504.

[66] S. Takerkart, M. Benharrosh, J. D. Cohen, and I. Daubechies (2003). Estimating spatially distributed patterns of brain activity in fMRI datasets by using ICA. *Neuroimage* 19.2, p. 946.

[65] T. D. Wager, E. E. Smith, A. Sokolik, J. D. Cohen, J. K. Rilling, R. J. Davidson, S. Kosslyn, R. M. Rose, and K. L. Casey (2003). Placebo reduces the BOLD fMRI response in the anticipation and experience of thermal pain. *Neuroimage* 19.2, p. 84.

[64] G. Aston-Jones, J. Rajkowski, W. Lu, Y. Zhu, J. D. Cohen, and R. Morecraft (2002). Prominent projections from the orbital prefrontal cortex to the locus coeruleus (LC) in monkey. *Society for Neuroscience Abstracts*. 86.9. 2002.

[63] C. S. Carter, V. van Veen, C. Holroyd, V. A. Stenger, and J. D. Cohen (2002). Errors and conflict but not error feedback engage the anterior cingulate cortex during event-related fMRI: implications for performance monitoring in the human brain. *Neuroimage* 16.2, p. 1002.

[62] E. Clayton, J. Rajkowski, J. Cohen, and G. Aston-Jones (2002). The primate locus coeruleus and discrimination/decision-making in the eriksen flanker task. *Society for Neuroscience Abstracts*. 86.8. 2002.

[61] L. E. Nystrom, N. P. Yeung, M. T. Kitazono, and J. D. Cohen (2002). Anterior cingulate activity: From conflict or motor planning? *Society for Neuroscience Abstracts*. 180.3. 2002.

[60] S. M. Polyn, K. A. Norman, and J. D. Cohen (2002). Connectionist modeling of source memory phenomena. *Society for Neuroscience Abstracts*. 582.5. 2002.

[59] J. Rajkowski, H. Majczynski, E. Clayton, J. Cohen, and G. Aston-Jones (2002). Phasic activation of monkey locus coeruleus (LC) neurons with recognition of behaviorally significant stimuli. *Society for Neuroscience Abstracts*. 86.1. 2002.

[58] J. K. Rilling, A. G. Sanfey, J. A. Aronson, L. E. Nystrom, and J. D. Cohen (2002). Mapping the brain's response to reciprocated and unreciprocated social cooperation. *Society for Neuroscience Abstracts*. 78.10. 2002.

[57] A. G. Sanfey, J. K. Rilling, J. A. Aronson, L. E. Nystrom, and J. D. Cohen (2002). An investigation of the neural dynamics of fairness and unfairness? *Society for Neuroscience Abstracts*. 78.11. 2002.

[56] N. Yeung, L. Nystrom, M. Kitazono, and J. Cohen (2002). An fMRI study of switching attention between tasks. *Neuroimage* 16.2, p. 692.

[55] M. S. Gilzenrat, B. D. Holmes, P. J. Holmes, J. Rajkowski, and J. D. Cohen (2001). A modified Fitzhugh-Nagumo system simulates locus coeruleus-mediated regulation of cognitive performance. *Society for Neuroscience Abstracts*. Vol. 26. 851.14. 2001.

[54] T. Li, J. K. Kroger, S. Kastner, and J. D. Cohen (2001). The neural substrate of haptic working memory. *Society for Neuroscience Abstracts*. Vol. 26. 782.1. 2001.

[53] T. Q. Li, L. Nystrom, and J. D. Cohen (2001). Ultra-fast trial-based fMRI using amplitude modulated stimuli. *Neuroimage*.

[52] T.-Q. Li, S. Skare, and J. D. Cohen (2001). White matter fiber tracking using the eigenvalue weighted average eigenvector. *Proceedings of the International Society for Magnetic Resonance in Medicine*. Vol. 9. 2001.

[51] W. M. Perlstein, S. K. Lageman, M. A. Cole, V. M. Jones, J. Demery, C. S. Carter, and J. D. Cohen (2001). Working memory and inhibition: prefrontal mediation and dysfunction in schizophrenia. *Society for Neuroscience Abstracts*. Vol. 26. 886.3. 2001.

[50] N. Yeung and J. D. Cohen (2001). The role of anterior cingulate in performance monitoring: response conflict and the error-related negativity. *Society for Neuroscience Abstracts*. Vol. 26. 849.6. 2001.

[49] G. Aston-Jones and J. D. Cohen (2000). Neuromodulation and Cognitive Function: Recent Studies of Noradrenergic Neurons in Behaving Monkeys. *Abstracts of Panels and Posters, 39th Annual Meeting of the American College of Neuropsychopharmacology*. 2000.

[48] R. Y. Cho, L. E. Nystrom, P. Holmes, E. Brown, B. J. Casey, and J. D. Cohen (2000). A connectionist model of conflict and control in a forced-choice task. *Society for Neuroscience Abstracts*. Vol. 26. 2000, p. 1317.

[47] J. D. Greene, R. B. Sommerville, L. E. Nystrom, J. M. Darley, and J. D. Cohen (2000). An fMRI investigation of emotion in moral judgment. *Society for Neuroscience Abstracts*. Vol. 26. 2000, p. 2022.

[46] J. K. Kroger, L. E. Nystrom, T.-Q. Li, B. D. Holmes, and J. D. Cohen (2000). Differential cortical responses to memory load and representational structure. *Journal of Cognitive Neuroscience*, S145.

[45] J. Rajkowski, W. Lu, Y. Zhu, J. D. Cohen, and G. Aston-Jones (2000). Prominent projections from the anterior cingulate cortex to the locus coeruleus in rhesus monkey. *Society for Neuroscience Abstracts*. Vol. 26. 2000, p. 2230.

[44] D. M. Barch, C. S. Carter, F. W. Sabb, A. MacDonald III, D. C. Noll, T. S. Braver, and J. D. Cohen (1999). Prefrontal cortex and context processing in medication naïve first episode patients with schizophrenia. *Society for Neuroscience Abstracts*. Vol. 25. 1999, p. 1289.

[43] C. S. Carter, A. MacDonald III, M. M. Botvinick, V. A. Stenger, and J. D. Cohen (1999). Anterior cingulate cortex and executive control: what is it really doing? *Society for Neuroscience Abstracts*. Vol. 25. 1999, p. 2168.

[42] M. E. Wheeler, F. W. Sabb, M. S. A. Graziano, and J. D. Cohen (1999). Imagined movement of real objects in space near the body differentially activates frontal and parietal cortex. *Society for Neuroscience Abstracts*. Vol. 25. 1999, p. 98.

[41] G. S. Berns and J. D. Cohen (1998). Dissociating brain regions for novel learning from response competition using implicit sequence learning. *Society for Neuroscience Abstracts*. 1998.

[40] J. M. Chein, D. C. Noll, J. D. Cohen, K. Fissell, L. E. Nystrom, F. Saab, and J. A. Fiez (1998). An fMRI study of verbal working memory: Effects of length, lexicality, and phonological similarity. *Society for Neuroscience Abstracts*. 1998.

[39] J. L. McClelland, J. A. Pollock, J. D. Cohen, R. N. Fisher, and T. Abdulaziz (1998). Tracking the human brain: An interactive multimedia presentation. *Society for Neuroscience Abstracts*. 1998.

[38] D. M. Barch, T. S. Braver, L. Nystrom, D. C. Noll, and J. D. Cohen (1997). Activation of prefrontal cortex by the representation and maintenance of context information. *Schizophrenia Research* 24.1,2, p. 163.

[37] D. M. Barch, C. S. Carter, T. S. Braver, and J. D. Cohen (1997). The effects of d-amphetamine on working memory and language deficits in schizophrenia. *Schizophrenia Research* 24.1,2, p. 129.

[36] D. M. Barch, C. S. Carter, J. D. Cohen, and F. W. Sabb (1997). The effects of D-amphetamine on language function in schizophrenia. *Society for Neuroscience Abstracts*. Vol. 23. 1997, p. 1952.

[35] T. S. Braver and J. D. Cohen (1997). Driving subjects to distraction: Cognitive control, prefrontal cortex, and schizophrenia. *Abstracts of the Psychonomic Society*. Vol. 2. 1997, p. 23.

[34] T. S. Braver, J. D. Cohen, and J. L. McClelland (1997). An integrated computational model of dopamine function in reinforcement learning and working memory. *Society for Neuroscience Abstracts*. Vol. 23. 1997, p. 775.

[33] C. S. Carter, D. M. Barch, and J. D. Cohen (1997). Disturbed language processing: disorganization, and attentional impairment in schizophrenia. *Schizophrenia Research* 24.1,2, p. 130.

[32] C. S. Carter, D. M. Barch, J. D. Cohen, and T. S. Braver (1997). CNS catecholamines and cognitive dysfunction in schizophrenia. *Schizophrenia Research* 24.1,2, p. 211.

[31] C. S. Carter, M. Mintun, and J. D. Cohen (1997). Abnormal cortical physiology associated with selective attention dysfunction in schizophrenia. *Schizophrenia Research* 24.1,2, p. 164.

[30] B. J. Casey, J. D. Cohen, S. W. King, P. L. Franzen, L. E. Nystrom, R. D. Badgaiyan, A. B. Schubert, and D. C. Noll (1997). A developmental functional MRI study of cortical activation during a spatial working memory task. *NeuroImage* 5.4, S69.

[29] B. J. Casey, S. D. Forman, P. L. Franzen, R. D. Badgaiyan, S. W. King, T. S. Braver, J. D. Cohen, and D. C. Noll (1997). Ventral and dorsolateral prefrontal activation as a function of target probability. *NeuroImage* 5.4, S92.

[28] J. D. Cohen, L. E. Nystrom, F. Sabb, and D. C. Noll (1997). Tracking the dynamics of fMRI activation under manipulations of duration and intensity of working memory processes. *Society for Neuroscience Abstracts*. Vol. 23. 1997, p. 1678.

[27] R. Condray, S. R. Steinhauer, J. D. Cohen, D. P. van Kammen, A. Kasparek, and M. E. Kelley (1997). Effects of haloperidol on the event-related potential in schizophrenics. *Schizophrenia Research* 24.1,2, p. 232.

[26] V. Silakov, J. Rajkowski, S. Ivanova, T. Watanabe, J. D. Cohen, M. Usher, and G. Aston-Jones (1997). Correlated impulse activities of locus coeruleus (LC) neurons in monkey: Dependency on level of task performance. *Society for Neuroscience Abstracts*. Vol. 23. 1997, p. 1587.

[25] M. Usher and J. D. Cohen (1997). Interference-based capacity limitations in active memory. *Abstracts of the Psychonomic Society*. Vol. 2. 1997, p. 11.

[24] G. Aston-Jones, P. Kubiak, J. Rajkowski, S. Ivanova, and J. Cohen (1996). Enhancement of locus coeruleus (LC) neuronal responses to visual targets by preceding auditory distractors. *Society for Neuroscience Abstracts*. Vol. 22. 1996.

[23] J. D. Cohen and M. Usher (1996). A neural network model of Stroop interference and facilitation effects in schizophrenia. *Biological Psychiatry* 39, p. 568.

[22] D. D. Barch, J. D. Cohen, D. Servan-Schreiber, and R. Ganguli (1995). The stability and interrelationships of CPT-AX and Stroop performance in schizophrenia. *Schizophrenia Research* 15.1-2, p. 108.

[21] T. S. Braver, J. D. Cohen, J. Jonides, E. E. Smith, E. Awh, E. Schumacher, E. Lauber, and D. C. Noll (1995). A parametric study of prefrontal cortex involvement in human working memory using functional MRI. *Society for Neuroscience Abstracts*. Vol. 21. 1995, p. 274.

[20] T. S. Braver, J. D. Cohen, and D. Servan-Schreiber (1995). Neural network simulations of schizophrenic performance in a variant of the CPT-AX: A predicted double dissociation. *Schizophrenia Research* 15.1-2, p. 110.

[19] C. S. Carter, D. Barch, W. Perlstein, J. Baird, R. Baker, J. D. Cohen, and N. Schooler (1995). A cognitive neuropsychological study of schizophrenia symptoms: Correlates of Stroop and semantic priming performance. *Schizophrenia Research* 15.1-2, p. 111.

[18] J. D. Cohen, R. Ganguli, C. Carter, J. Brar, T. Nichols, M. DeLeo, and M. Mintun (1995). Hypofrontality and working memory dysfunction in schizophrenia. *Biological Psychiatry* 37, p. 633.

[17] J. D. Cohen and R. C. O'Reilly (1995). A computational model of prefrontal cortical and hippocampal function, and their interaction in behavioral tasks. *Schizophrenia Research* 15.1-2, p. 112.

[16] J. D. Cohen, M. Usher, D. Servan-Schreiber, and G. Aston-Jones (1995). A computational model of the effects of locus coeruleus neuromodulation on attention. *Biological Psychiatry* 37, p. 618.

[15] S. D. Forman and J. D. Cohen (1995). Modeling saccadic eye movements in schizophrenia: Insights into memory mechanisms. *Schizophrenia Research* 15.1-2, p. 175.

[14] S. D. Forman, J. D. Cohen, T. S. Braver, and R. J. Orr (1995). Validation of functional magnetic resonance imaging: Application to schizophrenia research. *Schizophrenia Research* 15.1-2, p. 82.

[13] C. S. Carter, J. D. Cohen, and M. Mintun (1994). Interference, facilitation and strategy development in Stroop task performance: An $^{15}OH_2O$ PET study. *Abstracts of Panels and Posters, 33rd Annual Meeting of the American College of Neuropsychopharmacology*. 1994.

[12] S. D. Forman, J. D. Cohen, M. A. Mintun, and D. C. Noll (1994). Improved assessment of significant change in functional magnetic resonance imaging (fMRI): Use of the contiguity threshold. *Annual Meeting of the Society for Magnetic Resonance*. 1994.

[11] S. D. Forman, J. D. Cohen, D. C. Noll, M. A. Mintun, and B. J. Casey (1994). Within subject comparison of PET & fMRI to visualize activation of prefrontal cortex (PFC). *Society for Neuroscience Abstracts*. Vol. 20. 1994.

[10] T. Huston and J. D. Cohen (1994). Stimulus-induced shifts of task attention. *Abstracts of the Psychonomic Society, 35th Annual Meeting*. 1994.

[9] D. Servan-Schreiber, D. Noll, J. D. Cohen, M. Beuger, T. Olga, D. Swanson, and J. J. Mann (1994). fMRI studies of limbic activation. *Abstracts of Panels and Posters, 33rd Annual Meeting of the American College of Neuropsychopharmacology*. 1994.

[8] B. J. Casey, J. D. Cohen, D. C. Noll, S. D. Forman, and J. L. Rapoport (1993). Activation of the anterior cingulate during the Stroop conflict paradigm using functional MRI. *Society for Neuroscience Abstracts*. Vol. 19. 1993, p. 1285.

[7] J. D. Cohen, S. D. Forman, B. J. Casey, and D. C. Noll (1993). Spiral-scan imaging of dorsolateral prefrontal cortex during a working memory task. *Twelfth Annual Meeting of the Society of Magnetic Resonance in Medicine*. 1993.

[6] D. C. Noll, J. D. Cohen, S. D. Forman, W. Schneider, and C. H. Meyers (1993). Spiral k-space MRI of brain function. *Twelfth Annual Meeting of the Society of Magnetic Resonance in Medicine*. 1993.

[5] D. C. Noll, W. Schneider, and J. D. Cohen (1993). Artifacts in functional MRI using conventional scanning (1993). *Twelfth Annual Meeting of the Society of Magnetic Resonance in Medicine*. 1993.

[4] J. D. Cohen and D. Servan-Schreiber (1991). Computer simulation models of the relationship between disturbances of dopamine, prefrontal cortex and cognitive function in schizophrenia. *Abstracts of Panels and Posters, 30th Annual Meeting of the American College of Neuropsychopharmacology*. 1991.

[3] R. Halliday, E. Callaway, H. Naylor, K. Herzig, L. Yano, J. D. Cohen, and D. Servan-Schreiber (1991). Two empirical findings linked to neural network modeling: Amphetamine speeds reaction time (RT) but not P3. Yohimbine speeds P3 but not RT. *Abstracts of Panels and Posters, 30th Annual Meeting of the American College of Neuropsychopharmacology*. 1991.

[2] D. C. Noll, C. H. Meyer, J. D. Cohen, and W. Schneider (1991). Spiral-scan imaging of cortical activation. *11th Annual Meeting of the Society of Magnetic Resonance Imaging*. 1991.

[1] J. D. Cohen (1989). A network model of schizophrenic language deficits. *Book of Abstracts, 142nd Annual Meeting of the American Psychiatric Association*. 1989.

5. Preprints: Manuscripts Under Review / In Preparation

[19] N. Budny, K. Ghods, D. Campbell, R. Marjieh, A. Joshi, S. Kumar, J. D. Cohen, T. W. Webb, and T. L. Griffiths (under review). Visual serial processing deficits explain divergences in human and VLM reasoning. under review. URL: https://arxiv.org/abs/2509.25142.

[18] H. C. Conklin, T. Hosking, T. Yi-Chern, J. D. Cohen, S.-J. Leslie, T. L. Griffiths, M. Bartolo, and S. Goldfarb-Tarrant (under review). Learning is forgetting: LLMS as lossy compression. under review.

[17] C. Dawes, S. Segert, K. Krishnamurthy, and J. D. Cohen (under review). A group theoretic analysis of the symmetries underlying base addition and their learnability by neural networks. under review.

[16] S. M. Frankland, J. S. Phillips, T. Redford, and J. D. Cohen (under review). Excess capacity coding (not learning) & the curse of generalization: Commentary on Dubova and Sloman (2026).

[15] V. Geadah, J. Arbelaiz, H. Ritz, N. D. Daw, J. D. Cohen, and J. W. Pillow (under review). System identification and inverse optimal control for partially observed discrete-time stochastic linear quadratic regulators. under review.

[14] J. Huang, J. D. Cohen, and J. Busemeyer (under review). A quantum model of arousal and cognitive control in decision making. under review.

[13] B. J. Jongkees, M. T. Todd, K. Lloyd, P. Dayan, and J. D. Cohen (in prep). When it pays to be quick: dissociating control over task preparation and speed-accuracy trade-off in task switching. URL: https://psyarxiv.com/quhns.

[12] S. Musslick, A. M. Saxe, A. N. Hoskin, Y. Sagiv, D. Reichman, G. Petri, and J. D. Cohen (under review). On the rational boundedness of cognitive control: Shared versus separated representations. under review. URL: https://psyarxiv.com/jkhdf.

[11] A. J. Nam, T. L. Griffiths, J. D. Cohen, and S.-J. Leslie (under review). Understanding task representations in neural networks via Bayesian ablation. under review. URL: https://openreview.net/forum?id=SA8lLiWl8V.

[10] M. Nurisso, J. Fernando, R. Deshpandes, A. Perotti, R. Marjieh, S. M. Frankland, R. L. Lewis, T. W. Webb, D. Campbell, F. Vaccario, J. D. Cohen, and G. Petri (under review). Bound by semanticity: Universal laws governing the generalization-identification tradeoff. under review. URL: https://arxiv.org/abs/2506.14797.

[9] R. P. Pothukuchi, L. Lufkin, Y. J. Shen, A. Simon, B. E. Trevisan, M. Tu, M. Yang, B. Foxman, V. S. Pothukuchi, T. H. Kyaw, G. Epping, B. Jongkees, J. Busemeyer, J. D. Cohen, and A. Bhattacharjee (under review). Quantum cognitive modeling: New applications and systems research directions. under review. URL: https://arxiv.org/abs/2309.00597.

[8] R. Pyle, S. Musslick, J. D. Cohen, and A. B. Patel (under review). A quantitative approach to predicting representational learning and performance in neural networks. under review. URL: https://arxiv.org/abs/2307.07575.

[7] A. Qin, J. D. Cohen, and G. Reddy (in prep). Cognitive maps in large language models. URL: https://arxiv.org/abs/2610.25596.

[6] S. Ravi, S. Musslick, M. Hamin, T. L. Willke, and J. D. Cohen (under review). Navigating the trade-off between multi-task learning and learning to multitask in deep neural networks. under review. URL: https://arxiv.org/abs/2007.10527.

[5] H. Ritz, A. Jha, J. W. Pillow, N. D. Daw, and J. D. Cohen (under review). Active reconfiguration of neural task states. under review.

[4] A. Shenhav, S. Musslick, M. M. Botvinick, and J. D. Cohen (under review). Misdirected vigor: Differentiating the control of value from the value of control. under review.

[3] M. Shvartsman, N. Sundaram, V. Srivasta, and J. D. Cohen (under review). A normative theory of decision making from multiple stimuli.

[2] R. C. Wilson, S. Wang, H. Sadhegiyeh, and J. D. Cohen (under review). Deep exploration as a unifying account of explore-exploit behavior. URL: https://psyarxiv.com/uj85c.

[1] A. Chatterjee, N. K. Jha, J. D. Cohen, T. L. Griffiths, H. Lu, D. Marculescu, A. Rasul, and K. K. Parhi (2025). Energy-Efficient Domain-Specific Artificial Intelligence Models and Agents: Pathways and Paradigms. *arXiv preprint arXiv:2510.22052*.

PROFESSSIONAL ACTIVITIES

TEACHING:

1. Courses

1989-96 Introduction to Cognitive Psychology (undergraduate survey course)
Department of Psychology, Carnegie Mellon University

1989-96 Cognitive Neuroscience section of Cognitive Core (graduate survey course)
Department of Psychology, Carnegie Mellon University

1990-93 Co-coordinator, Fellowship Training Program in Schizophrenia Research
Western Psychiatric Institute and Clinic, University of Pittsburgh

1992-93 Research Methods in Cognitive Neuroscience (advanced undergraduate seminar)
Department of Psychology, Carnegie Mellon University

1992-93 Functional Neural Circuits (graduate and advanced undergraduate seminar)
Department of Psychology, Carnegie Mellon University

1994-95 Neural and Psychological Mechanisms of Working Memory (graduate and advanced undergraduate seminar)
Department of Psychology, Carnegie Mellon University

1996-97 Advanced Topics in Cognitive Neuroscience (graduate and advanced undergraduate seminar)
Department of Psychology, Carnegie Mellon University

1996-97 Biological and Psychological Mechanisms of Attention (graduate and advanced undergraduate seminar)
Department of Psychology, Carnegie Mellon University; co- taught with Gary Aston-Jones.

1999-00 Neural Bases of Cognitive Control (undergraduate course)
Department of Psychology, Princeton University

1999-01 Topics in Molecular and Cognitive Neuroscience (graduate seminar)
Departments of Psychology and Molecular Biology, Princeton University

1999-01 Introduction to Neural Networks (undergraduate course)
Department of Psychology, Princeton University

2001-02 Advanced Topics in Neural Network Models of Psychological Function (advanced undergraduate / graduate seminar)
Department of Psychology, Princeton University

2002-03 Statistical Methods in Psychological Research (advanced undergraduate / graduate course)
Department of Psychology, Princeton University

2004-07 Graduate Proseminar in Cognitive Psychology
Department of Psychology, Princeton University

2009-16	Core Course for Ph.D. Program in Neuroscience Princeton Neuroscience Institute, Princeton University
2017	Introduction to Cognitive Psychology (undergraduate survey course, with laboratory component) Department of Psychology, Princeton University
2018-20	Computational Models of Psychological Function (undergraduate course, with laboratory component) Princeton Neuroscience Institute and Department of Psychology, Princeton University
2022	The Computational Basis of Natural Intelligence in the Human Brain (undergraduate seminar) Princeton Neuroscience Institute, Princeton University

2. Tutorials and Workshops

May, 1990-93	Cohen JD, Servan-Schreiber D. Course co-directors, A primer on neural modeling in psychiatry. 144-7th Annual Meetings of the American Psychiatric Society, New York.
July, 1991	Invited faculty member. James S. McDonnell Summer Institute in Cognitive Neuroscience, Dartmouth College, Hanover.
October, 1993	Applications of Functional MRI to Studies of Human Memory. Invited tutorial, Memory Disorders Research Society, Boston.
November, 1993	Functional neuroimaging. Invited tutorial, Neural Information Processing Society, Boulder.
August, 1996	Neuroimaging and Behavior. Invited workshop, XXVI International Congress of Psychology, Montreal.
January, 1997	The Role of Neuromodulation in Cognition: Physiological and Computational Approaches. Panel organizer, 30th Winter Conference on Brain Research, Breckenridge, Colorado.
July, 1997	Invited faculty member. James S. McDonnell Summer Institute in Cognitive Neuroscience, Dartmouth College, Hanover.
September, 2000	International Workshop on Neural Bases of Executive Functions and Performance Monitoring, Jena, Germany.
July, 2001	Invited faculty member. James S. McDonnell Summer Institute in Cognitive Neuroscience, Dartmouth College, Hanover.

3. Trainees

Graduate advisees:

Therese Huston, Ph.D. (1990-1995)

CMU Department of Psychology

Behavioral and computational modeling studies of selective attention

Director, Center for Excellence in Teaching & Learning, University of Seattle

Todd Braver, Ph.D. (1992- 97)

CMU Department of Psychology

Computational and neuroimaging studies of prefrontal cortex and cognitive control

Professor of Psychology, Washington University, St. Louis

Matthew Botvinick, M.D., Ph.D. (1995-2001)

CMU Department of Psychology

Modeling and fMRI studies of the role of anterior cingulate cortex in conflict monitoring and control

Professor of Psychology and Neuroscience, Princeton University

Mark Gilzenrat, Ph.D. (1996-2006)

CMU Department of Psychology (1996-1998)

Princeton Department of Psychology (1998-2006)

Computational models and pupillometric studies of neuromodulatory influences on selective attention

Software architect, Navaraga Corporation

Raymond Cho, M.D. (1999-2003)

Department of Psychology, Princeton University

Assistant Professor of Psychiatry, University of Pittsburgh

Eric Shea-Brown, Ph.D. (1999-2004)

Program in Applied and Computational Mathematics, Princeton University

Co-advisor with Philip Holmes

Neural oscillators and integrators in the dynamics of decision tasks

Associate Professor of Applied Mathematics, University of Washington, Seattle

Sean Polyn (2000-2005)

Department of Psychology, Princeton University

Computational modeling of context updating, reinforcement learning and dopamine function

Associate Professor of Psychology and Psychiatry, Vanderbilt University

Aaron Schurger (2001-2008)

Department of Psychology, Princeton University

Electrophysiological and fMRI studies of perceptual awareness

Associate Professor, Inserm-CEA

Agatha Lenartowicz (2002-2008)

Department of Psychology, Princeton University

Behavioral, electrophysiological and fMRI studies of task switching

Postdoctoral Fellow, UCLA

Kimberly D'Ardenne McClure (2005-2008)

Department of Chemistry, Princeton University

fMRI studies of brainstem neuromodulatory nuclei

Postdoctoral Fellow, Montague Lab, Virginia Tech

Susan Robison (2005-2009; co-advised with Ken Norman)

Department of Chemistry, Princeton University

Behavioral and fMRI studies of cognitive control and episodic memory

Emily Chakwin (2006-2008)

Department of Psychology, Princeton University

Behavioral and fMRI studies of moral reasoning

Michael Todd (2006-2012)

Department of Psychology, Princeton University

Computational modeling studies of cognitive control

Data Scientist, Netflix

Adam Moore (2006-2011; co-advised with Andy Conway)

Department of Psychology, Princeton University

Behavioral and fMRI studies of moral reasoning

John White (2008-2013)

Department of Psychology, Princeton University

Behavioral and fMRI studies of economic decision making

Data Scientist, Netflix

Sarah Getz (2008-2013; co-advised with Andy Conway)

Department of Psychology, Princeton University

Behavioral and fMRI studies of economic decision making

Andra Geana (2010-2016)

Department of Psychology, Princeton University

Behavioral and fMRI studies of exploration and exploitation in decision making

Postdoctoral Fellow, Brown University

Jane Keung (2011-2016)

Princeton Neuroscience Institute, Princeton University

Behavioral and fMRI studies of prefrontal cortex and cognitive control

Postdoctoral Fellow, University of Arizona

Yida Wang (2011-2016)

Department of Computer Science, Princeton University

Co-advisor with Kai Li and Nick Turk-Browne

Full-correlation matrix analysis of fMRI data

Applied Scientist, Amazon

Olga Lositsky (2012-2017)

Princeton Neuroscience Institute, Princeton University

Behavioral and fMRI studies of decision making

Postdoctoral Fellow, Brown University

Gary Kane (2012-2018)

Department of Psychology, Princeton University

Behavioral and neurophysiological studies of foraging behavior and LC function in rodents

Postdoctoral Fellow, Harvard University

Sachin Ravi (2014-2019)

Department of Computer Science, Princeton University (Co-advisor with Kai Li)

Meta-learning and controlled vs. automatic processing

Machine Learning Research Engineer, Apple

Laura Bustamante (2014-2022)

Princeton Neuroscience Institute, Princeton University

Behavioral and fMRI studies of the cost of cognitive control

Postdoctoral Fellow, Washington University

Sebastian Musslick (2014-2022)

Princeton Neuroscience Institute, Princeton University

Behavioral and fMRI studies of cognitive control

Assistant Professor, Osnabrück University

Abigail Novick (2014-2022)

Department of Psychology, Princeton University

Behavioral and fMRI studies of representational sharing and multitasking

Lena Rosendahl (2016-2022)

Department of Mechanical and Aerospace Engineering

Co-advisor with Naomi Leonard

Quantum probabilistic models of cognitive control

Data Analyst, Mathematica

Mayank Agrawal (2017-2023)

Department of Psychology, Princeton University

Computational and Behavioral studies of learning and cognitive control

Zach Dulberg (2019-2025)

Princeton Neuroscience Institute, Princeton University

Learning mechanisms for acquiring representations that support generalization

Sreejan Kumar (2019-2025)

Princeton Neuroscience Institute, Princeton University

Neural mechanisms underlying learning, representation and reasoning

Simon Segert (2019-2024)

Princeton Neuroscience Institute, Princeton University

Neurally-plausible mechanisms of relational reasoning

Tyler Giallanza (2020-2025)

Department of Psychology, Princeton University

Neural mechanisms underlying learning, representation and processing of semantics

Samyak Gupta (2020-present)

Department of Computer Science, Princeton University

Co-advisor with Kai Li

Compilation and efficient computational of heterogeneous cognitive models

Shanka Mondal (2020-present)

Electrical and Computer Engineering, Princeton University

Neural mechanisms underlying learning, representation and generalization

Declan Campbell (2022-present)

Princeton Neuroscience Institute, Princeton University

Computational models of human intelligence

Yukang Yang (2023-present)

Electrical and Computer Engineering, Princeton University

Computational models of human intelligence

Udith Haputhanthrige (2024-present)

Electrical and Computer Engineering, Princeton University

Computational models of human visual processing

Younes Strittmatter (2024-present)

Department of Psychology, Princeton University

Automated methods for scientific discovery

Ph.D. Thesis Co-Advisor:

Cliona Golden (2004, Ingrid Daubechies), PACM, Princeton University

Adi Livnat (2005, Simon Levin), Ecology and Evolutionary Biology, Princeton University

Ilya Fischoff (2006, Daniel Rubenstein), Ecology and Evolutionary Biology, Princeton University

Juan Gao (2007, Phil Holmes), Program in Applied and Computational Mathematics, Princeton University

Yuan (Sophie) Liu (2007, Phil Holmes), Physics, Princeton University

Caitlin Newberry (2007, Wolf Richter), Chemistry, Princeton University

Phil Eckoff (2008, Phil Holmes), Program in Applied and Computational Mathematics, Princeton University

Andrea Nedic (2011, Phil Holmes), Electrical Engineering, Princeton University

Samuel Feng (Phil Holmes), Program in Applied and Computational Mathematics, Princeton University

Stephanie Goldfarb (2013, Naomi Leonard), Program in Applied and Computational Mathematics, Princeton University

Eran Eldar (2014, Yale Niv), Princeton Neuroscience Institute, Princeton University

Paul Reverdy (2014, Naomi Leonard), Mechanical and Aerospace Engineering, Princeton University

Postdoctoral trainees:

Steve Forman, M.D., Ph.D. (1992-1994)

University of Pittsburgh Department of Psychiatry

fMRI studies of prefrontal function

Associate Professor of Psychiatry, University of Pittsburgh

Medical Director of the Center for Treatment of Addictive Disorders, Pittsburgh VA

Marius Usher, Ph.D. (1993-1995)

CMU Department of Psychology

Computational models of catecholaminergic neuromodulation and selective attention

Professor of Psychology and Neuroscience, Tel Aviv University

Deanna Barch, Ph.D. (1993-1995)

University of Pittsburgh Department of Psychiatry

Professor of Psychology and Radiology, Washington University, St. Louis

William Perlstein, Ph.D. (1993-1996)

University of Pittsburgh Department of Psychiatry

Electrophysiological and fMRI studies of working memory in schizophrenia

Associate Professor of Clinical and Health Psychology and Psychiatry, University of Florida, Gainsville

Gregory Berns, M.D., Ph.D. (1995-1998)

University of Pittsburgh Department of Psychiatry

Functional neuroimaging studies of novelty detection

Professor of Economics, Emory University

Randy Gobbel, Ph.D. (1997-1998)

Carnegie Mellon University Department of Psychology

Computational modeling studies of basal ganglia function in control of sequential action

Computer Scientist, Artificial Intelligence Center, SRI International

James Kroger (1998-2001)

Princeton University Department of Psychology

FMRI studies of prefrontal cortex organization

Professor of Psychology, New Mexico State University

Nicholas Yeung, Ph.D. (1999-2004)

Princeton University Department of Psychology

Modeling, ERP and fMRI studies of conflict monitoring and cognitive control

University Lecturer in Experimental Psychology, University of Oxford

Gesine Dreisbach, Ph.D. (2000-2001)

Princeton University Department of Psychology

fMRI studies of task switching

Professor of Psychology, University of Regensburg

Clay Holroyd, Ph.D. (2001-2004)

Princeton University Department of Psychology

Neural network modeling, ERP, and fMRI studies of performance monitoring and reinforcement learning

Professor of Psychology, University of Victoria

James Rilling, Ph.D. (2001-2003)

Center for the Study of Brain, Mind & Behavior, Princeton University

Neural mechanisms of economic decision making; neural mechanisms in placebo responding

Associate Professor of Anthropology and Psychiatry and Behavioral Sciences, Emory University

Alan Sanfey, Ph.D. (2001-2003)

Center for the Study of Brain, Mind & Behavior, Princeton University

Neural mechanisms of economic decision making; neural mechanisms in placebo responding

Associate Professor of Psychology, University of Arizona

Principal Investigator, Donders Institute for Brain, Cognition and Behavior, Radboud University

Rafal Bogacz, Ph.D. (2002-2004)

Princeton University Department of Psychology

Neural network modeling and ERP studies of task switching and performance monitoring

Associate Professor of Clinical Neuroscience, University of Oxford

Sander Nieuwenhuis, Ph.D. (2002-2003)

Princeton University Department of Psychology

ERP studies and neural network modeling of performance monitoring, task switching and the attentional blink

Assistant Professor, Cognitive Psychology Unit, Leiden University

Joshua Greene, Ph.D. (2001-2006)

Princeton University Department of Psychology

Neural bases of moral reasoning

Professor of Psychology, Harvard University

Samuel McClure, Ph.D. (2003-2007)

Princeton University Department of Psychology

Neural network modeling and neuroimaging studies of reinforcement learning and decision making

Assistant Professor of Psychology, Stanford University

Jean-Baptiste Pochon, Ph.D. (2003-2005)

Princeton University Department of Psychology

Neuroimaging studies of decision making, conflict monitoring and cognitive control

Postdoctoral Fellow, L'Hôpital de la Salpêtrière in Paris

Patrick Simen, PhD. (2003-2007)

Princeton University Program in Applied & Computational Mathematics

Computational modeling, mathematical analysis, behavioral and neuroimaging studies of decision making and cognitive control

Assistant Professor, Oberlin College

Jason Chein, Ph.D. (2004-2005)

Princeton University Department of Psychology

Neuroimaging studies of prefrontal cortex organization and function

Assistant Professor of Psychology, Temple University

Brent Field, Ph.D. (2004-2015)

Center for Study of Brain, Mind and Behavior

Center for Health and Well-Being, Woodrow Wilson School of Public Policy

Behavioral and neuroimaging studies of attention and emotional regulation among meditation practitioners

Angela Yu, Ph.D. (2004-2008)

Princeton University Department of Psychology

Computational modeling and mathematical analysis studies of decision making and cognitive control

Associate Professor of Cognitive Science, University of California, San Diego

Damon Tomlin, Ph.D. (2006-2013)

Princeton University Department of Psychology and Princeton Neuroscience Institute

Neuroimaging studies of economic and social decision making and cognitive control

KongFatt Wong-Lin, Ph.D. (2006-2009)

Princeton University Department of Mechanical and Aerospace Engineering

Computational modeling and mathematical analysis studies of decision making and cognitive control

Lecturer, Ulster University

Yael Niv, Ph.D. (2007-2008)

Princeton University Department of Psychology

Neuroimaging and computational modeling studies of decision making and cognitive control

Associate Professor of Psychology and Neuroscience, Princeton University

Benjamin Eppinger, Ph.D. (2007-2010)

Princeton University Department of Psychology Center for Health and Well-Being, Woodrow Wilson School of Public Policy

Neuroimaging studies of age-related differences in economic decision making and cognitive control

Researcher, MPI for Human Development, Berlin

Marieke van Vugt, Ph.D. (2008-2010)

Princeton University Department of Psychology

Neuroimaging and computational modeling studies of decision making and cognitive control

Assistant Professor, University of Groningen

Fuat Balci, Ph.D. (2008-2010)

Princeton University Department of Psychology

Theoretical and behavioral studies of interval timing and decision making

Assistant Professor, Department of Psychology, Koc University, Istanbul

Robert Wilson, Ph.D. (2009-2014)

Princeton University Department of Psychology and Princeton Neuroscience Institute

Theoretical, behavioral and neuroimaging studies of cognitive control & locus coeruleus function

Assistant Professor, University of Arizona

Michael Schwemmer, Ph.D. (2010-2012)

Princeton Neuroscience Institute

Theoretical analyses of capacity constraints on cognitive control

Postdoctoral Fellow, Mathematical Biosciences Institute, Ohio State University

Jarrod Lewis-Peacock, Ph.D. in Psychology, University of Wisconsin-Madison

Princeton Neuroscience Institute (2011-2013; co-advised with Ken Norman)

Neuroimaging studies of cognitive control and prospective memory

Assistant Professor, University of Texas, Austin

Elliot Ludvig, Ph.D., Psychological and Brain Sciences, Duke University

Princeton Neuroscience Institute (2011-2013)

Theoretical model and behavioral studies of learning, memory and cognitive control

Professor, University of Warwick

Amitai Shenhav, Ph.D. in Psychology, Harvard University

CV Starr Fellow, PNI (2012-2016; co-advised with Matthew Botvinick)

Theoretical and neuroimaging studies of the costs of cognitive control

Associate Professor, Brown University

Aaron Bornstein, Ph.D. in Neuroscience, NYU

Princeton Neuroscience Institute (2013-2019; co-advised with Ken Norman)

Neuroimaging studies of episodic memory and decision making

Assistant Professor, University of California, Irvine

Ida Momennajad, Ph.D.
Princeton Neuroscience Institute (2013-2018; co-advised with Ken Norman & Nathaniel Daw)
Neuroimaging and theoretical modeling studies of prospective memory

Michael Shvartsman, Ph.D. in Cognitive Science, University of Michigan
Princeton Neuroscience Institute (2014-2018)
Theoretical analysis of decision making; Bayesian hierarchical analysis of neuroimaging data
Occulus Research

Hasan Kayhan Ozcimder, Ph.D. in Mechanical Engineering, Boston University
Princeton Neuroscience Institute (2015-2017; co-advised with Naomi Leonard)
Mathematical modeling of capacity constraints in controlled (interactive parallel) processing
Mathworks

Michael Lesnick, Ph.D in Applied Mathematics, Stanford University
Princeton Neuroscience Institute (2016-2018)
Tools for topological data analysis (TDA) and their application to neuroscientific data analysis
Department of Mathematics, SUNY Albany

Biswadip Dey, Ph.D. in Mechanical Engineering, University of Maryland, College Park
Princeton Neuroscience Institute (2015-present; co-advised with Naomi Leonard)
Mathematical modeling of capacity constraints in controlled (interactive parallel) processing

Marius Cătălin Iordan, Ph.D. in Computer Science, Stanford University
Princeton Neuroscience Institute (2016-present; co-advised with Daniel Osherson)
Theoretical and neuroimaging studies of semantic representations and cognitive control

Simon Cullen, PhD. in Philosophy, Princeton University
Princeton Neuroscience Institute (2017-2018)
Theoretical and experimental studies of moral reasoning
Assistant Professor, Carnegie Mellon University

Greg Henselman, Ph.D. in Applied Mathematics, University of Pennsylvania
Princeton Neuroscience Institute (2017-present)
Tools for topological data analysis (TDA) and their application to neuroscientific data analysis

Steven Frankland, Ph.D. in Psychology, Harvard University
Princeton Neuroscience Institute (2017-2023)
Neural network modeling of abstract reasoning
Assistant Professor, Dartmouth University

Taylor Webb, Ph.D. in Psychology, Princeton University
Princeton Neuroscience Institute (2018-2019)
Neural network modeling of analogical reasoning

Bryant Jonkes. Ph.D. in Cognitive Psychology, Leiden University, the Netherlands

Princeton Neuroscience Institute (2019-2020)

Theoretical modeling and behavioral studies of the dynamics of cognitive control

Assistant Professor in Cognitive Psychology Unit, Leiden University, the Netherlands

Javier Masis, Ph.D. in Psychology, Harvard University

Princeton Neuroscience Institute (2020-present)

Neural network modeling of learning and control

Kamesh Krishnamurthy, Ph.D. in Computational Neuroscience, University of Pennsylvania

CV Starr Fellow, PNI and Center for the Physics of Biological Function Fellow (2018-present)

Neural network models of symmetry discovery

Harrison Ritz, Ph.D. in Psychology, Brown University

CV Starr Fellow, PNI (2022-present)

Theoretical and neuroimaging studies of the dynamics of cognitive control

Andrew Nam, Ph.D. in Psychology, Stanford University

Fellow in Natural and Artificial Minds, (2024-present)

Computational models of human intelligence

RESEARCH AND PROFESSIONAL ACTIVITIES

1. Grants

NIMH Physician Scientist Award: Context Disturbance in Schizophrenia: Models and Measures; PI, 1987-92, MH00673

NIMH P50 PI: Cortical Circuitry and Cognition in Schizophrenia (Edward Stricker, PI); Project 4 (1990-96), Project 7 (1997-02): The Role of Prefrontal Cortex in the Cognitive Dysfunctions of Schizophrenia; Project 3 (2003-07): Neuroendophenotypes and the expression of illness liability in schizophrenia; PI, Projects 4, 7 & 3, 1990-07, MH45156

NIMH FIRST Award; RO1: Mechanisms of Context Processing in Schizophrenia; PI, 1991-2012, MH47073

NIMH Program Project: Toward Models of Normal and Disordered Cognition (James L. McClelland, PI); Project 2 (1991-96): Neuromodulation and the Processing of Context in Schizophrenia; Project 4 (1997-02): Mechanisms of Cognitive Control; PI Projects 2 & 4, 1991-2002, MH47566

NIMH P50: Center for Functional Brain Imaging (Robert Moore & Mark Mintun, Co-PIs) Cognitive Studies Core; Co-Director, Cognitive Core, 1992-97, MH49815

McDonnell Foundation: Neural Bases of Rehearsal and Maintenance in Working Memory; PI, 1994-96, JSMF 94-32

NSF CRI: Computational and Statistical Methods for the Analysis of Neuroimaging Datasets; PI, 1995-96, IBN9418982

NIMH RO1: fMRI Studies of Prefrontal Cortex; PI, 1996-2009, MH52864

NIMH Program Project: Toward Models of Normal and Disordered Cognition (James L. McClelland, PI); Project 2 (1991-96): Neuromodulation and the Processing of Context in Schizophrenia; Project 4 (1997-02): Mechanisms of Cognitive Control; PI Projects 2 & 4, 1997-02, MH47566

NIDA/HBP RO1: Advanced Methods for Neuroimaging Data Analysis; PI, 1997-99, DA11469

NSF ESI: Tracking the Human Brain: An Interactive Planetarium Exposition (Bryan Rogers, PI); Co-Investigator, 1997-99, ESI9705491

NARSAD Independent Investigator Award: An fMRI Study of the Role of Anterior Cingulate in Working Memory Dysfunction in Schizophrenia; PI, 1997-99

NIMH RO1: Neurophysiological and Modeling Studies of Locus Coeruleus; Co-PI (Gary Aston-Jones, Co-PI), 1998-2001, MH33194

NSF MRI: Acquisition of Core Equipment for Princeton Cognitive and Behavioral Neuroscience Initiative (Marcia Johnson and Charles Gross, Co-PIs); Co-PI, 1998-2001, MRI/ OSTI9871186

NJCST: New Jersey Brain Imaging Consortium: Acquisition of high field MRI scanner PI, 1999

NIMH/HBP RO1: Usability and Interoperability of Neuroimaging Software; PI, 2000-03, MH62006

NIMH RO1: Pathophysiology of Cognitive Disability in Schizophrenia (Cameron Carter, PI); Co-Investigator, 2000-04, MH59883

NIMH P50: Conte Center for Neuroscience Research: Cognitive and Neural Mechanisms of Conflict and Control; PI, 2000-10, MH62196

Seaver Institute: Neural Economics: Understanding the brain mechanisms underlying cognitive-emotional interactions in decision making; PI, 2001-02

NIDA R21: Hyperscan: Simultaneous fMRI Across the Internet (Emory University; Greg Berns, PI) Co-Investigator, 2001-03, DA014883

MacArthur Foundation: Neural Bases of Placebo Effect and the Expectation of Pain; PI, 2001-03

NIMH P50: IBSC: Toward a Neurobiologically Constrained Framework for Modeling Human Cognition (James L. McClelland, PI); Project 4: Mechanisms of Cognitive Control; PI Project 4, 2002-07, MH64445

NIMH RO1: New Wavelet-Based and Source Separation Methods for fMRI (Ingrid Daubechies, PI); Co-Investigator, 2002-07, MH067204

NIMH T32: Training Program in Quantitative Neuroscience; PI, 2002-present, MH65214

NJCST: Center for Molecular and Biomolecular Imaging (Warren Warren, PI); Co-Investigator, 2002-09

DURIP: Computing Environment for Computational Modeling of Brain Functions; PI, 2003, ONR

NSF BCS: Social Cognitive Neuroscience of Category-based Responses (Susan Fiske, PI); Co-Investigator, 2004-05

NIDA RO1: Neural Mechanisms and Social Influence in Delay Discounting and Impulsive Behavior; PI, 2006-11, DA022564

NIDA T90: Training Program in Quantitative and Computational Neuroscience (David Tank, Co-PI); Co-PI (David Tank, Co-PI), 2006-11, DA022770

MURI: Dynamic Decision Making in Complex Task Environments: Principles and Neural Mechanisms (James L. McClelland, PI); Co-Investigator, 2006-11, AFOSR

MURI: Behavioral Dynamics in the Cooperative Control of Mixed Human/Robotic Teams (John Baillieul, PI); Co-Investigator, 2006-11, AFOSR

DURIP: A Second Generation Flexible Computing Environment for Computational Modeling of Brain Function and Neuroimaging Data Analysis; PI, 2008, AFOSR

NCRR: Expansion of a Computing Facility for fMRI and Neuroimaging Analysis; PI, 2008, RR023532

NSF MRI: Acquisition of High Performance Compute Cluster for Multivariate Realtime; PI, 2012, BCS1229597

John Templeton Foundation: Toward a Scientific Understanding of the Human Capacity for Cognitive Control; PI, 2012-2022

Intel Corporation: Advanced Methods for Realtime Analysis of Human Brain Imaging Data; PI, 2014-2019

Templeton World Charity: System-Level Modeling of Intelligent Behavior; PI, 2018-2020, Beyond Turing

NIH CTSA: New Jersey Alliance for Clinical and Translational Science Co-Investigator, 2019-2024, UL1 TR003017

NIH R21: PsyNeuLink: A Block Modeling Environment for Cognitive Neuroscience; PI, 2019-2021, MH117548

NSF Convergence Accellerator — Track D: A Standardized Model Description Format for Accelerating Convergence in Neuroscience, Cognitive Science, Machine Learning and Beyond; PI, 2020-2021

DURIP: A Balanced, shared Computational Resource for Multidisciplinary Neuroscience Research; PI, 2020, AWD1006863

Vannevar Bush Faculty Fellowship: Toward a Brain-Inspired Model of the Flexibility and Autonomy of Human Behavior; PI, 2021-2026, N00014-22-1-2002

2. Invited Lectureships

American Association for the Advancement of Science (2002)

American Association of Directors of Psychiatry Residency Training (AADPRT), Annual Meeting, Schein Lecture (2012)

American College of Neuropsychopharmacology, Panels (1994, 1995, 1997, 1998, 1999, 2005)

American Economic Association, Symposia (2003, 2005, 2006)

American Psychological Association, Distinguished Scientific Contribution Award Lecture (2010)

American Psychological Society (1994, 1998)

ARVO (2000)

Association for Research in Nervous and Mental Disease, Annual Conference Special Lecture (2006)

Association for Psychological Science, William James Award Public Address (2018)

Attention and Performance XV, XVIII (1992, 1998)

AutoRA Workshop and Tutorial, Princeton University (2025)

Baylor College of Medicine, Neuroscience Colloquium (1999); Keynote speaker, Annual Neuroscience Retreat and Rush and Helen Record Forum (2008)

Beckman Institute for Advanced Science and Technology, University of Illinois, Smith, Hinchman & Grills Distinguished Lecture (2003)

Behavioral Neurology Society, Keynote Address (1998)

Biological Psychiatry Society, Presidential Symposium (2002, 2008)

Boston University, Department of Cognitive and Neural Systems Colloquium (2001)

Brandeis University, Department of Biology, Colloquium (1997, 2003)

Brown University, Shlossberg Colloquium (2017)

California Institute of Technology 2nd Annual Chen Center Distinguished Lecture (2018)

Cambridge University and the Royal Society, Symposium on Executive and Cognitive Functions of Prefrontal Cortex (1996)

Cardiff University, Cardiff Cognitive Neuroscience Seminar Series (2005) Carmel Conference XV (1997)

Carnegie Mellon University, Psychology Department Colloquium (1994, 2009)

CENTAI, Turin (2023), Navigating the Research Frontier of AI and Complexity

Cognitive Science Society (1992, 2025)

Cognitive Neuroscience Society (1995, 1996, 2000, 2002, 2006)

Cognitive Neuroscience Treatment Research to Improve Cognition in Schizophrenia Meeting, Invited Talk (2007)

Cold Spring Harbor Laboratory, Computational and Systems Neuroscience Workshop (2004)

College de France, Colloque de Rentrée, Invited Talk (2007)

Columbia Presbyterian Hospital, Joseph Zubin Memorial Fund Award Lecture (1994)

Columbia University, College of Physicians and Surgeons, Department of Psychiatry, Grand Rounds (1990)

Columbia University, College of Physicians and Surgeons, Department of Economics, Cognition and Decision Seminar Series (2016)

Computational Psychiatry 2017 Keynote Address (2017)

Computational Psychiatry 2018 Keynote Address (2018)

Cornell Medical School, Sackler Institute Colloquium (2002)

CUNY, Department of Psychology Colloquium (2000)

DARPA ISAT Toward Optimal Learning Workshop, Invited Address (2014)

Defense Basic Research Exchange Forum (2023)

Dynamical Systems in Neuroscience, Annual Meeting (1999)

Eden Institute Foundation, Lecture Series Fellow (2001)

Emory School of Medicine, Department of Psychiatry, Grand Rounds (1999)

Ellison Medical Foundation, Workshop of the Biological Assessment of Mental Processes (2006)

Eunice Kennedy Shriver Center for Developmental Cognitive Neuroscience, Colloquium (2000)

FENS and The Brain Prize, Brain Conference on New Insights into Psychiatric Disorders through Computational, Biological and Developmental Approaches, Keynote Address (2016)

Flatiron Analytic Connectionism Summer School (2025), Workshop on Computational Models of Human Intelligence

Florida State University, Department of Psychology, Colloquium (1998)

Frankfurt Institute for Advanced Studies, Ernst Strüngmann Forum (2007)

Future Science Prize Ceremony and Future Forum Science Symposium Keynote Address (2018)

Harvard University, Department of Psychology, Colloquium (1996, 2002)

Harvard University, Department of Economics, Labor Economics Seminar (2003)

Human Brain Project, Annual Conference (1998, 1999)

Indiana University, William Lowe Bryan Memorial Lecture on Cognitive Science (1992)

Institute for Advanced Studies, Department of Mathematics, Symposium (2003)

Institute of Psychiatry, King's College, London, Paul Janssen Lecture (2010)

Intel Corporation Annual Developers' Conference, Keynote Address (2016)

Intel Corporation, 2018 Consumer Electronics Show Spotlight Presentation (2018)

Intel Corporation Technology Strategy and Leadership Meeting: Outsider Perspective (2018)

Intel Labs Open Innovation Leadership Forum, Invited Address (2015)

Intel Labs, Mini-Symposium: The Mind's Eye Project (2016)

Interface 95 - The 27th Symposium on the Interface: Computing Science and Statistics (1995)

International Conference on Cognitive and Neural Systems, 10th Annual Meeting (2006), Invited Address

International Conference on Cognitive Neuroscience, Keynote Address (1996)

International Congress on Schizophrenia Research (1997), Invited Address

International Meeting on Fully Three-Dimensional Image Reconstruction in Radiology and Nuclear Medicine (1997)

International Neuropsychological Society (1992), Invited Address

James S. McDonnell Summer Institute in Cognitive Neuroscience (1995, 1997, 2001)

Japanese Neuropsychological Association, Keynote Address (1997)

Jena International Workshop on Executive Functions and the Brain (2000)

Kern Medina Seminar on Humanities and Science for State and Federal Judges (2014)

Lehigh University, Annual Neuroscience Retreat, Keynote Address (2015)

Library of Congress / NIMH Annual Decade of the Brain Public Program (1999)

Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Distinguished Guest Lecture Series (2011)

McGill University, Department of Psychiatry, Grand Rounds (1991)

Memory Disorders Research Society (1994, 1997, 1999)

Mind-Life Institute / M.I.T. (2003)

National Foundation for Functional Brain Imaging 1st Annual Meeting (1999)

New York Academy of Medicine, Annual Salmon Lecture (2006)

New York Academy of Sciences, Imaging Discussion Group Meeting (2005)

NIDA, Invited Seminar (2011)

NINDS, Cognitive Neuroscience Section, Grand Rounds (1993)

NIMH, St. Elizabeth's Hospital, Grand Rounds (1997)

NIMH Extramural program, Colloquia and Workshops (1999, 2000, 2001)

NIMH Intramural program, Neuroscience Colloquium (1999)

Nordic Center of Excellence and the Stockholm Brain Institute, Invited Talk (2007)

Northern California Psychiatric Society, Award Address (1986)

Northwestern University, Department of Psychology, Colloquium (1998)

NYU, Departments of Psychology and Neuroscience, Colloquia (1999, 2000)

Ohio State University, Mathematical Biosciences Institute Workshop on Systems Level Modeling (2002)

President's Council on Bioethics (2004)

Princeton Conference on Cerebral Vascular Disease (1994)

Princeton Plasma Physics Laboratory, Colloquium (2004)

Princeton University, Department of Psychology, Colloquium (1996)

Princeton University, Council on Science and Technology Public Lecture Series (2000)

Psychonomic Society, Invited Symposium Lectures (1996, 2002)

Queens College, CUNY, Annual Neuropsychology Symposium, Keynote Address (2007)

Reinforcement Learning and Decision Making, First Annual Meeting, Invited Address (2013)

Research Society on Alcoholism, Plenary Address (2002)

Rockefeller University, Neuroscience Colloquium (1999)

Rotman Research Institute, 10th Annual Conference on the Frontal Lobes (2000)

Royal Society, UK, Mental Processes in the Human Brain (2006)

Rutgers University, Department of Psychology & Center for Molecular and Behavioral Neuroscience Colloquium (1999, 2000)

Rutgers University Brain Health Institute, Invited Colloquium Address (2015)

Sierra Ventures 13th Annual CXO Summit Keynote Address (2018)

Simons Foundation SFARI Annual Scientific Meeting Keynote Address (2018)

SISA, Trieste Encounters in Cognition (1992)

Smithsonian Institute Public Lecture Series (1999)

Society for Psychophysiological Research, Invited Address (2006)

Society for Research on Psychopathology (1993)

Stanford University, Neurobiology Department, Frontiers in Neuroscience Lecture Series (2009)

Templeton Foundation, Annual Members Meeting Keynote Address (2016)

Templeton World Charities Fund Diverse Intelligences Grantee Meeting (2018)

TPG Annual Retreat, Featured Speaker (2007)

University of California, Berkeley, Helen Wills Neuroscience Institute Inaugural Lecture (2000)

University of California, Berkeley, Neuroscience Student Seminar Series (2010, 2016)

University of California, Davis, Keynote Address, Opening of Brain Imaging Center (2005)

University of California, Davis, Department of Psychiatry Grand Rounds (2005)

University of California, San Francisco, Department of Psychiatry Grand Rounds (2001)

University College London and Welcome Functional Imaging Laboratory (1997, 2000)

University of Colorado Boulder, Department of Psychology, Symposium (1997, 2002)

University of Colorado Boulder, Determinants of Executive Function & Dysfunction Conference (2013)

University of Illinois, Program in Neuroscience, Colloquium (1998)

University of Michigan, Departments of Psychology and Psychiatry Colloquia (1994, 2000)

University of Michigan, Marshall Weinberg Cognitive Science Symposium (2013)

University of Maryland, Psychiatric Research Center, 25th Anniversary Symposium (2002)

University of Maryland, Cognitive Science Colloquium (2016)

University of Medicine and Dentistry of New Jersey, Graduate Program in Physiology and Neurobiology, Special Lecture (1999)

University of Medicine and Dentistry of New Jersey, Dept. of Neurology Grand Rounds (2002)

University of North Carolina at Greensboro, Kendon Smith Annual Lecture Series (2004)

University of Oregon, Institute of Cognitive and Decision Sciences, Symposia (1990, 1996)

University of Pennsylvania, Department of Psychology, Cognitive Science Program, and Institute for Neural Sciences Colloquia (1996, 2001)

University of Pennsylvania, Institute of Neurological Sciences, James M. Sprague Annual Lecture (2006)

University of Pennsylvania and Philadelphia Psychoanalytic Center, Evening Program (2006)

University of Rochester, Department of Brain and Cognitive Sciences, Colloquium (2006)

University of Texas Austin, Cognitive Neuroscience & Imaging Research Center Seminar, Invited talk (2016)

University of Texas Southwestern Medical Center, Dept. of Psychiatry, Colloquium (2003)

University of Vermont, Department of Psychiatry, Grand Rounds (1992)

University of Waterloo, Centre for Theoretical Neuroscience, 5th Annual Brain Day (2011)

University of Waterloo, Neuroscience Colloquium Series (2021)

University of Wisconsin, Department of Psychology, Colloquium (1987, 2002)

University of Wisconsin Medical School, 5th Annual Symposium on Emotion (1999)

Vanderbilt University, Annual Neuroscience Retreat Keynote Address (2001)

Vanderbilt University, Stroopfest (2002)

Virginia Tech Carilion Research Institute Maury Strauss Distinguished Public Lecture (2018)

Washington University, Department of Psychiatry, Grand Rounds (2003)

Winter Conference on Brain Research (1993, 1996, 1997, 1998)

Workshop on Neural Modeling of Brain and Cognitive Disorders (1995, 1998)

Yale University School of Medicine, Department of Neurobiology, Colloquium (2002)

Yale University School of Medicine, Department of Psychiatry, Abraham Ribicoff Annual Lecture (2004)

3. Other research-related activities

Patents and Licenses

Title: ARTICLES AND METHODS FOR QUANTUM SCHEDULING, U.S. Provisional application 63/689,480 filed 8/29/25 and PCT/US2025/044232 application filed 8/30/25.

Advisory Boards and Councils

Allegheny County Neuropsychiatric Survey, Executive Advisory Board (1996-1998)

University of Michigan, Department of Psychology, External Advisory Board (1997)

National Alliance for Research on Schizophrenia & Depression (NARSAD), Scientific Council (1998-present)

NIMH Board of Scientific Counselors, Advisory Panel on Intramural Research Program (1999)

Yale-New Haven VAMC Schizophrenia Research Center, Scientific Advisory Board (1999)

International Organization of Human Brain Mapping, Governing Council (1998-2002), Treasurer (2000-2001), Chair of Neuroinformatics Committee (1998-2001), Chair, Nominations Committee (2001)

National Foundation for Functional Brain Imaging, Advisory Board (1999-2004)

Center for Magnetic Resonance Research, University of Minnesota, Advisory Board (2000)

Harvard Initiative in Systems Neuroscience, Advisory Board (2000)

American Psychiatric Association / NIMH DSM-V Workgroup on Neuroscience (2000-2002)

NIMH Workgroup on Strategic Plan for Mood Disorders (2000-2002)

International Association for the Study of Attention and Performance, Advisory Council (2001-present)

University of Pennsylvania NIMH Silvio O. Conte Center for Neuroscience Research, "The Neurobiology of Stimulus Encoding in Schizophrenia," External Advisory Board (2003, 2008)

Harvard University, Department of Psychology, External Review Committee (2003)

NIMH Measurement and Treatment Development Activities on Cognition in Schizophrenia (MATRICS), Neurocognition Committee (2002-2006)

Council of Princeton University, Executive Committee (2004-2005)

National Advisory Mental Health Council (NAMHC) (2004-2008)

The Society for Neuroeconomics, Board of Directors (2004-2005)

Gatsby Computational Neuroscience Unit, UCL, Quinquennial Review Panel (2005)

National Advisory Mental Health Council Workgroup on MRI Safety (2005-2007)

Brookhaven National Laboratory, Science and Technology Steering Committee (2005-2014)

Institute for Advanced Studies, Princeton, Decadal Visiting Committee for School of Social Sciences (2007)

National Advisory Mental Health Council Workgroup on Neuroscience Training (2007-2008)

University of Colorado, Boulder NIMH Interdisciplinary Behavioral Science Center, "Executive Function and Dysfunction," External Advisory Board (2009)

Johns Hopkins University, Psychological Brain Sciences Department and Mind Brain Institute External Review Committee (2011)

Princeton University, Research Computing Advisory Council, Member (2011-present)

Harvard University, Mind, Brain and Behavior Initiative, External Review Committee (2013)

Ecole Normale Supérieure, Scientific Advisory Committee of the Département d'Etudes Cognitives (2014-present)

National Academy of Medicine, Forum on Neuroscience (2015-2023)

Yale University, Wu Tsai Institute, External Advisory Board (2022-present)

Max Planck Society, Appointment Commission for MPI for Human Cognitive and Brain Sciences, Member (2024-present)

Princeton University, Senior Advisor for Computing and Data, Office of the Dean for Research (2025-present)

Mercator Fellow, German Research Foundation (2025-present)

Editorial Boards

American Journal of Psychiatry, Consulting Editor (2001-2006)

Biological Psychiatry, Board of Editors (1999-2009)

Brain Research, Senior Editor for Computational Neuroscience (2005-2010)

Cognitive Neuropsychology, Advisory editor (1997-2002)

Journal of Experimental Psychology: General, Consulting Editor (1996-2005)

Journal of Neurophysiology (2003-2004)

Neuroimage, Board of Editors (2002-2003)

Neuroinformatics, Board of Editors (2002-present)

Neuropsychopharmacology, Board of Editors (1999-2008)

Neuroscience, Board of Editors (1999-2003)

NMR in Biomedicine, Board of Editors (2003-2006)

Proceedings of the Royal Society, Biological Sciences, Board of Editors (2003-2008)

Science, Board of Reviewing Editors (1998-2014)

Trends in Cognitive Science, Advisory Editorial Board (2004-present)

Computational Psychiatry, Editorial Board (2014-present)

Grant Review

Integrative Cognitive Functional Neuroscience Study Section, NIH

Clinical Psychopathology Study Section, NIMH

Human Development and Aging Study Section, NIH

Human Frontier Science Program

Medical Research Council (MRC), UK

National Center for Research Resources, NIH

NIMH Intramural Research Program, NIH

NSF Review Panel

Wellcome Trust

Conference Organization

New Directions in Health Care and Education Annual Colloquium. University of Pennsylvania Medical School, May, 1980. Founder and Co-organizer.

25th Annual Carnegie Symposium on Cognition: Scientific Approaches to the Question of Consciousness. Carnegie Mellon University, May, 1993. Co-organizer.

Center for Neuroscience and Mental Disorders bi-annual workshop: Cognitive Neuroscience Approaches to Schizophrenia. University of Pittsburgh, May, 1994. Organizer.

International Congress on Schizophrenia Research. Colorado Springs, April, 1997. Program Consultant.

Society for Research in Psychopathology. Palm Springs, October, 1997. Program Committee.

Neural Processes & Economics Workshop, Woodrow Wilson School, Princeton University, 2000. Co-organizer.

Organization for Human Brain Mapping, New York City, 2003, Chair, Local Organizing Committee.

Computational Cognitive Neuroscience Conference, Co-Founder (with Randall O'Reilly); 2005-2008, Program Committee.

Membership in Professional Organizations

American Academy of Arts and Sciences

American Association for the Advancement of Science

American Psychological Society

Cognitive Science Society

Psychonomic Society

Society for Neuroscience

Software Development

PsyScope [14]: Designer and Co-Producer with Brian MacWhinney, Psychology, Carnegie Mellon University — this is a graphical, interactive program for the design and implementation of cognitive experimental tasks on Macintosh computers. It provides the ability to present stimuli in text, graphic, and acoustic form, and can be used to record manual or voice responses with millisecond accuracy. It incorporates a fully general scripting language, as well as a graphic interface, and is extensible through the use of plug-and-play add-on modules. PsyScope was originally designed for Mac OS prior to and through System 9. It was independently ported to MacOS X, and continues to be supported by the community, freely available, and widely used (with over 3,000 downloads) for experimental research and as a teaching instrument in research centers throughout the world. The design of PsyScope also provided one of the foundations for E-Prime, a PC/Windows-based

commercially supported package that was developed in collaboration with Psychology Software Tools (PST) Inc. and is also in widespread use.

Brain Image Analysis Kit [245]: Project Co-Director, with Ted Willke, Brain Inspired Computing Lab, Intel Labs; Ken Norman, Neuroscience and Psychology, Princeton; and Nicholas Turk-Browne, Psychology, Yale University — this is a Python-based, open source software package, developed in collaboration with Intel Labs, that supports the application of advanced methods from machine learning and multivariate statistics to the analysis of neuroimaging data. It is tightly integrated with SciKit-Learn, and includes modules for Full Correlation Matrix Analysis (FCMA; Wang et al. 2015), Multi-voxel Pattern Analysis (MVPA), a suite of methods for Shared Response Modeling (SRM) (including hyper alignment and Inter-Subject Functional Correlation [IFSC]), Topographic Factor Analysis (TFA), and Bayesian-derived methods for Representational Similarity Analysis (RSA). Within the first year and a half of its development it has attracted over 9,000 downloads.

RT-Cloud [251]: Project Co-Director, with Ken Norman, Princeton Neuroscience Institute. This is an open-source software package, integrated into the BrainIAK environment, that makes it easier to build and deploy real-time fMRI experiments. The framework provides a coordination hub between the experimenter's script, a subject feedback script, the scanner data, and experiment control. It streams scanner data (in real-time) to an experimenter's script and forwards the results for use in subject feedback (optionally using tools like PsychoPy, jsPsych, or PsychToolbox). It provides a web-based user interface that allows for starting and stopping runs, changing settings, and viewing output. It can be configured to run in the cloud, on a cluster, or in the control room. This framework is under active development with funding from NIMH to further extend its capabilities, including support for standards such as BIDS and OpenNeuro.

PsyNeuLink: Designer and Lead Developer, with Abhishek Bhattacharjee, Computer Science, Yale University and Amitai Shenhav, Brain and Cognitive Sciences, Brown University; and support from Templeton World Charitable Foundation and NIMH — this is a "block modeling environment" designed for use by neuroscientists, psychologists, computational psychiatrists and others interested in building system-level models of the computational mechanisms underlying brain function and its expression in psychological processes and behavior, and in exploring their relationship to developments in research on machine learning and artificial intelligence. It allows components to be constructed that implement various, possibly disparate functions, at potentially different levels of analysis and/or timescale of operation, and integrate these into a coherent modeling environment that can be used to simulate and study their interaction. PsyNeuLink is written in Python, is open source, and meant to be extended. Its goal is to provide an environment for implementing models that are expressed in a concise and easy to read form, and that can be executed, shared, compared, and integrated with one another. PsyNeuLink maintains a publicly accessible library of its components and models, to which users can contribute, providing a common repository for model-sharing in a manner paralleling data-sharing efforts in empirical research.

SweetPea [255]: Co-Designer, with Matthew Flatt and Vivek Srikumar, Computer Science, University of Utah — this is an experimental design environment that simplifies and standardizes the format in which experimental factors are expressed and combined, and used to generate balanced samples of valid trial sequences. As experimental designs in psychology and neuroscience — as well as the theories they are designed to test — become more complex, the ability to insure appropriately balanced sampling of relevant experimental and control conditions becomes increasingly difficult, and designs that do so increasingly difficult to express and/or interpret, which poses challenges to reliability and/or reproducibility of results. Similar problems arise in computational modeling, where uncontrolled or poorly understood differences in the inputs to a simulation can confound interpretations of its behavior, just as it does empirical data. SweetPea addresses this problem by providing a declarative language for expressing a design— in terms of factors (experimental variables), levels (values for those variable to be sampled), crossings among factors, and constraints on trial sequences (e.g., number or types of repeats). The design is then translated into a form (currently, conjunctive normal form) that can be sampled using a SAT solver, to insure that valid trial sequences are sampled in a balanced fashion.

ModECI (Model Exchange and Convergence Initiative) [269]: Lead Developer with Padraig Gleason, Neuroscience, Physiology and Pharmacology, University of College London and NeuroML; in collaboration with Abhishek Bhattacharjee, Computer Science, Yale University; Sharon Crook, Arizona State University and NeuroML, Ted Willke, Brain Inspired Computing Lab, Intel Labs; and support from the NSF Convergence

Accelerator Program — this project seeks to accelerate the convergence of progress in the brain, cognitive and computer sciences by facilitating the exchange of computational models through the development of a standardized model description format that expresses models in the form of a computational graph in which each node is a computational component and edges specify connections between them that determine the flow of computation. The format is implemented in serialized forms (e.g., JSON, YAML) that allows the exchange of models in machine-readable form among a broad range of existing environments — from biologically detailed ones (e.g., NeuroML, Emergent, and Nengo) and system-level level packages (such as PsyNeuLink) to abstract symbol processing environments used in cognitive science (such as ACT-R) and neural network modeling ones served by the ONNX standard used in the machine learning community (e.g., PyTorch and TensorFlow).